Virtuoso® Layout Pro: T1 Environment and Basic Commands

Course Version IC 25.1

Lecture Manual **Revision 1.0**

Cadence Design Systems, Inc. (Cadence), 2655 Seely Ave., San Jose, CA 95134, USA.

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Virtuoso Layout Pro: T1 Environment and Basic Commands

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Version IC 25.1

Revision **1.0**

Estimated time: **1 Day**

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Module 1

About This Course

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Course Prerequisites

Before taking this course, you need to have

- Experience with layout design
- Knowledge of schematic symbols and MOS devices
- A basic knowledge of UNIX/Linux

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Course Objectives

In this course, you

- Manage database libraries
- Analyze and customize the Virtuoso® graphic environment
- Use the Palette Assistant
- Use the custom bindkeys
- Use the Toolbar Manager
- Use the basic commands available in the Virtuoso Layout Suite

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Course Agenda

- The Design Environment
  - Opening Your Designs
  - Using the Library Path Editor
  - Using the Library Manager
  - Setting User Preferences
  - Using History and Bookmarks
  - Using World View
  - Floating the Palettes Away from Cellviews and Saving Workspaces
- User Interface
  - Using the Layers, Objects, and Grids Assistants
  - Loading and Verifying the Bindkeys
  - Saving the Defaults
  - Using the Toolbar Manager
- Basic Layout Commands
  - Viewing Your Designs
  - Selecting and Deselecting the Objects
  - Using the Alignment Commands
- Virtuoso iPegasus DRC and Fill for Virtuoso Studio
  - Submodule: iPegasus DRC for Virtuoso Studio
  - Submodule: iPegasus FILL for Virtuoso Studio
  - Setting Up and Running iPegasus Signoff DRC
  - Setting Up and Running iPegasus Signoff Fill

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Software and Licenses

For the software and licenses used in the labs for this course, go to:

https://www.cadence.com/en_US/home/training/all-courses/85087.html

If there is additional information regarding the specific software, it is detailed in the lab document and/or the README file of the database provided with this course.

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How do I register to take the exam?

- Log in to our Learning Management System, click on the course in your transcript, and go to the Content tab to locate the exam.

How long will it take to complete the exam?

- Most exams take 45 to 90 minutes to complete. You may retake the exam multiple times to pass the exam.

How do I access and use the digital badge?

- After you pass the exam, you get a digital badge and instructions on how to place it on social media sites.

How is the digital badge validated?

- Credly validates the digital badge as issued to you by Cadence and includes the details of the criteria you completed to earn the badge.

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Icons Used in This Class

Throughout this class, we use icons to draw your attention to certain kinds of information. Here are the icons we use, and what they mean.

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Module 2

The Design Environment

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Module Objectives

In this module, you

- Start the Virtuoso® Layout Suite with the command *virtuoso &*
- Use the Command Interpreter Window (CIW)
- Use the Library Path Editor
- Use the Library Manager
- Set user preferences
- Use history, bookmarks, and world view
- Set and save workspaces

Terms and Definitions

Library

- A collection of related design elements

Cell

- The name of a design element

View

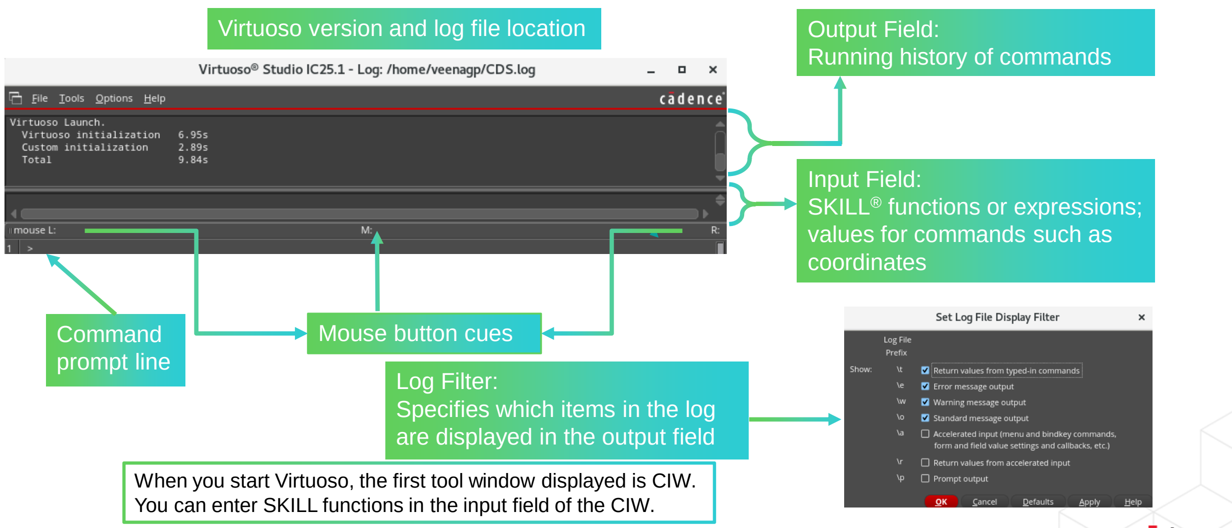
- Location of the actual design data

CIW

- Command Interpreter Window, the main interface into the Virtuoso environment

The Command Interpreter Window (CIW)

To start the Cadence® Virtuoso software, enter the **virtuoso &** command in the terminal window. The Command Interpreter Window (CIW) appears.



Opening the CIW

When you start Virtuoso, the first tool window displayed is CIW. You can enter SKILL functions in the input field of the CIW.

Menu Banner

You can display command menus from the menu banner running across the top of the CIW. For example, you often use the File menu to access a library or to edit a specific schematic or layout design in a library.

Input Field

Using the input field, you can enter and edit SKILL expressions, commands, and user-defined scripts. By using window numbers, you can direct SKILL commands to other tool windows. Usage of the expanded input field for editing SKILL code is available.

Prompt Line and Mouse Button Cues

The prompt line at the bottom of the CIW reminds you of what you need to do next to continue a command. The mouse button cues (L, M, R) display what you can do with each mouse button as you execute commands.

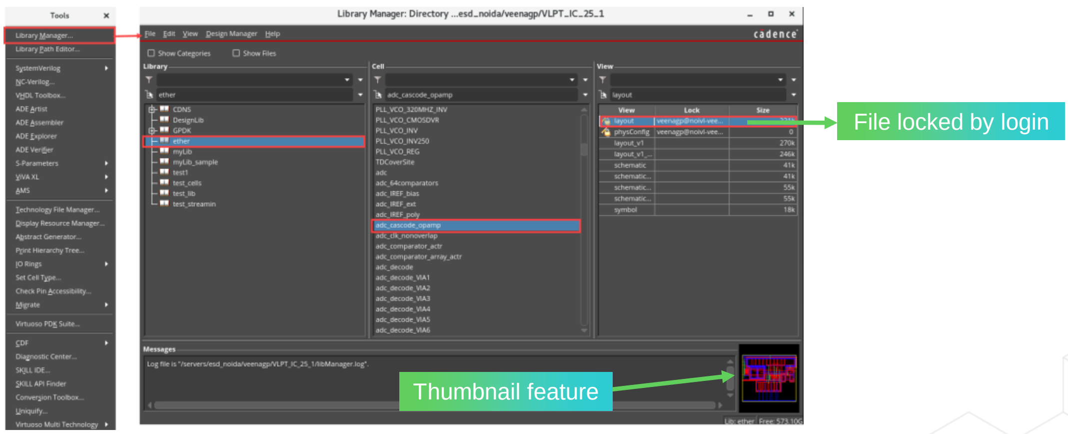
Output Field

The output field displays a running history of the commands you have executed and their results. All this information is recorded in a log file for later replay. To filter the information, choose the **Options – Log Filter** command.

The Library Manager Overview

To open the Library Manager, choose **Tools – Library Manager** in the CIW. The Library Manager window appears.

A library is a collection of cells. It also contains all the different views associated with each cell.



Library

A library is a collection of cells. The library also contains all the different views associated with each cell.

Cells and Views

A cell is a low-level building block you use to create a chip or logical system.

A view is a specific representation of a cell. Each cell can have a layout view, a schematic view, a symbolic view, and so forth.

Cellviews

A cellview is a combination of a cell and a view. A cellview designates both the cell and view, such as the *inverter layout*.

Canvas

A canvas is a viewport where the design is opened.

Thumbnails

To generate thumbnails, use the following command:

```
hiGenerateThumbnails(?lib “Lib_Name” ?cell “Cell_Name” ?view “layout”)
```

If you want to run just a library that generates *all* cell and view thumbnails, eliminate the *?cell* and *?view* arguments:

```
hiGenerateThumbnails(?lib “Lib_Name“)
```

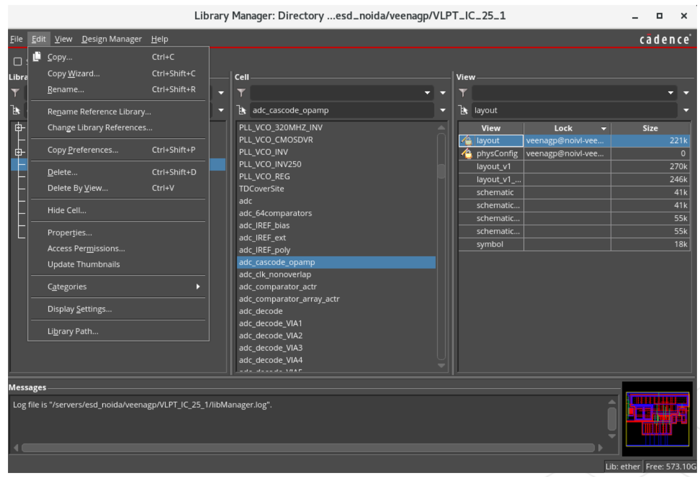
Refer to the *Cadence User Interface SKILL Reference, Product Version IC25.1*, for more details on this feature.

Editing Data with the Library Manager

To open the Library Manager, choose **Tools – Library Manager** in the CIW. The Library Manager window appears.

The **Library Manager** is the interface you use to create, add, copy, delete, or organize the cells and libraries you use in a design project.

- Copy Cells
- Rename a Library
- Rename a Reference Library
- Edit Properties
- Delete a Cell
- Hide a Cell
- Delete a Library
- Delete a View
- Create Categories



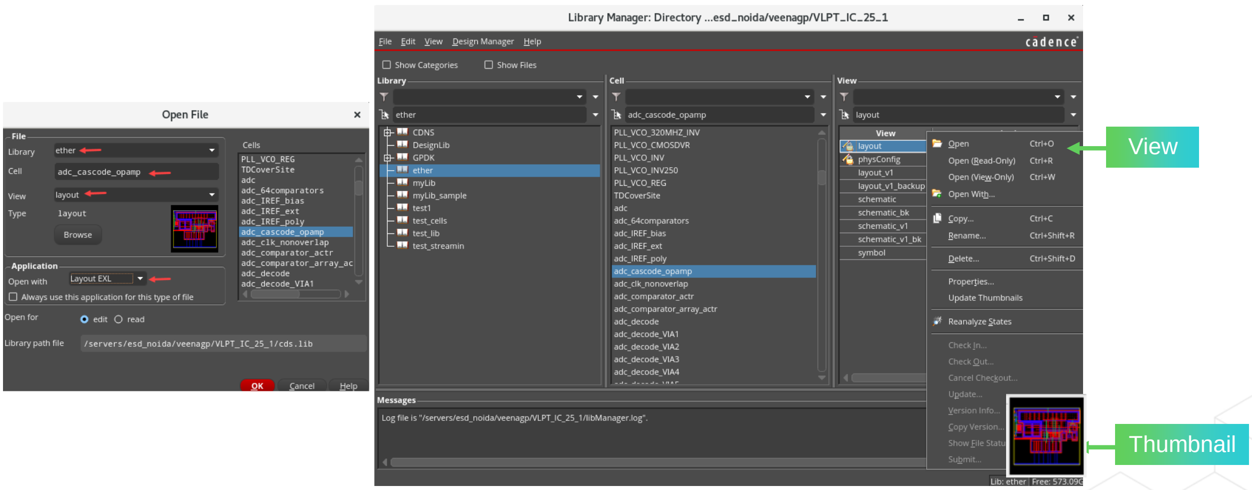
The **Library Manager** is the interface you use to create, add, copy, delete, hide, or organize the cells and libraries you use in a design project. The Library Manager also includes the **Library Path Editor**, which you use to define the path to the libraries you want all your Cadence design tools to access in the *cds.lib* file.

The Library Manager includes functions that you use to:

- See and traverse the directories, libraries, cells, and cellviews in your directory structure.
- Put cells in categories for better organization.
- Open a terminal window to check the location of your files and hierarchies.

Opening a Design

In the CIW, choose **File – Open** to display the Open File form. Use this form to open the design. Or **double**-click the desired view or the thumbnail in LM to open the design.



Opening a Design

To access an individual cell for editing, you need to specify the library, cell, and view names.

Using the Open File Form

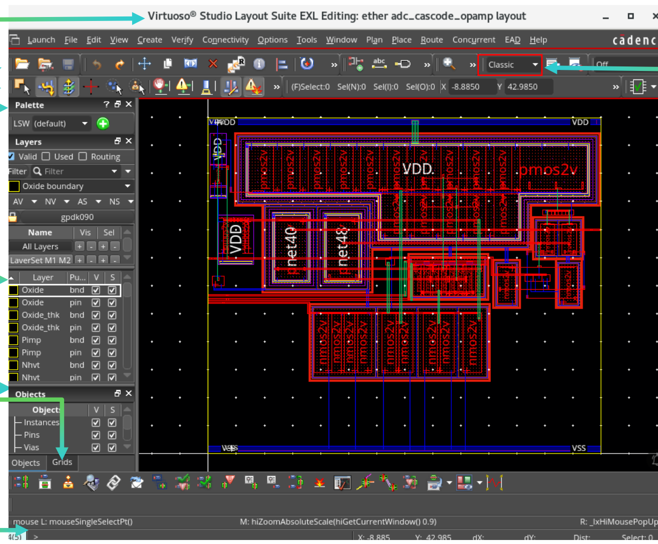
An easy way to open a design is to use the Open File form. Here, you specify the library name from a cyclic field. This information is taken from the *cds.lib* file. You can either enter the cell name or click the **Browse** button to display the Library Browser form. In the Library Browser form, you can click the cell and view, which updates the Open File form. You can also get the cell and view name information from the Open File form.

Using the Library Manager Form

With the Library Manager, you can not only open a design but also edit the information inside of your library. You still need to specify the library, cell, and view name. If there are any cells that reside in categories, you can list the categories. Categories are a way of grouping cells together so that you can find them faster. You create the category in the Library Manager and then specify which cells reside in that category. If you do not know the full name of a cell, you might want to use the **Filter** field. You can use wildcards in the Library Manager. For instance, if you enter **d***, then you receive a listing of all cells that begin with *d*.

1 The Classic Design Window

The Palette assistant is activated by default in the Classic and Basic workspaces.



The Virtuoso Layout Suite has two initial workspaces:

1. Classic
2. Basic

When you select a cellview from a specific library, the design window opens with your design. Each window is numbered in the order in which it is opened.

Title banner – Specifies the library name, cell name, and view name. The window banner also specifies whether the cell is open for reading or editing purposes.

Status banner – Displays information about the cursor, selection, points, and command.

Menu banner – Displays the Layout Editor menus. You can click on an item in the banner to display a menu of commands.

Icon menu – Appears on the edges of design windows. You can start common commands quickly by clicking on an icon.

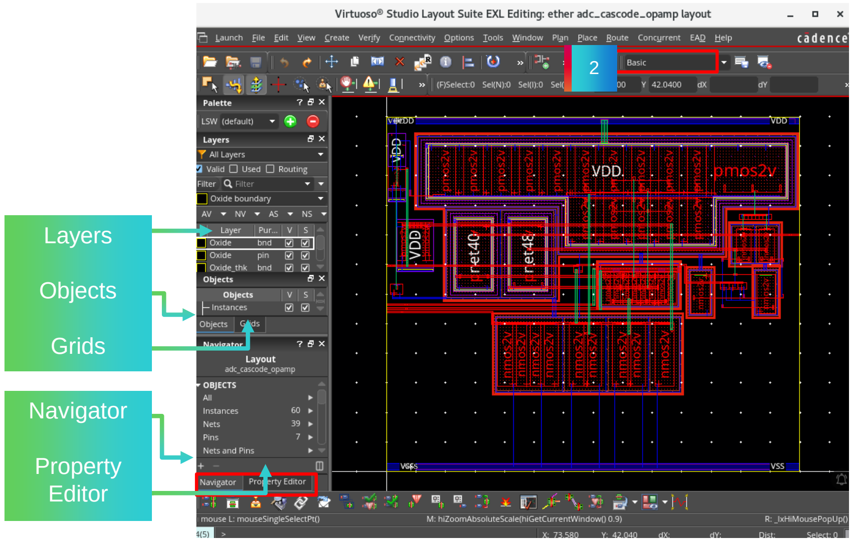
Mouse settings – At the bottom of the window, shows the command state when clicking the left, middle, or right mouse button.

Prompt line – At the very bottom of the window, shows instructions from the current command.

Design Canvas – Lets you work in the familiar Classic workspace or switch to the Basic workspace, complete with Navigator and Property Editor assistants. Additionally, you can create personalized workspaces by docking or undocking the Navigator, Property Editor, World View, and Annotation Browser assistants, tearing off menus, etc., and saving them in various configurations for later sessions.

2 The Basic Design Window

The Palette assistant is activated by default in the Classic and Basic workspaces.



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This is the **Basic** workspace. It offers the Layers, Objects, Grids, Navigator, and Property Editor assistants. You can modify this workspace to include other assistants, such as World View.

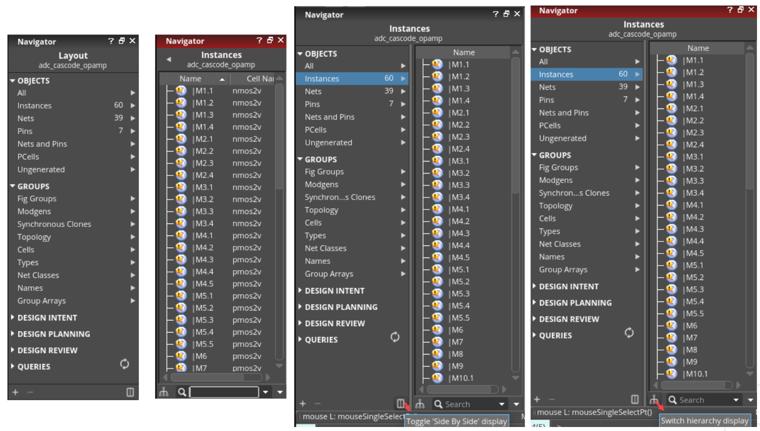
"Flightlines" simulate the connectivity between pins when you select a net, pin, or instance from the Navigator, or from the canvas itself. There is NO physical connectivity in Layout XL as there is in EXL or MXL; however, through the **Highlight** function, you will now have an additional interactive routing aid when in Layout EXL.

Assistants can be dismissed (and recalled) by toggling the **F11** function key on your keyboard. This will maximize your canvas whenever you do not require assistants.

The Navigator Changes

In the layout window, change the workspace to Basic. It offers Layers, Objects, Grids, Navigator, and Property Editor assistants.

- Supports two major categories, **OBJECTS** and **GROUPS**, and further detail views.
  1. **OBJECTS** has All, Instances, Nets, Pins, Nets and Pins, PCells, and Ungenerated. Shows object counts upfront.
  2. **GROUPS** has Fig Groups, Modgens, Synchronous Clones, Topology, Cells, Types, Net Classes, Names, and Group Arrays.
- **Current and Below** filters are replaced with the **Switch hierarchy display** button.
- **Toggle 'Side By Side' display** is used to arrange the sets and their contents sideways.
- Selection and toggle settings are saved in the *./.cadence/dfII/Navigator/Options.xml* file.



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Toggle 'Side By Side' display is used to arrange the sets and their contents sideways.

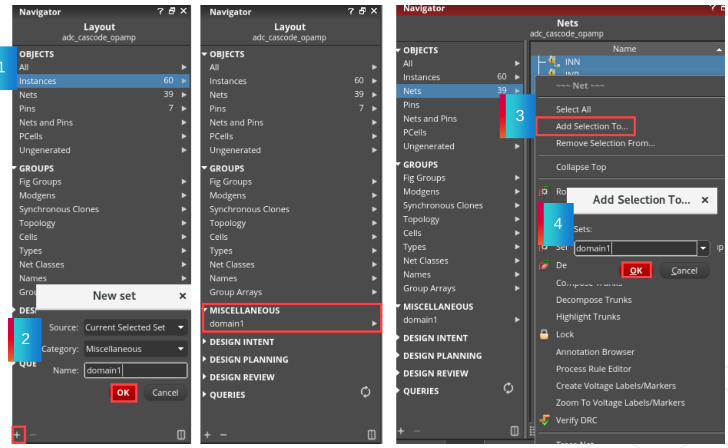
The **Switch hierarchy display** button is used to control whether to display the design hierarchy along with the members of the set. This is useful when you are operating at the lower hierarchy levels in the design.

The parameters for parameterized cells (Pcells), i.e., Pcell Parameters, appear in the **Create Instance** form as you place an instance of the Pcell.

Creating and Using New Sets in the Navigator

Add Set and *Remove Set* enables you to add a new set and delete an existing set, respectively.

- Click **Add Set** to open the **New set** dialog box and select the **Category** under which the new user-defined set is to be created.
- Once the new user-defined set is created, you can add objects to this set using the **Add Selection To** option.



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Add Set or *Remove Set* enable you to add a new set or delete an existing set, respectively. Click **Add Set** to open the **New set** dialog box and select the Category under which the new user-defined set is to be created.

Once the new set is created, you can use one of the following methods to add or remove the objects from the set:

- **Right**-clicking in the Summary pane to view the context menu and selecting **Add Selected Set**
- *Select Content*: Selects all the members in a set
- *Add Selected Set*: Adds the objects/sets, selected in Navigator, layout canvas, or Property Editor, to the current user-defined set
- *Rename*: Renames the set
- *Delete*: Deletes the set

Click **Add Selection To** in the context menu of the Details pane to view the **Add Selection To** dialog box. You can choose from the existing user-defined sets in the Sets drop-down list box or specify a new set in the text field where the selected objects are required to be added.

- *Select All*: Selects all the members in a set
- Select the objects in the design and click + to add them to a set

Creating and Using New Sets in the Navigator (continued)

- New sets created by the user will, by default, show up under the category **Miscellaneous**.
- Objects can be added to the user set using the RMB option "**Add Selection To**".
- The user-added sets are **saved in an OA database**, so it will be cellview specific.
- *opcAddListToSet* and other *opc** SKILL APIs are provided for the Navigator assistant.

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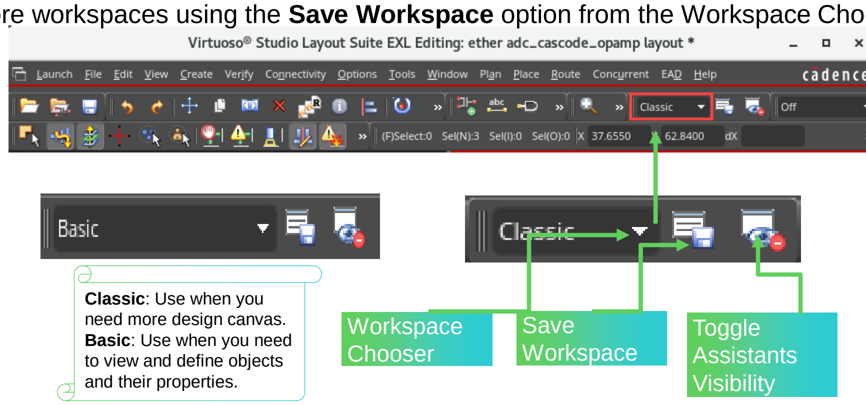
What Is the Workspace Chooser?

In the Virtuoso Layout Suite, the Workspace Chooser is where you select from two initial workspaces:

1. **Classic** – Just the Design Canvas
2. **Basic** – Basic Design Canvas + Navigator + Property Editor Assistants

You can modify each of these workspaces and add your own personalized workspace(s) for later use. You can create more workspaces using the **Save Workspace** option from the Workspace Chooser.

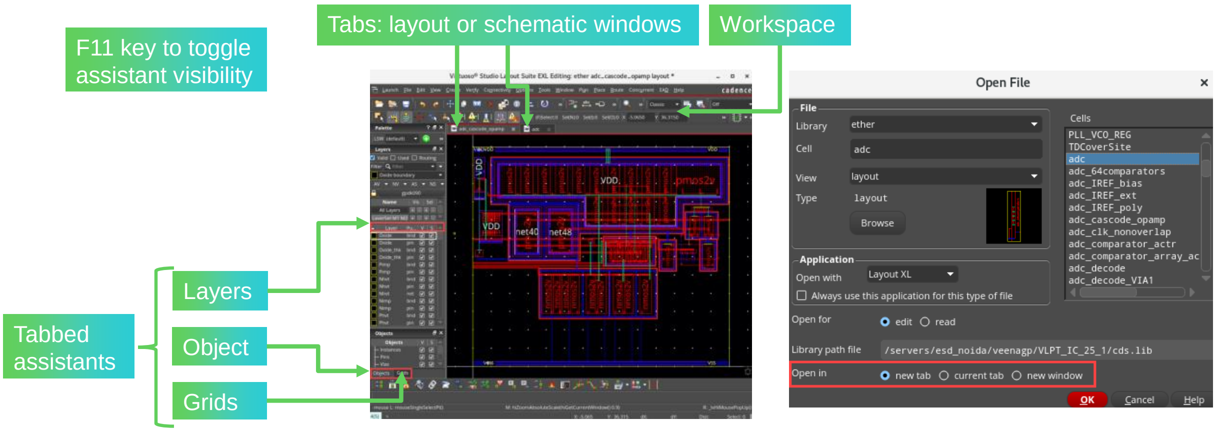


Although the Classic and Basic workspaces are originally coded to come with no assistants (Classic) and only the Navigator and Property Editor assistants (Basic), you can customize any workspace to include/exclude any of the assistants. You can even create a new assistant using the **Save Workspace** option from the Workspace Chooser.

Additionally, you can move the assistants off the Virtuoso canvas altogether and place them in your terminal window, then save that configuration as an entirely new workspace.

Dockable Assistants and Tabs

In the layout window, choose **File – Open** and choose the *new* tab or *current* tab option in the **Open in** field to open more than one cellview in the same window. Use the tabs to bring the cellviews forward.



You can open more than one cellview in the same window and use the tabs to bring them forward.

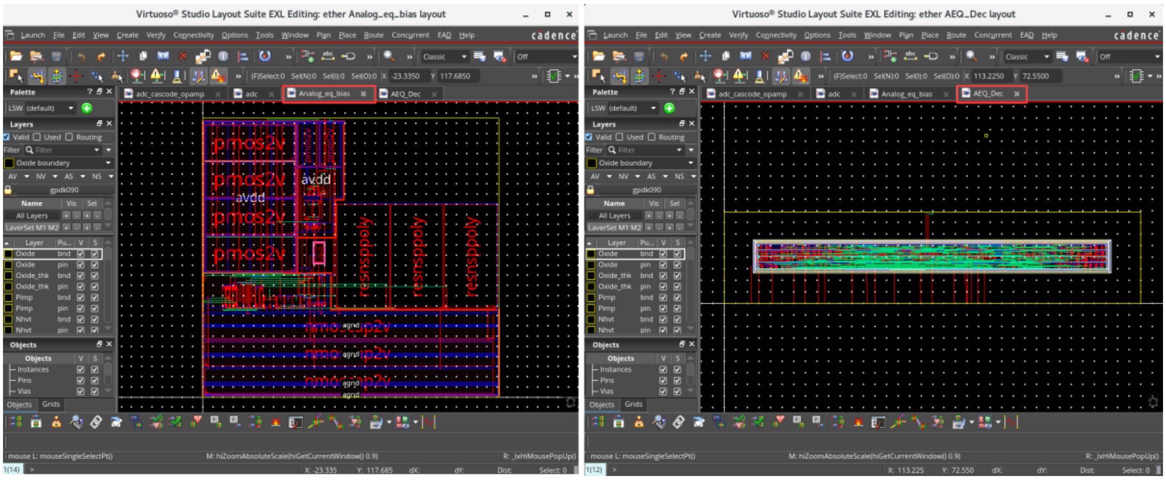
The assistants are configurable for choice and location. You can also save the configuration as a workspace, which appears in the cyclic field.

The assistants highlight the selected design element.

You can open more than one cellview in the same window, and use the tabs to bring them forward.

Tabbed Cellviews

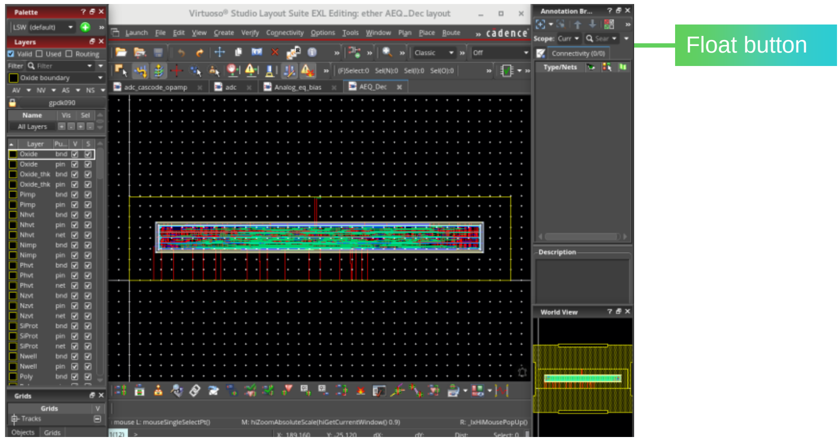
You can open more than one cellview in the same window and use the tabs to bring them forward, as shown here.



You can open more than one cellview in the same window and use the tabs to bring them forward.

Undocked Assistants

In the layout window, click the **Float** button of the assistants to tear them off from the layout and move them to the required location.

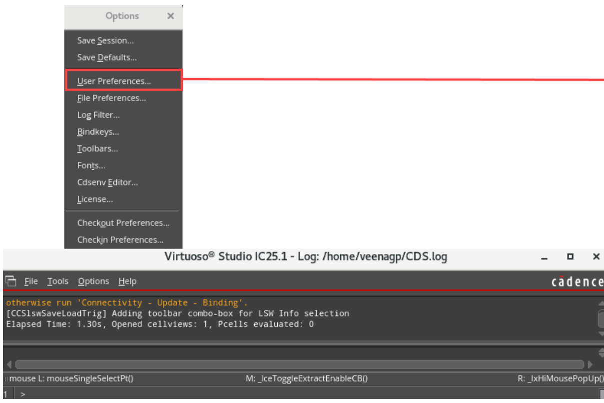


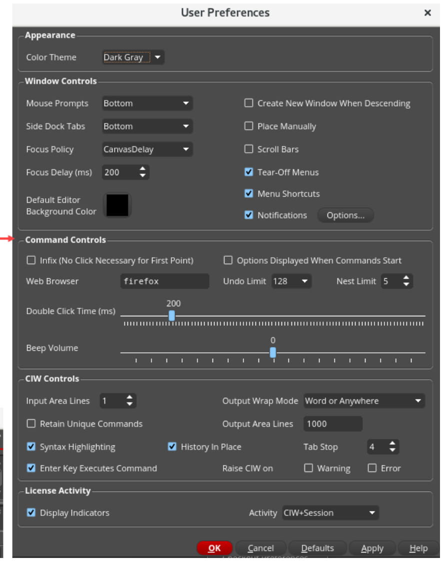
You can undock the assistants and place them outside the canvas.

When you configure the design space, you can also choose to expand the design area by undocking the assistants and moving them outside the design space. You can tear them off and drag them outside the canvas.

Global Settings

From the CIW, choose **Options – User Preferences** to open the User Preferences form.





The **User Preferences** form sets common interface choices.

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Setting Window Controls

To set the window controls for your design windows, use the User Preferences form. You can specify whether the design windows have scroll bars, fixed menus, prompt lines, or status lines.

Setting Specific Options

You can set infix from this form. Infix is used only when you invoke commands with bindkeys. The cursor location is taken as the first point of the command, and you are prompted to give the second point, thus reducing mouse clicks. If you want the Options forms displayed at startup, instead of **double-**clicking with the middle mouse button or pressing the **F3** key, turn on the **Options Displayed When Commands Start** option.

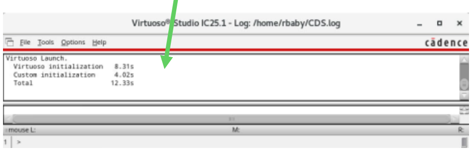
Saving Window Controls

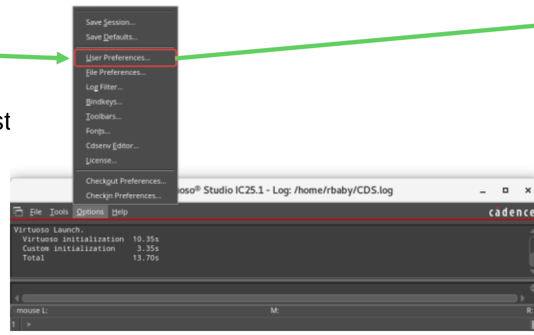
If you click **OK** or **Apply** in the User Preferences form, the changes are saved for that editing session only. To save this information as the default environment, save it to the *.cdsenv* or *.cdsinit* file.

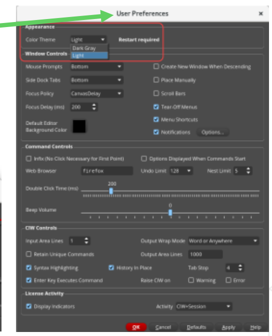
Switching the Color Theme of the Virtuoso Studio Session

From the CIW, select **Options – User Preferences** and use the *Appearance* section at the top of the form to choose the color theme for the Virtuoso Studio session.

- In IC25.1, you can select between two themes:
  - Dark Gray (default)
  - Light
- The theme change is applied after you rest the session, as shown here.







The appearance setting is user-specific and stored in *~/.cadence/ui/theme.json* file.

Virtuoso Studio Themes

Virtuoso Studio provides the flexibility to change color themes. For some, visual performance tends to be better when viewing the design in a Light theme and for some using the Dark Gray theme generates a better visual experience.

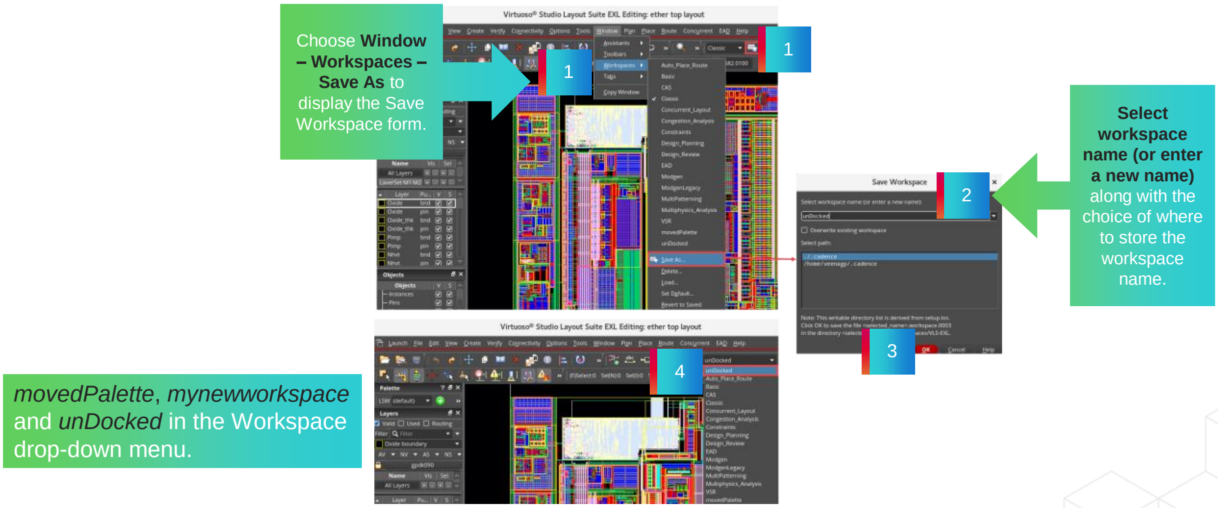
Switching to the Dark Gray theme reduces power usage, improves visibility, and makes it easier for designs to be read. In this theme, the background is Dark Gray, and the text uses a contrasting lighter color. The Dark Gray theme also introduces modern visual effects such as mouse hover, smaller sidebar and flat design.

By default, the CIW opens in dark mode. To switch between color themes, use the Color Theme option in the Appearance section of the User Preferences form.

The Dark Gray theme is introduced for VSE, VLS and CIW, and for all Cadence-style windows such as Library Manager, SKILL API Finder, ADE Maestro, and Pegasus.

Saving the Workspace

You can save the workspace configuration for improved performance.



Choose **Window – Workspaces – Save As** to display the Save Workspace form. In this form, **Select workspace name (or enter a new name)** along with the choice of where to store the workspace name. After clicking **OK**, you are able to recall the **saved** configuration with the Workspace cyclic field.

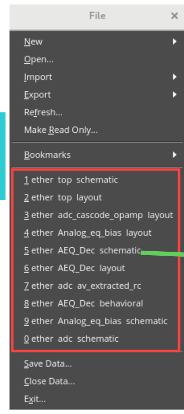
Using History

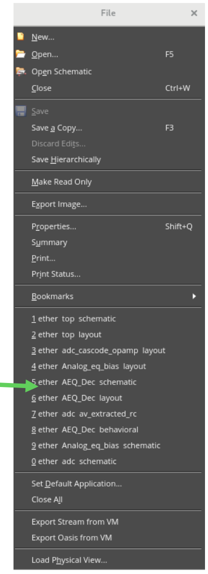
From the CIW or layout canvas, choose the **File** menu to view the **History** of the recently opened files.

History is available from the **File** pull-down menu in the CIW window.

History is also available from the layout canvas in the **File** pull-down menu.

List of recently opened cellviews in the CIW and in the layout canvas.

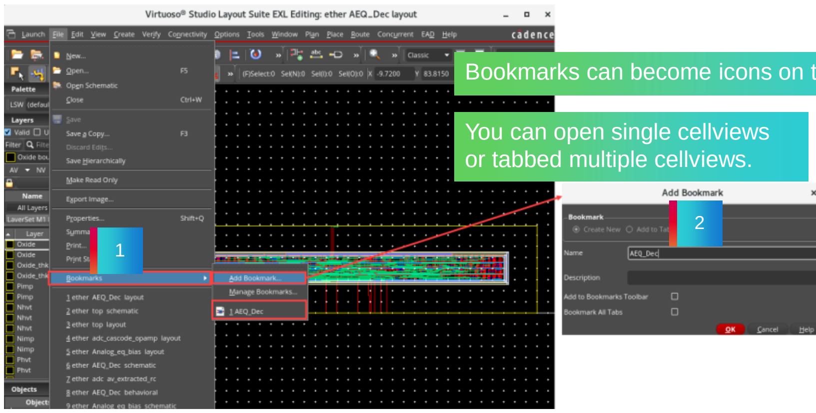




The files that have been recently opened are kept in these locations, so you can find them and open them more easily and quickly.

Adding Bookmarks

Bookmarks are designed to keep particular sets of cellviews available for quick editing. The bookmarks can be located in the toolbar as icons or in the **File** pull-down menu in the layout canvas, so you can quickly open commonly used cellviews.



Bookmarks can become icons on the toolbar.

You can open single cellviews or tabbed multiple cellviews.

You can add bookmarks.
Choose **File – Bookmarks – Add Bookmark** to display the Add Bookmark form.

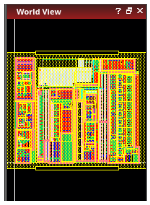
Bookmarks are designed to keep particular sets of cellviews available for quick editing. The bookmarks can be located in the toolbar as icons or located in the **File** pull-down menu in the layout canvas, so you can quickly open commonly used cellviews.

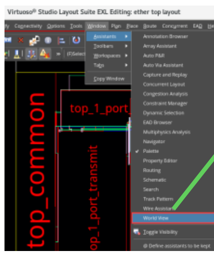
Note: Information about Bookmarks, History, Navigator, and Workspaces are stored in the *.cadence/dfII* directory of your working library.

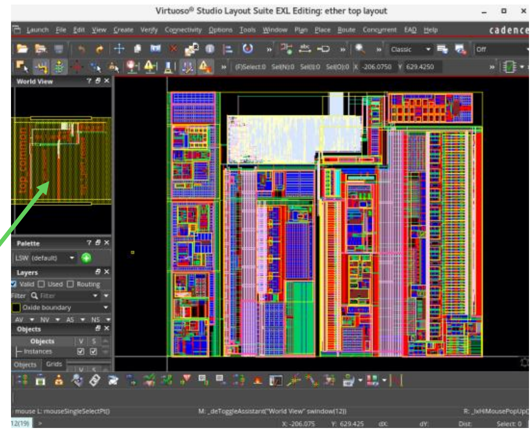
Using World View Assistant

In the layout canvas, choose **Window – Assistants – World View** to display the World View assistant.

World View assistant is undocked.







Use the assistant to pan and zoom the main canvas.

World View

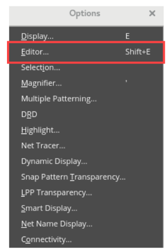
You can access the World View assistant by choosing **Window – Assistants – World View**.

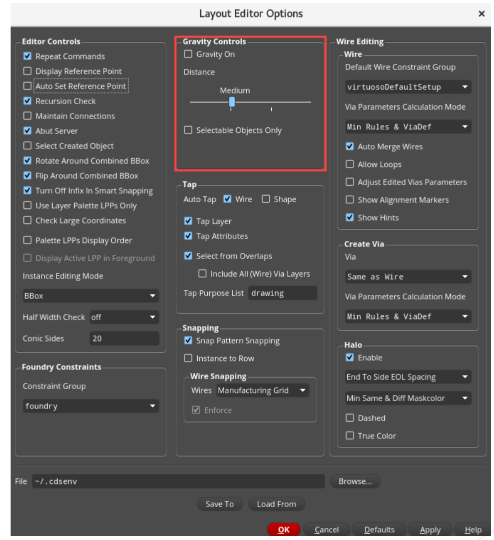
The World View assistant is a navigation tool, especially useful in large designs. It shows you a complete picture of your entire design, and marks the part of the design that is currently displayed in the drawing area.

Gravity Controls in the Layout Editor Options Form

In the layout canvas, choose **Options – Editor** or press the bindkey **Shift+E** to open the Layout Editor Options form.

- New snapping algorithm and improved snapping engine for fast and accurate cursor snapping.
- Simplified controls: One control to turn on/off the Gravity and one control to change the snap distance from the cursor to an object.
- Cursor snaps to all visible objects in any hierarchy level by default.
- Same control for Gravity and Smart Snapping.





Gravity Controls

Gravity On sets the drawing pointer to automatically snap to objects in the cellview. The pointer snaps to all the visible objects at any hierarchy level.

- Environment variable: *gravityOn*

Distance slider sets the snap distance between the pointer and an object. You can choose from *Small*, *Medium*, *Large*, or *Very Large* using the slider.

- Environment variable: *gravityDistance*

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1* for details on these features.

Module Summary

In this module, you learned how to

- Start the Virtuoso Layout Suite with the command *virtuoso &*
- Use the Command Interpreter Window (CIW)
- Use the Library Path Editor
- Use the Library Manager
- Set user preferences
- Use history, bookmarks, and world view
- Set and save workspaces

This page does not contain notes.

Labs

Lab 2-1 Opening Your Designs

Lab 2-2 Using the Library Path Editor

Lab 2-3 Using the Library Manager

Lab 2-4 Setting User Preferences

Lab 2-5 Using History and Bookmarks

Lab 2-6 Using World View

Lab 2-7 Floating the Palettes Away from Cellviews and Saving Workspaces

This page does not contain notes.

Quiz: The CIW Window

When the Virtuoso software is started, which tool window is displayed first?

Select the correct choice.

A. Layout
B. Schematic
C. Palette
D. CIW
E. None of the above

Answer: D

When you start the Virtuoso software, the first tool window displayed is the CIW window.

Answer: D

When you start the Virtuoso software, the first tool window displayed is the CIW window.

Quiz: The Initial Workspaces in VLS

What are the initial workspaces in the Virtuoso Layout Suite?

Select all that apply.

A. Classic
B. Constraints
C. Basic
D. Floorplan
E. All of the above

Answers: A and C

There are two initial workspaces in the Virtuoso Layout Suite: Classic, which has just the Design Canvas, and Basic, which has the Basic Design Canvas + Navigator + Property Editor Assistants. However, you can modify each of these workspaces and add your own personalized workspace(s) for later use.

Answers: A and C

There are two initial workspaces in the Virtuoso Layout Suite: Classic, which has just the Design Canvas, & Basic, which has the Basic Design Canvas + Navigator + Property Editor Assistants. However, you can modify each of these workspaces and add your own personalized workspace(s) for later use.

Quiz: The Tabs in Layout

You cannot open more than one cellview in the same session window.

A. True

B. False

Answer: False

You can open more than one cellview in the same window and use the tabs to bring them forward.

Answer: False

You can open more than one cellview in the same window and use the tabs to bring them forward.

Quiz: The History in Layout

History: The files that have been recently opened are kept in these locations, so you can find them and open them more easily and quickly. History is available in the following locations:

Select all that apply.

A. History is available from the Library Manager.
B. History is available from the File pull-down menu in the CIW window.
C. History is available from the Library Path Editor.
D. History is available from the layout canvas in the File pull-down menu.
E. All of the above.

Answers: B and D

History is available from the File pull-down menu in the CIW window and from the layout canvas in the File pull-down menu.

Answers: B and D

History is available from the File pull-down menu in the CIW window and from the layout canvas in the File pull-down menu.

Quiz: The Thumbnails Feature in Layout

What SKILL function is used to generate thumbnails for the given library cellviews?

The **hiGenerateThumbnails()** SKILL function generates thumbnails for the given library cellviews.

What is the SKILL function which is used to generate thumbnails for the given library cellviews?

- The **hiGenerateThumbnails()** SKILL function generates thumbnails for the given library cellviews.

Quiz: The Library Manager in Virtuoso

What interface is used to create, add, copy, delete, hide, or organize the cells and libraries you use in a design project?

You can use the **Library Manager** interface to create, add, copy, delete, hide, or organize the cells and libraries you use in a design project.

What is the interface which is used to create, add, copy, delete, hide, or organize the cells and libraries you use in a design project?

- The **Library Manager** is the interface you can use to create, add, copy, delete, hide, or organize the cells and libraries you use in a design project.

Quiz: Setting the Window Controls in Virtuoso

Which command is used to set the window controls in the layout canvas?

You can use the **User Preferences** command from the CIW window to set the window controls in the layout canvas.

Which command is used to set the window controls in the layout canvas?

- You can use the **User Preferences** command from the CIW window to set the window controls in the layout canvas.

Quiz: The Path Information for the Libraries in Virtuoso

When the Virtuoso software is started, which file is read to provide the path information for the libraries available?

When you start the Virtuoso software, the **cds.lib** file is read to provide path information for the libraries available.

When the Virtuoso software is started, which file is read to provide path information for the libraries available?

- When you start the Virtuoso software, the ***cds.lib*** file is read to provide path information for the libraries available.

Module 3

User Interface

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Module Objectives

In this module, you

- Use the Layer, Object and Grid Assistants
- Customize the drawing layers
- Define and use layer sets
- Use custom bindkeys
- Save user-specified options as defaults
- Build your own customized toolbar
- Use Dynamic Display: Info Balloon, Dynamic Measurement, and Highlight Aligned Edges

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What Is the Palette Assistant?

Palette is a tool used to view and manage the display context of layer, object, and grid items by editing their properties. The Palette Assistant comprises **Layers**, **Objects**, and **Grids** panels.

The key features of the Palette Assistant are listed below:

- Enables you to configure the display context by setting the visibility and selectability properties of layer, object, and grid items.
- Provides access to various layer properties, including routing direction, GDS number, and technology file properties, such as priority, function, description, and mask.
- Enables you to edit the routing direction of layers.
- Provides options to filter layer, object, and grid items. These options include layer set activation, search filter, and used/routing/valid layers filtering.
- Enables management of layer properties individually or as groups.
- Supports both synchronized and independent display contexts for palettes that are attached to different windows.
- Includes the Options form, which is used to configure Palette default behavior.
- Enables configuration of Palette settings in the Virtuoso® workspace. The Palette configuration settings are saved as part of the Virtuoso workspace.

Note: The *Palette Assistant* is activated by default in the Classic and Basic workspaces.

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You can configure the Palette Assistant to dispatch the three panels in three different types of standalone containers. Refer to the Palette Containers section in the *Virtuoso Layout Viewer User Guide*, *Product Version IC25.1* for more information.

Features of the Palette Assistant

- Enables you to specify different palette settings for different layout windows
- Can be docked to a layout window or undocked from the layout window
- Provides improved sorting and filtering capabilities
- Lists only the layer purpose pairs present in the current cellview
- Enables you to group layer purpose pairs in the Layers Assistant using *View By*
- Displays additional information, such as routing direction and functions
- Enables you to specify the routing direction for layer purpose pairs
- Enables you to specify a predefined filter to suppress layer purpose pairs
- Enables you to save the Palette Assistant in a workspace
- Enables you to group layer purpose pairs in the Layers Assistant

Palette Assistant/Container/Panel

The Virtuoso Layout Suite supports the following types of Palette containers:

1. Single Assistant Container (Default)
2. Multiple Assistants Container
3. Single Window Container

The default Palette container type is SingleAssistant. You can change the Palette container type by setting the CDS_PALETTE_TYPE UNIX environment variable. This UNIX environment variable value must be set before running Virtuoso.
Enter one of the following commands in the shell environment to set the Palette type:

- `setenv CDS_PALETTE_TYPE SingleAssistant`
- `setenv CDS_PALETTE_TYPE MultiAssistant`
- `setenv CDS_PALETTE_TYPE SingleWindow`

Important: The *CDS_PALETTE_TYPE* UNIX environment variable value must be set before running Virtuoso. Its value cannot be changed during a Virtuoso session.

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Setting the Palette Type

The default Palette container type is *SingleAssistant*. You can change the Palette container type by setting the *CDS_PALETTE_TYPE* UNIX environment variable.

Enter one of the following commands in the shell environment to set the Palette type:

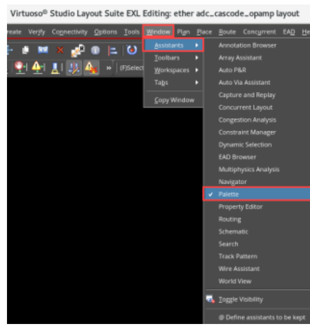
- `setenv CDS_PALETTE_TYPE SingleAssistant`
- `setenv CDS_PALETTE_TYPE MultiAssistant`
- `setenv CDS_PALETTE_TYPE SingleWindow`

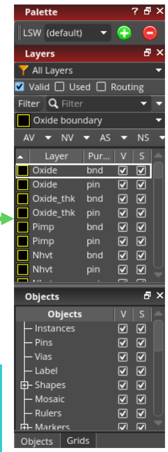
Important

The *CDS_PALETTE_TYPE* UNIX environment variable value must be set before running Virtuoso. Its value cannot be changed during a Virtuoso session.

1 Single Assistant Container (Default)

- Enter the following command in the shell environment to set the Palette type:
  - **setenv CDS_PALETTE_TYPE SingleAssistant**





The three panels can be combined together. For example, Objects and Grids panels can be placed together in separate tabs below the Layers panel.

The Single Assistant Container encapsulates the Layers, Objects, and Grids panels into a Single Palette Assistant. Choose **Window – Assistants – Palette**.

Single Assistant Container (Default)

The Single Assistant Container encapsulates the *Layers*, *Objects*, and *Grids* panels into a Single Palette Assistant. Therefore, when you hide the Palette Assistant, all three panels are hidden from view. If required, you can hide each panel individually. You can also dock or undock the entire Palette or each of the three panels individually.

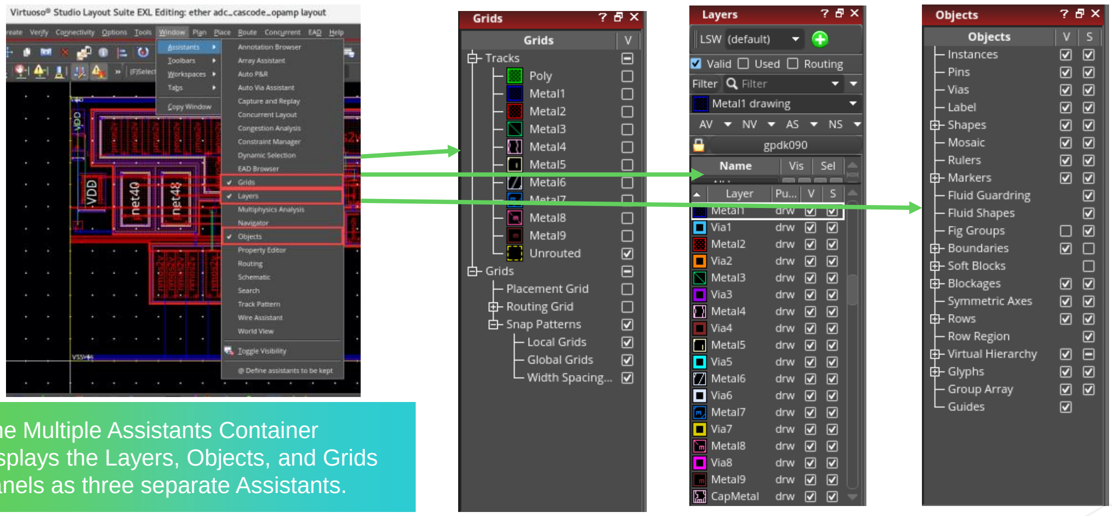
Note: The Single Assistant Container is linked to the layout window. Therefore, the Assistant is automatically closed when the layout window is closed.

Enter the following command in the shell environment to set the Palette type:

```
setenv CDS_PALETTE_TYPE SingleAssistant
```


2 Multiple Assistants Container

- Enter the following command in the shell environment to set the Palette type:
  - `setenv CDS_PALETTE_TYPE MultiAssistant`



The Multiple Assistants Container displays the Layers, Objects, and Grids panels as three separate Assistants.

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Multiple Assistants Container

The Multiple Assistants Container displays the *Layers*, *Objects*, and *Grids* panels as three separate Assistants. These are also listed as three separate Assistants on the Assistants menu that is displayed when you **right**-click in the menu bar area of the layout window. You can hide or unhide each panel individually. Each panel can also be docked or undocked individually.

Note: The three panels are linked to the layout window. Therefore, the panels are automatically closed when the layout window is closed.

Enter the following command in the shell environment to set the Palette type:

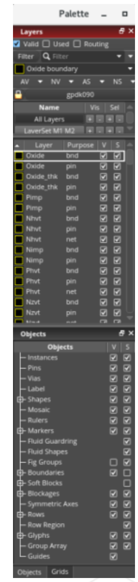
```
setenv CDS_PALETTE_TYPE MultiAssistant
```

3 Single Window Container

- Enter the following command in the shell environment to set the Palette type:
  - `setenv CDS_PALETTE_TYPE SingleWindow`

The Single Window Container is similar to the Layer Selection Window (LSW). It encapsulates the Layers, Objects, and Grids panels in a single window that is independent of the layout window.

No Palette Assistant here.



The Single Window Container permits the same modifications as the Single Assistant Container (combination of the three panels).

The Single Window Container is not linked to the layout window. Therefore, the Single Window Container is NOT closed when the layout window is closed.

Single Window Container

The Single Window Container is similar to the **Layer Selection Window** (**LSW**). It encapsulates the *Layers*, *Objects*, and *Grids* panels in a single window that is independent of the layout window. In this mode, a single Palette window is used to manage all open designs. The three panels can be docked or undocked individually. You can also hide or unhide them individually.

Note: The Palette window is closed automatically when you exit the Virtuoso session.

Enter the following command in the shell environment to set the Palette type:

```
setenv CDS_PALETTE_TYPE SingleWindow
```

To resize the window, use **pteResizeSingleWindowPalette**.

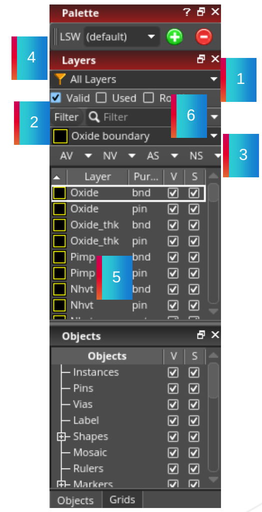
To set the location of the window, use **pteMoveSingleWindowPalette**.

Analyzing the Layers Panel

The Layers panel lists the layer-purpose pairs defined in the *leLswLayers/techLayerPurposePriorities* subsection of the technology file.

The Layers panel options are given below:

1. Adjust the *Scope* toolbar to shorten the Layers List.
2. Use *Filter* to search for items.
3. Choose **All Visible (AV), None Visible (NV), All Selectable (AS)**, or **None Selectable (NS)**.
4. Create/Manage *Layer Sets*.
5. *Layers List* with User Options.
6. This Layers List can be shortened to suit the designer through the options provided, such as setting *Used* and *Routing Scope* instead of the Valid Layers choice.



The Layers panel is displayed by default on the left of the layout window. The Layers panel lists the layer-purpose pairs (LPPs) that are defined in the *leLswLayers* section of the technology file. By default, only valid layer-purpose pairs are displayed if the *Valid* checkbox is selected in the Scope toolbar. The layer-purpose pairs are displayed in the Layers panel in the order of their priority.

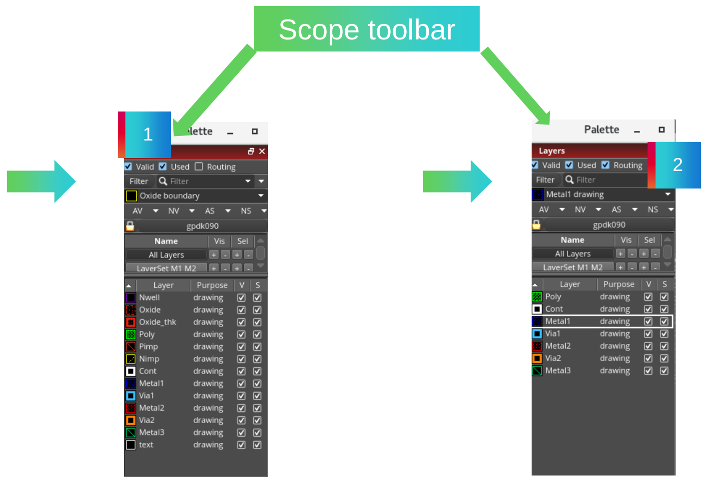
The “LSW layers” SKILL functions operate on the *leRules* section of the technology database. The *leLswLayers* subsection of the *leRules* section lets you specify the layers that are initially displayed in the Palette assistant. In the *leLswLayers* subsection, you list the layers in the order in which you want them to appear in the Palette assistant.

If you do not define the *leLswLayers* subsection in the technology database, the *techLayerPurposePriorities* subsection of the *layerDefinitions* section determines the layers that are initially displayed and the order in which they are displayed in the Palette assistant.

The Layers Panel: Layers List

1 With Valid and Used Layers List

2 With Valid, Used and Routing Layers List



The **Layers Panel** *With Valid & Used Layers List* and *With Valid, Used & Routing Layers List* are shown.

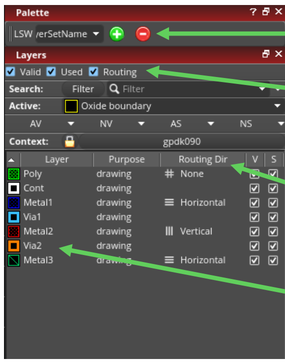
Scope

The Scope toolbar provides options for filtering layers, namely, *Valid*, *Used*, and *Routing*.

- **Valid:** If selected, displays only valid layers.
- **Used:** If selected, displays only the layers that contain design objects of type shape. If you want to view layers that contain all types of design objects, such as blockages, boundaries, instances, mosaics, markers, and rows, set the *pteShowUsedSystemLpps* environment variable to **t**, and then select the **Used** checkbox. You need to reapply the filter to refresh the display in the Layers panel if you add a design object to a previously unused layer while the filter is still applied.
- **Routing:** If selected, displays only the routing layers. Routing layers are identified at the start of a Virtuoso session. By default, the following layers are identified as routing layers:
  - All layers with function *metal*, *cut*, and *poly* and purpose *drawing* through *drawing9*

    Note: You can exclude *cut* layers from routing layers using the *pteIncludeCutLayers* environment variable. The value of this variable must be set before opening a design. Once set, it is applicable for the entire session.
  - All layers with purpose *drawing* through *drawing9* present in the *validRoutingLayers* constraint.
  - All combinations of the layers and purposes present in the *validRoutingLPPs* constraint.

The Layers Panel: Used and Routing Layers



Layer selections are stored and managed in the “MylayerSetName” Layer Set.

The Layers List is shortened to include ONLY working/routing layers (Used and Routing).

All used layers are viewable and selectable.

Preferred Routing Direction is also shown.

You can see that this list is greatly shortened due to the Used and Routing options selected in the Scope section.

Layer Set – A Palette supports creation and manipulation of layer sets. A layer set contains a subset list of layer, object, and grid items. By default, all layers described in the technology file and all existing objects and grids are listed in the Palette. Activation of a layer set reduces the list of items displayed in the Palette. It is possible to select multiple layer sets at the same time. A layer set can be activated and modified in a Palette, and saved to a layer set file. Additionally, the visibility and selectability statuses of layers can be saved in each layer set. Use the **Layer Set Manager** to activate layer sets and to control the display contexts on a layer set basis. Use the Palette menu to save, reload, or delete layer sets.

Synchronized Palettes – Palettes are *synchronized* by default if they point to the same technology file. If two palettes are synchronized, changes in one Palette are reflected in the other Palette. For example, changing the visibility of a layer-purpose pair in one Palette changes the visibility of that layer-purpose pair in the other Palette that points to the same technology file.

Palettes are *desynchronized* if they are linked to different technology files or if one of the Palettes is explicitly made independent. For example, a change in one independent Palette is not reflected in the other palettes. Similarly, if the palettes in layout windows 1 and 2 point to different technology files, namely techfile 1 and techfile 2, the palettes are independent.

New Features for Palette Simplification

From IC6.1.7, the following features are introduced:

- Layer Set Manager is hidden by default.
- The list of objects can be customized and saved in the layersets.
- The objects and grids settings can be saved in the layersets (protected with an *envvar*).
- The Options form is revamped and follows the form's guidelines.
- The RMB contextual menu is available in the Assistants.
- Quick access to different functions is available from the **AV/NV/AS/NS** buttons.

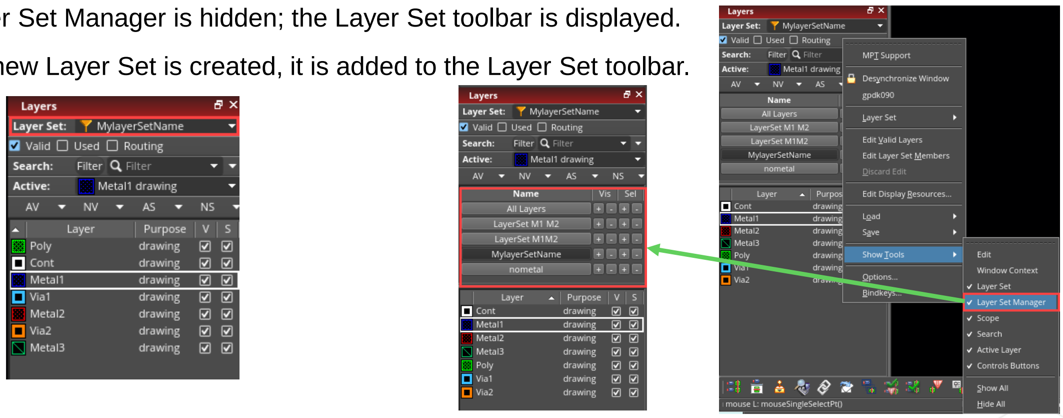
Refer to the *Virtuoso Layout Viewer User Guide, Product Version IC25.1*, for details on these features.

Displaying the Layer Set Manager

In the Layers panel, **right**-click and select the **Show Tools** menu to display the **Layer Set Manager**.

New Use Model:

- The Layer Set Manager is hidden; the Layer Set toolbar is displayed.
- When a new Layer Set is created, it is added to the Layer Set toolbar.



Layer Set Manager

The Layer Set Manager lists the layer sets that are enabled in the current design window.

Use the Layer Set Manager to perform the following tasks:

- Reorder layer sets by dragging them up and down the list.
- Select one or more layer sets by using the following methods:
  - To select a single layer set, click that layer set.
  - To select multiple adjacent layer sets, click the first layer set in the sequence, and then hold down the **Shift** key and click the last layer set in the sequence.
  - To select multiple non-adjacent layer sets, select a layer set, and then hold down the **Ctrl** key and click the other layer sets that you want to select.

All layer sets selected in the Layer Set Manager are considered active and a layer-purpose pair is displayed in the Layers panel if it is a member of at least one of the selected layer sets.

- Turn on or off the visibility or selectability status of all the layers present in a layer set by clicking the corresponding + or – button, respectively. These buttons work with respect to the filter settings defined on the Scope and Filter toolbars if the *pteLSManagerRespectFilters* environment variable is set to **t**.
- Set a layer set to active and turn on the visibility and selectability of all its member layers by clicking the layer set with the middle mouse button. The visibility and selectability for all other layers is turned off.
- Disable a layer set by deselecting the corresponding checkbox in the column labeled *E*. This column is displayed when you **right**-click a column header in the Layer Set Manager and choose **Columns – Enable Layer Set** from the shortcut menu.

Note: The layer set that is currently active cannot be disabled.

- Save the layer set order and status – whether enabled or disabled – in the *layerset.order* file, which, by default, is created in the local *.cadence* directory.

Note: Names of layer sets that are not synchronized with corresponding layer set files are italicized.

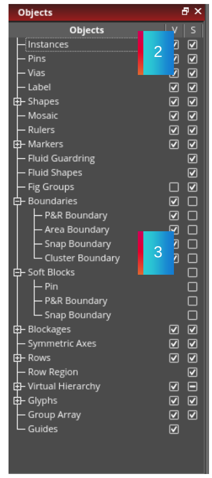
What Is the Objects Panel?

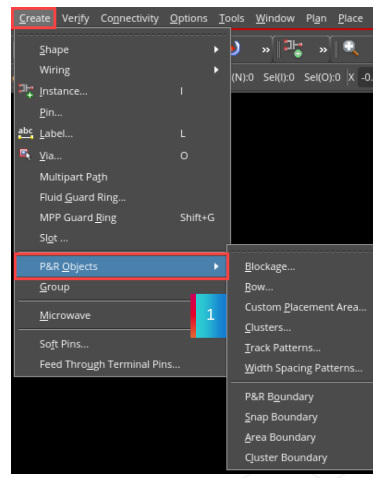
The **Objects** panel displays objects such as instances, pins, and vias.

It is a dockable panel that replaces the LSW **Objects** pane.

1. Blockages and Boundaries can be defined by choosing **Create – P&R Objects** on the layout editor menu.
2. The Objects Assistant controls the visibility and selectability of Open Access Objects.
3. The Objects panel displays objects such as shapes, instances, pins, vias, boundaries, fluid guardring, mosaic, fluid shapes, fig groups, soft blocks, blockages, etc.

By default, the Objects panel is displayed below the Layers panel docked to the current session window.





Objects Panel

The *Objects* panel displays objects such as instances, pins, and vias. It is a dockable panel that replaces the LSW *Object* pane.

By default, the Objects panel is displayed below the Layers panel docked to the current session window.

This Objects Assistant offers an enhancement to the previous functionality of the LSW/Objects tab with the addition of the Groups section.

The Objects Assistant displays three columns:

- The **Objects** column displays OpenAccess objects (instances, pins, markers, vias, fig groups, rows, boundaries, soft blocks, and blockages). The objects in the Objects column are listed in a hierarchical tree structure. Instance and pin selection and visibility are controlled by Objects panel.
- The **Visibility** (**V**) column, which lets you control the visibility of objects by selecting and deselecting the checkbox next to the object name. If a parent object is selected or deselected, the child objects should update accordingly.
- The **Selectability** (**S**) column, which lets you control the selection of objects by selecting and deselecting the checkbox next to the object name. If a parent object is selected or deselected, the child objects should update accordingly.

Displaying and Hiding Objects Panel

Use one of the following methods to display the Objects panel:

- **Right**-click any panel title bar and choose **Objects**.
- Use the *pteMapWindow* function to specify the name of the Palette panel to be displayed.

Use one of the following methods to hide the Objects panel:

- **Right**-click any panel title bar and choose **Objects**.
- Use the *pteUnmapWindow* function to specify the name of the Palette panel to be hidden.

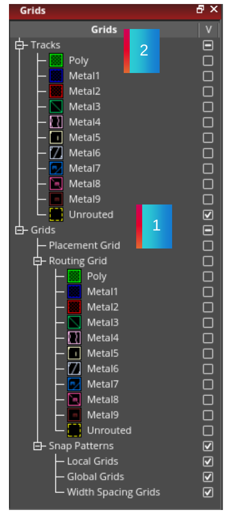
Refer to the *Virtuoso Layout Suite SKILL Reference*, *Product Version IC25.1*, for more details of these functions.

What Is the Grids Panel?

The **Grids** panel displays *Tracks* and *Grids*.

It is a dockable panel that replaces the LSW **Grids** pane.

1. The Grids panel displays elements such as Tracks, Placement Grid, and Routing Grid.
2. By default, the Grids panel is displayed below the Layers panel in the current session window.
3. If both Objects and Grids panels are displayed, the panels appear on separate tabs, and you can switch between the two panels by clicking the tabs.



Grids Panel

The *Grids* panel displays *Tracks* and *Grids*. It is a dockable panel that replaces the LSW Grids pane.

The Grids panel is displayed below the *Layers* panel. If both Objects and Grids panels are displayed, they appear on separate tabs and you need to click the Grids tab to view its contents.

Displaying and Hiding the Grids Panel

Use one of the following methods to display the Grids panel:

- **Right**-click any panel title bar and choose **Grids**.
- Use the *pteMapWindow* function to specify the name of the Palette panel to be displayed.

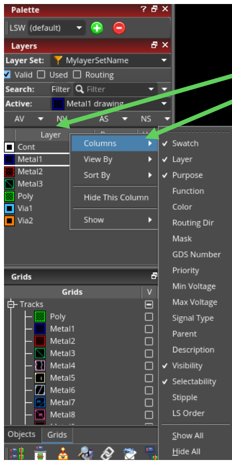
Use one of the following methods to hide the Grids panel:

- **Right**-click any panel title bar and choose **Grids**.
- Use the *pteUnmapWindow* function to specify the name of the Palette panel to be hidden.

Refer to the *Virtuoso Layout Suite SKILL Reference*, *Product Version IC25.1*, for more details of these functions.

Customizing the Drawing Layers

You can use the **Columns** menu in the **Layer** header to selectively display columns in the Layers panel.



By **right**-clicking on the **Layer** header, and expanding the **Columns** option, you can control the aspects of each layer, including:

- Swatch
- Layer
- Purpose
- Function
- Color
- Routing directions of the layers
- Mask
- GDS Number
- Priority
- Min Voltage
- Max Voltage

Columns

The Layers panel *Columns* menu lists the following column names:

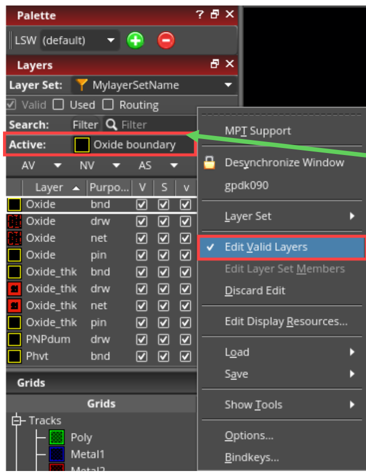
> *Swatch*, *Layer*, *Purpose*, *Function*, Color, *Routing Dir*, *Mask*, *GDS Number*, *Priority*, *Min Voltage*, *Max Voltage*, *Signal Type*, *Parent*, *Description*, *Visibility*, *Selectability*, *Stipple*, and *LS Order*

Use this menu to selectively display columns in the Layers panel. You can hide or display all columns, except *Swatch*, by selecting the **Hide All** or **Show All** commands, respectively.

Note: All columns except **V** (Visibility) and **S** (Selectability) can be resized by dragging the column margin separator.

Setting the Valid Drawing Layers

Layers are enabled by **right**-clicking on the **Active** layer in the **Layers** panel, then by choosing **Edit Valid Layers**.



Click in the **Active** field, and select **Edit Valid Layers**.

About Valid Drawing Layers

The Layers Assistant shows the valid drawing layers, which are all the user layers you defined in your technology file with a layer-purpose of *drawing*. When you open a cellview from another library, the Layers Assistant changes to show the layers for the new library.

You may not want to scroll through all the layers defined in the technology file, so you can greatly shorten the Layers List by turning **off** the Valid switch, and turning **on** the Used and/or Routing switches.

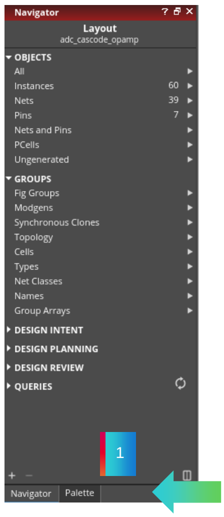
Edit Valid Layers

Use this command to display the Membership (*m*) and Validity (*v*) columns in the Layers panel.

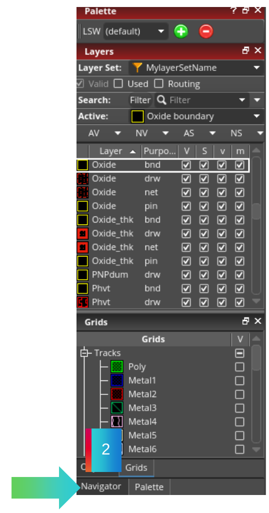
Corresponding SKILL Functions: pteEditLayerSetValidity, pteCloseLayerSetEdition

Example: Tabbed Assistants

1 With Navigator Assistant in front in the tabbed Navigator and Layers Assistants.



2 With Layers Assistant in front in the tabbed Navigator and Layers Assistants.



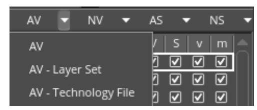
The Layers Assistant can “float” off the canvas or become embedded into the Navigator by “dropping” the Layers Assistant directly over the Navigator. You will notice that you now have tabs with which you can quickly switch back and forth between the two Assistants.

The Right Mouse Button (RMB) Contextual-Menu of Layers Panel

Controlled with an *envvar pteEnableMouseBindings*

The impact is as listed below:

- The MMB and the RMB don’t control the visibility and selectability anymore.
- The visibility and selectability of the LPPs are controlled either through the checkboxes or through the AV/NV/AS/NS buttons.
- By default, clicking on the buttons always control the visibility or selectability of the LPPs present in the Palette.



- The user can control the entire list of LPPs in the layerset or in the technology file using the new functions.

pteEnableMouseBindings

- `layout pteEnableMouseBindings boolean { t | nil }`

Description: Causes a **right**-click in the **Layers** panel to display the context-sensitive Palette menu if set to **nil**. If set to **t**, a **right**-click in the **Layers** panel turns on or off the selectability of a layer-purpose pair. The default is **t**.

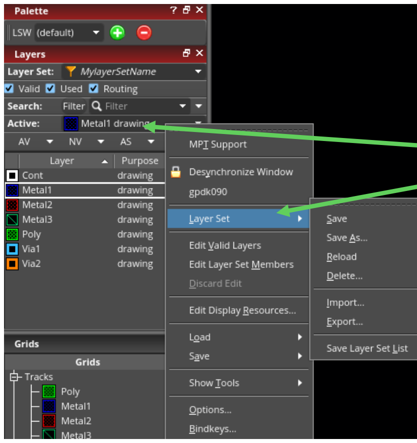
GUI Equivalent: None

Examples

```
envGetVal("layout" "pteEnableMouseBindings")
envSetVal("layout" "pteEnableMouseBindings" 'boolean t)
envSetVal("layout" "pteEnableMouseBindings" 'boolean nil)
```


Creating and Using the Layer Sets

A Layer Set is a list of layers to be displayed in the **Layers** Assistant.
For example, a predefined set shows only the **Used** and **Routing** layers.



Click in the **Active** field, and select **Layer Set** to Save, Save As, Reload, and Delete the layer sets.

Using Layer Set

The *Layer Set* option lets you create, save, edit, and delete user-defined files containing an ordered list of layers which you can load into the Layers Palette. You can use these layer set files only when you launch the software from the same working directory in which you created them. The files (once created) are stored in the current working directory, which must be writable.

The path to the layer sets files is:

./.cadence/dfII/layerSets/<technology library name>/<layerSetName>.layerSet

Creating a Default Layer Set

To create a default layer set:

1. In the Layers Assistant, choose the layers and associated options.
2. Click in the Active field, and choose **Layer Set**.
3. Click **Save As** to name the layer set, as for example, *MylayerSetName*.

This saves the current layers settings in a file named *MylayerSetName* in your working directory at:

- *./.cadence/dfII/layerSets/<technology library name>/<layerSetName>.layerSet*

Displaying the Highlight Options Form

In the layout canvas, choose **Options – Highlight** to display the Highlight Options form.

Highlight controls the objects highlighted when a net is selected or probed.

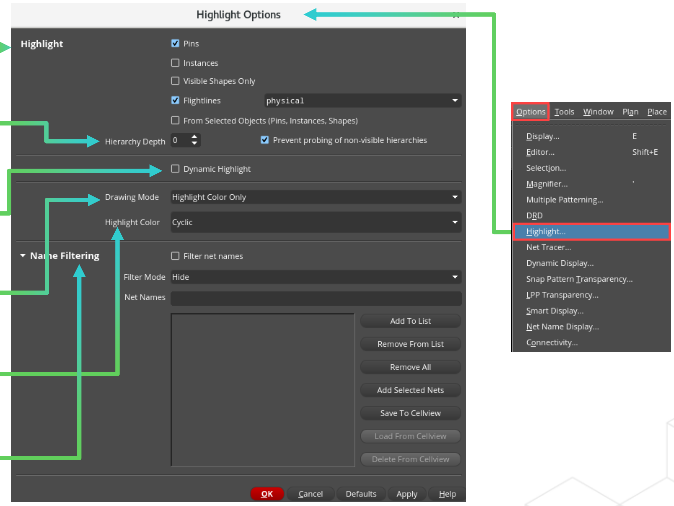
Hierarchy Depth specifies the hierarchy level you want to display for net probing.

Dynamic Highlight specifies that any object over which the pointer hovers is probed.

Drawing Mode allows you to choose the drawing mode for the next probe.

Highlight Color shows highlight drawing layers to be used on the next probe that will be drawn.

Name Filtering lets you selectively display (or hide) the probes only for the specified list of nets.



Highlight Options Form

Highlight controls the objects highlighted when a net is selected or probed.

Pins specifies that the pins on probed nets are highlighted (environment variable: probeNetDisplayCheckPins).

Instances specifies that the instances on probed nets are highlighted (environment variable: probeNetDisplayCheckInstance).

Visible Shapes Only specifies that shapes, pins, and vias on the visible layers are highlighted (environment variable: probeVisibleLpp).

FlightLines specifies that either all the shapes and opens on the probed net are highlighted (physical) or segments linking all pins on the probed net are highlighted (logical) (environment variable: probeNetDisplayFlightLines). **Note:** Flight lines are visible in VLS EXL and VLS MXL.

From Selected Objects (Pins, Instances, Shapes) specifies that the nets associated with probed or selected pins, instances, and shapes are highlighted (environment variable: probeTermDisplayRadio, probeInstDisplayRadio, probeShapeDisplay).

Hierarchy Depth specifies the hierarchy level you want to display for net probing (environment variable: probeNetDisplayHierarchyLevel).

Dynamic Highlight specifies that any object over which the pointer hovers is probed (environment variable: dynamicProbe).

Drawing Mode, allows to choose the drawing mode for the next probe. If you choose:

- *True Color and Highlight Color*, the probed shapes are redrawn with their layerpurpose pair and with the highlight color.
- *Highlight Color Only*, the probed shapes are redrawn only with the LPP chosen from Highlight Color list.
- *True Color Only*, the probed shapes are redrawn only with their LPP. The drawing mode with the True Color option enables you to visualize the different layers of the connected shapes in the cellview (environment variable: turnInfixOffWhenSmartSnapping).

Highlight Color shows the highlight drawing layers to be used on the next probe that will be drawn. You can choose *Cyclic* or select the LPP (environment variable: probeHighlight).

Name Filtering lets you selectively display (or hide) the probes only for the specified list of nets.

Filter net names lets you enable or disable probe filtering by net name (environment variable: probeNetNameFilterByName).

Filter Mode lets you show or hide the probes for specified nets (environment variable: probeNetNameFilterByNameMode).

Net Names lets you specify multiple net names and add them to the list. The net names should be separated by a space. This is a case-sensitive field (environment variable: probeNetNameFilterByNameDefaultList).

Add Selected Nets lets you add nets to the filter list by selecting the nets in the Navigator assistant or from the layout canvas. To remove net names from an already existing filter list, select the net name(s) and click the **Remove From List** button. To remove the entire filter list, click **Remove All**.

Save To Cellview lets you save the current filtering options to the cellview that you can reload later in a different window by clicking the **Load From Cellview** button. You can also delete the saved filtering options from the cellview by clicking the **Delete From Cellview** button.

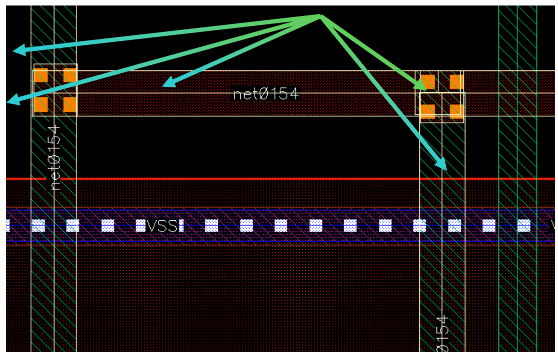
Note: The Save To Cellview option is enabled only when the current cellview is editable.

True Color Probe

Probe net command now supports hierarchical true color probe:

- True color probe allows users to visualize a net using the true colors of the objects implementing the net in the canvas.
- Dimming is automatically enabled when *True Color Only* option is enabled in the *Highlight Options* form.
  - This helps with properly visualizing the probed net in the layout canvas.

net0154 probed in true color hierarchically



True Color Only, if selected, redraws the probed shapes with their layer-purpose pair. This option enables you to visualize the different layers of the connected shapes in the cellview. If you select the **True Color Only** option in *Drawing Mode*, the *Highlight Color* field is automatically *greyed out*.

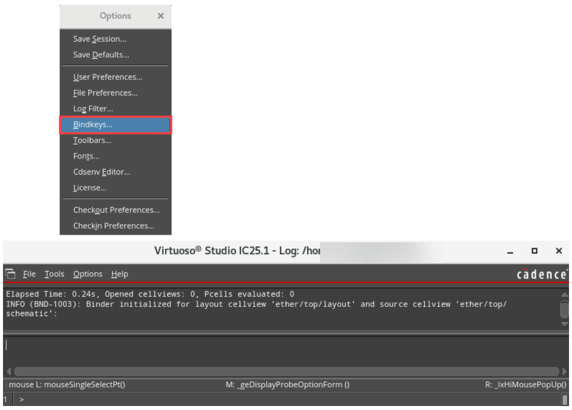
Tip: To easily highlight the probed nets, it is recommended that the *Enable Dimming* option in the Display Options form is selected when you select the **True Color** option.

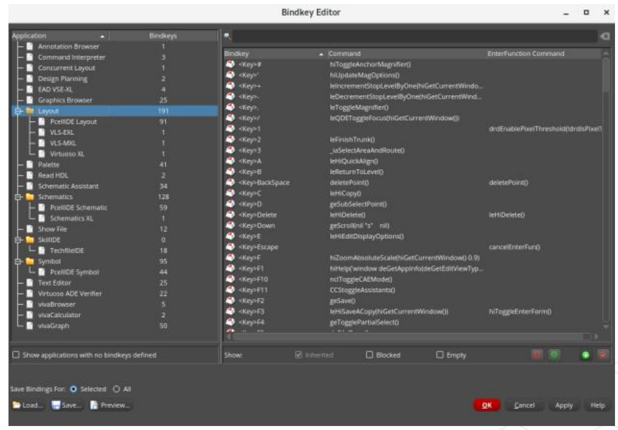
Dimming is automatically enabled when **True Color Only** is enabled in the *Highlight Options* form. This helps with properly visualizing the probed net in the layout canvas.

Opening the Bindkey Editor

From the CIW, choose **Options – Bindkeys** to open the Bindkey Editor.

Click on each category to see the bindkey mapping. Bindkeys are sorted by application.





Creating Bindkeys

You can create your own bindkeys by:

- Editing the *leBindKeys.il* file.
- Finding the SKILL syntax through the output field of the CIW and adding the SKILL definition to the *leBindKeys.il* file.
- Choosing the **CIW – Options – Bindkeys** command.

You can enter the key you want to define and the SKILL definition for the bindkey.

.cdsinit File

You can also define bindkeys by entering the definition directly in the *.cdsinit* file.

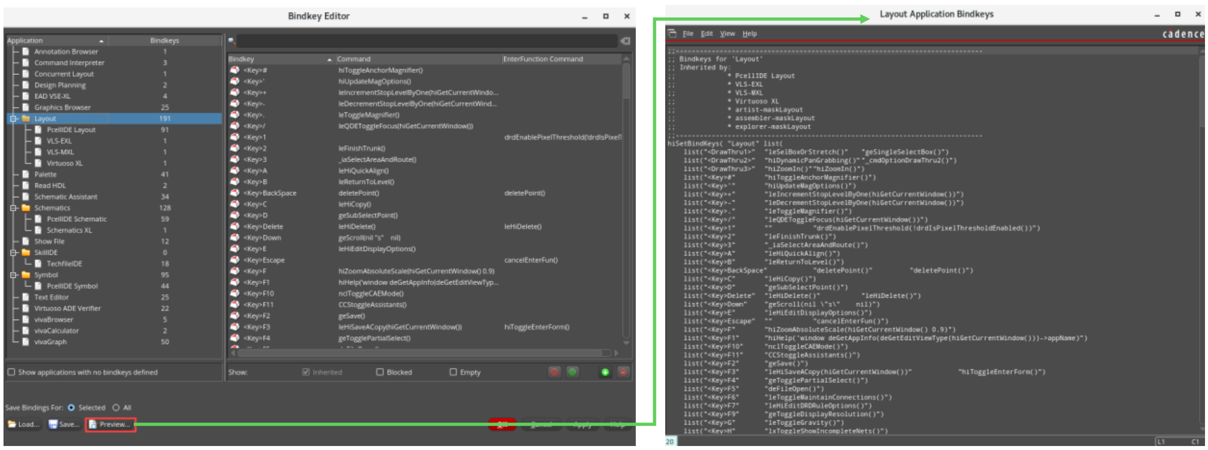
A sample SKILL procedure for defining a bindkey is:

```
hiSetBindKey("Layout" "<Key>Delete" "leHiDelete()")
```

This sets the **Delete** key equivalent to the **Edit – Delete** menu command. The menu command calls the SKILL function *leHiDelete().*

Bindkey Editor: Preview

You can override the bindkeys inherited from the root application by defining a bindkey for the inherited application.



Bindkey Editor: Preview

While the Bindkey Editor is open, you can click on the Preview button to preview the command that is mapped to each of the bindkeys.

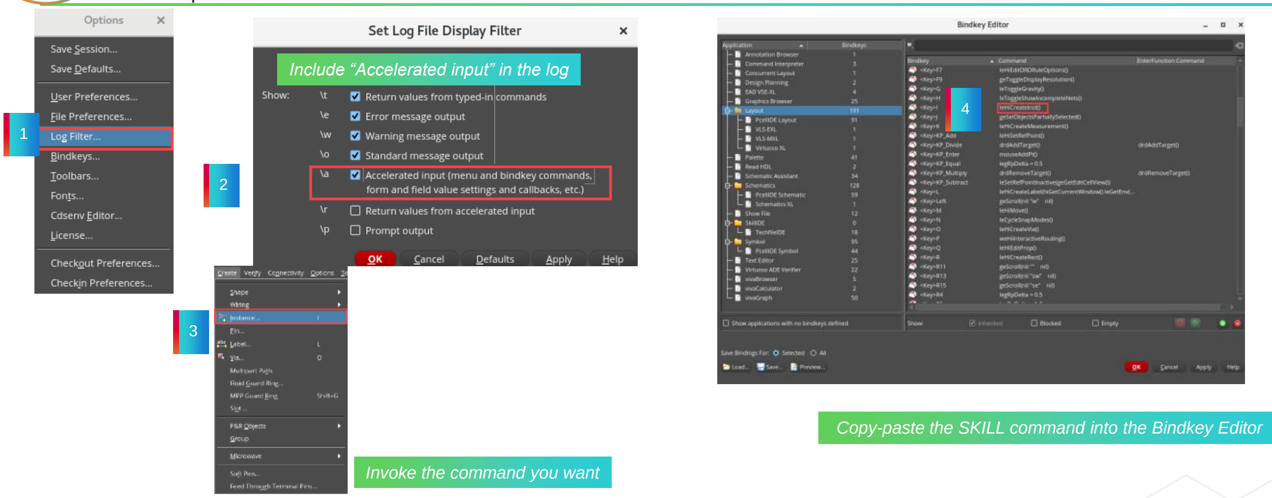
Preview lets you to see the commands associated with the keys and allows you to modify or edit them.

You can override the bindkeys inherited from the root application, by defining a bindkey for the inherited application.

Creating a New Bindkey

In most cases, you don't know exactly the SKILL® command, which corresponds to the action you want to assign to a given bindkey. In these cases, you can use the **Log Filter** option to include all the commands in the CIW.

You can use this information and copy-paste the SKILL command into the **Bindkey Editor** to create the bindkey for the required action.



In most cases, you do not know exactly the SKILL command, which corresponds to the action you want to assign to a given bindkey.

In these cases, you can use the **Log Filter** option to include all the commands in the CIW window.

This option is usually turned off to avoid having too much information displayed in the CIW window.

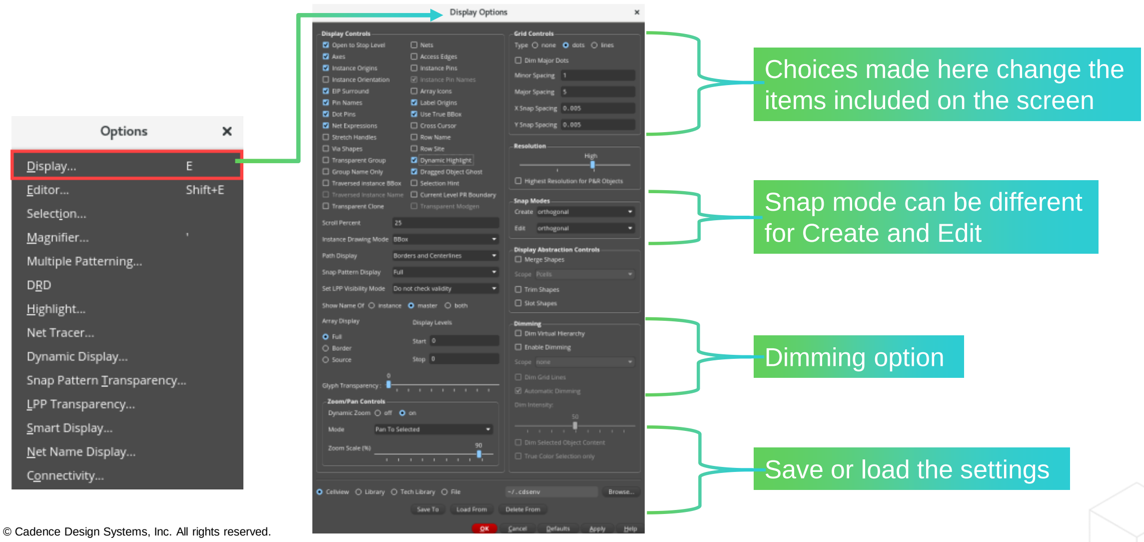
You can use the **right** mouse button on a selected part of the text to invoke the Copy Text command.

After including the Accelerated input command in the log, you can invoke the command for which you want to create the bindkey.

You can use this information and Copy-Paste the SKILL command from the CIW output field into the Bindkey Editor to create the bindkey for the required action.

Invoking the Display Options Form

In the layout, choose **Options – Display** or bindkey **E** to open the Display Options form.



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The *Display Options* form controls the appearance of objects and the behavior of commands in the cellview.

The form comprises the following sections:

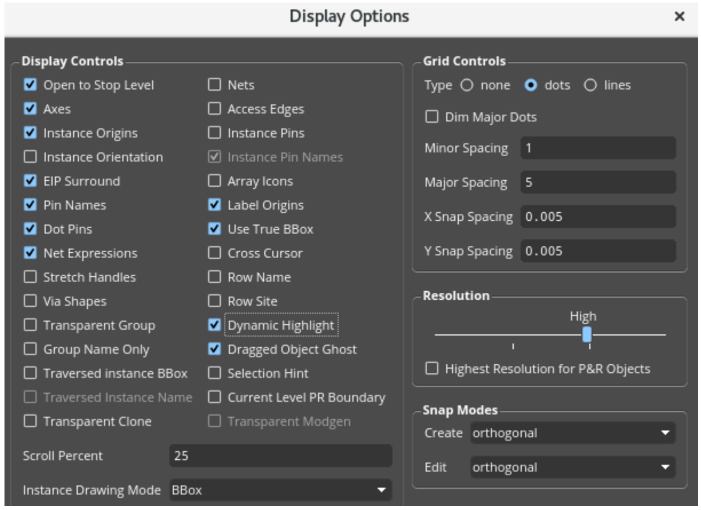
- Display Controls
- Zoom/Pan Controls
- Grid Controls
- Resolution
- Snap Modes
- Display Abstraction Controls
- Dimming

The **Display Options** is the form that controls how the screen looks and acts. **Display Controls** mostly work with how the screen appears and what options are visible. **Grid Controls** take care of snap resolution and visibility of dots. **Resolution** lets you change the display resolution. The resolution determines how much detail of a design is displayed in the cellview. **Snap Modes** will set the snap modes for creating and editing shapes and instances. **Display Abstraction Controls** help to make the display more readable by hiding all the extra edges in the design. **Dimming** acts when you select items on the canvas. Unselected objects are dimmed by the specified amount.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1*, for details on these features.

The Display Options: Top of the Form

In the layout, choose **Options – Display** or bindkey **E** to open the Display Options form.



Display Controls

Open to Stop Level sets the display level to the Display Levels Stop level. **Nets** show flight lines between objects on the same net. **Axes** display the cellview X and Y axes. **Access Edges** shows the routing edges of pins. **Instance Origins** show the origin of the instances. **Instance Pins** shows pins in instances. **Instance Pin Names** show pin names in instances. **Instance Orientation** shows the orientation of the instances.

Array Icons show outlines of array cells when *Display Levels* suppresses cell details.

EIP Surround (edit in place) displays the surrounding design when you edit a cell in place.

Label Origins marks the origins of labels with diamond markers.

Pin Names show terminal names of pins that have pin name text displays or labels.

Use True BBox when on; displays the instance master bounding box.

Dot Pins display via origins with diamond markers when you set *Display Levels* to show instance content.

Cross Cursor displays the cursor as dot-shaped or as dotted-line cross shaped.

Net Expressions displays the net expression instead of the terminal name of a pin. When there are net expressions in instances, the terminal name is displayed, not the net expression, even when *Net Expressions* is set on.

Row Name displays the names of placed rows, which are displayed on the rows in the horizontal or vertical direction depending upon the rotation of the rows.

Stretch Handles displays the handles on a Relative Object Design parameterized cell (Pcell) that indicates that the Pcell can be stretched. A *stretch handle* is a named set of coordinates assigned to a specific parameter of the Pcell. Stretch handles look like small diamonds.

Row Site displays the position of row sites, which are represented by a cross marker on the row.

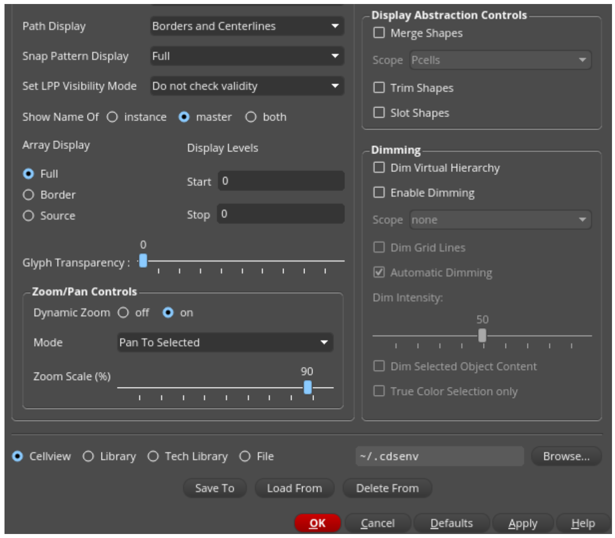
Via Shapes displays via attributes.

Dynamic Highlight marks the edge, object, or point that is selected if you enter a point selection.

Transparent Group enables editing a group without the need for EIP in that group.

The Display Options: Bottom of the Form

In the layout, choose **Options – Display** or bindkey **E** to open the Display Options form.



Display Abstraction Controls help to make the display more readable by hiding all the extra edges in the design.

Merge Shapes make the display more readable by hiding all the extra edges in the design.

Scope lets you select the hierarchies for which the *Enable Merge* command applies. The choices are *Pcells*, *Except current*, and *All*.

Slot Shapes let you create holes in wires. You overlap the shape to slot with another shape in slot purpose that is displayed as a hole in the shape. Environment variable: slotEnable

Dimming acts when you select items on the canvas. Unselected objects are dimmed by the specified amount.

At the bottom, you can save the setting of the form to the individual cellview so that each person who opens the cell will have those settings available to them. This is good for large top-level cells. You can set the start level to **0** and the stop level to **0** and save it to the cellview. Each time anyone opens the cellview, it will open quickly because it does not have to draw the hierarchy. You can then use **Area Display** to open specific areas for editing.

Enable Dimming, if selected, lets you change the color luminosity of objects in the cellview.

Scope choices are *none*, *outside*, *all*, *eipSurround*.

Dim Grid Lines dims the grid lines on the canvas.

Automatic Dimming dims the design automatically either when you select an object or when the design contains highlight sets or probes.

Dim Intensity lets you set the luminosity of dimmed objects.

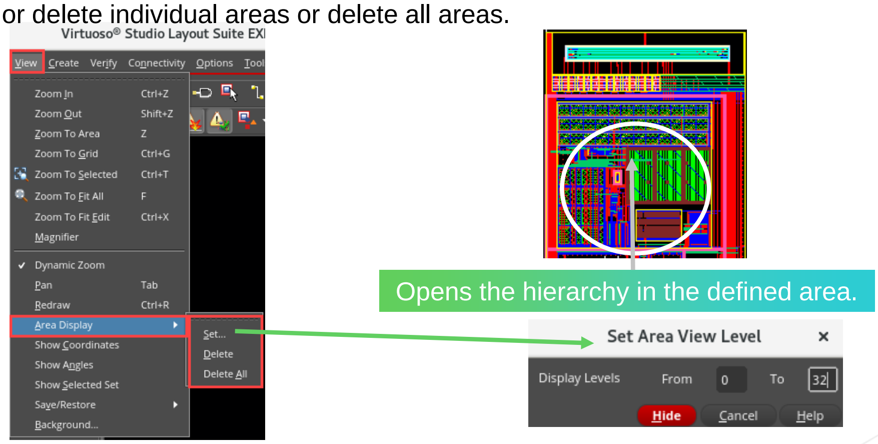
Dim Selected Object Content dims the selected object.

True Color Selection only displays the selected objects in their original colors when other objects are dimmed.

Setting the Area Display

In the layout, choose **View – Area Display – Set** to display the Set Area View Level form.

Area Display command opens the cellview data only within the area you describe by entering the selection rectangle. You may set or delete individual areas or delete all areas.



Area Display command is useful when you have a large amount of data. By having the cellview properties set to display levels of *Start 0* and *Stop 0*, you will be able to open a cellview quickly. To work in specific areas of the cellview, you can open only the area where you wish to work.

Area Display opens the cellview data only within the area you describe by entering the selection rectangle. You may set or delete individual areas or delete all areas.

This is very efficient when working at the top level of dense circuits. You will only bring in the amount of data necessary to do the work in a specific area.

Set Area View Level Form

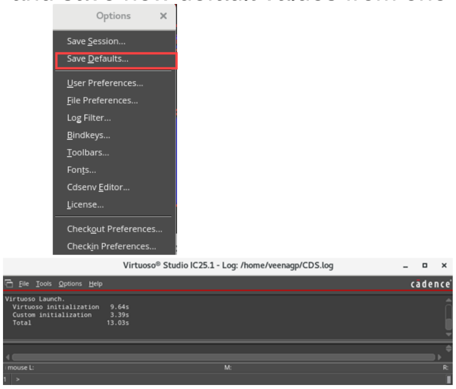
Display Levels specify the first (*From*) and last (*To*) levels in the design hierarchy that can be seen in detail. The hierarchy levels are numbered 0 to 32. Shapes in the current cellview are at level 0, shapes in the masters of instances inside of it are at level 1, and so forth.

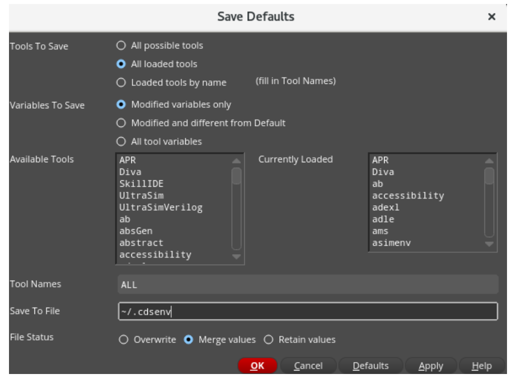
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for details on these features.

Saving the User-Specified Options as Defaults

From the CIW, choose **Options – Save Defaults** to open the Save Defaults form.

To specify and save new default values from one session to the next, use the **Save Defaults** option.





Used to specify and save default values from one session to the next.

Specifying New Default Values

To save the view properties for the different tools, such as layout, schematic, and place and route, choose **Options – Save Defaults**. Instead of going to the default *.cdsenv* and pulling out the environment variables, you can set up the properties for each tool using the interface associated with that tool's options and then save them to your *.cdsenv* file.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1*, for details on these features.

Environment Variable Examples: *.cdsinit* and *.cdsenv* Files

Refer to the information on *.cdsinit* and *.cdsenv* files.

Environment variables can be loaded from *.cdsinit* and *.cdsenv* files.

.cdsinit file – Contains the SKILL language file.

```
envSetVal("layout" "xSnapSpacing" 'float 0.5)
envSetVal("layout" "ySnapSpacing" 'float 0.5)
```

.cdsenv file – Contains application environment variable defaults.

```
layout arrayDisplay string "Full"
layout drawAxesOn boolean t
```

Samples provided:

```
.cdsenv   <installPath>/tools/dfII/samples/.cdsenv
.cdsinit   <installPath>/tools/dfII/cdsuser/.cdsinit
```

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cadence

Setting Variables

You store view properties in either the *.cdsenv* or *.cdsinit* files. You must specify the view name before the property value. If you are setting the variables in the *.cdsinit* file, use the *envSetVal* function. The variables reflect how your design will be displayed. In the example here, the entire array is displayed, the X,Y axis is turned on, and the grid is displayed as dots. The window size and location are defined for the layout view. In the example, the *.cdsinit* file also sets environmental variables, and then loads the *.cdsenv* in the working directory.

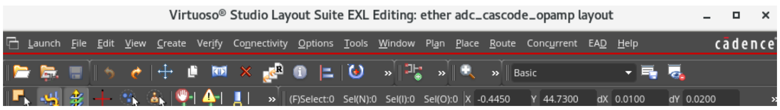
The *.cdsinit* file is a Cadence® SKILL language file executed when the Virtuoso Design Environment product starts.

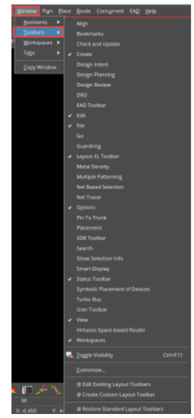
The *.cdsenv* file is an environment setup file that holds application defaults for environment variables.

Toolbars and Icons in the Layout Canvas

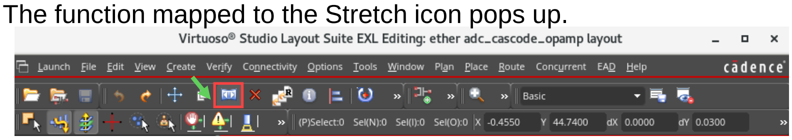
In the layout, from **Window – Toolbars** menu, you can activate the toolbar menu icons.

- Icons for **single**-click shortcuts to commands can be moved into rows or to the sides of the canvas and stored in any location as part of your customized workspaces.





- The function mapped to the Stretch icon pops up.



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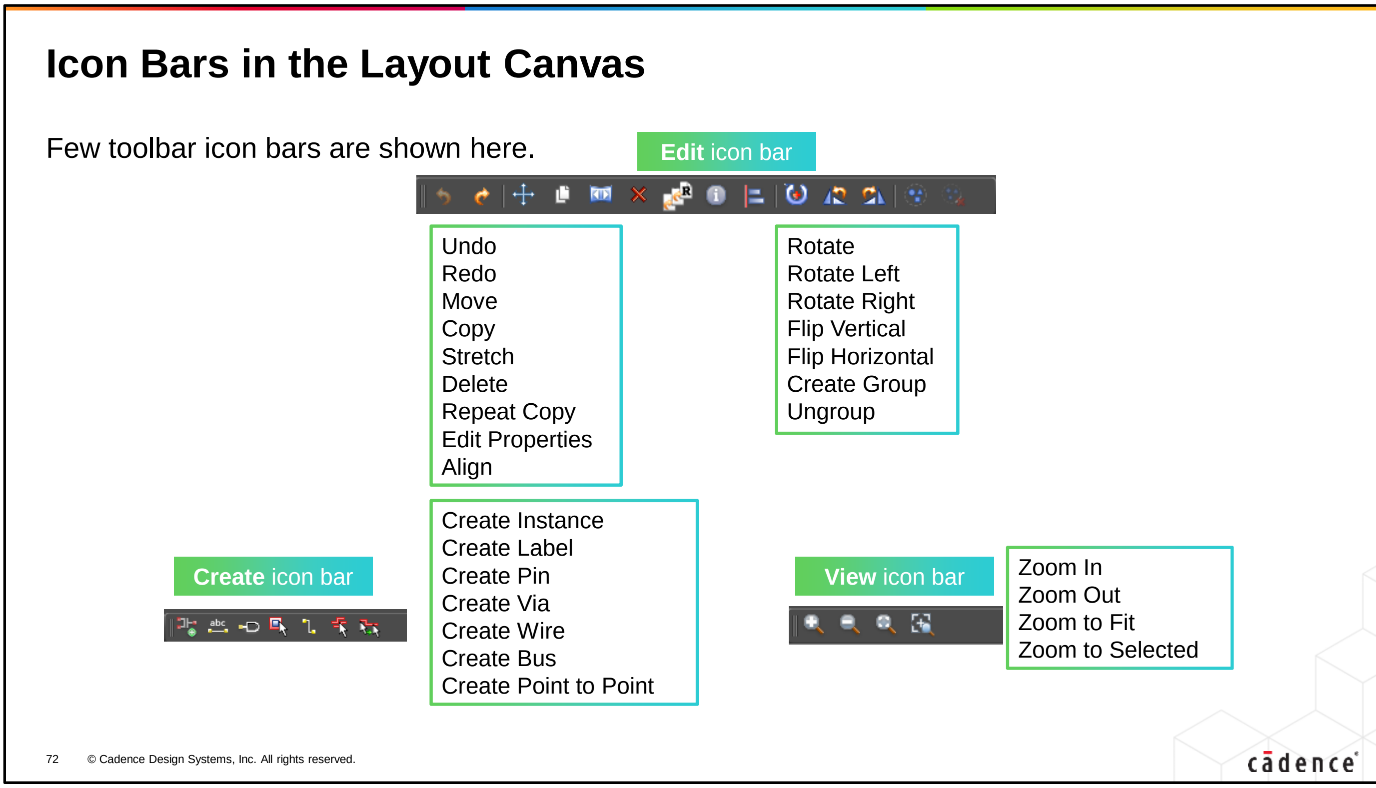
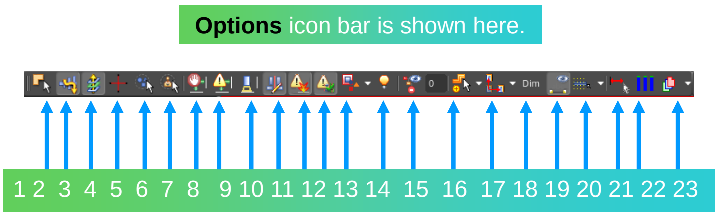
Before clicking on any icon, and while hovering over it, the function mapped to that icon will pop up, as shown with the **Stretch** icon.



Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for details on these features.

Options Icon Bar in the Layout Canvas



1. Full/Partial Select
2. Path Spine ON/OFF
3. Via Stack ON/OFF
4. Create/Edit Snap Mode
5. Transparent Group ON/OFF
6. Transparent Clone ON/OFF
7. DRD Enforce ON/OFF
8. DRD Notify ON/OFF
9. DRD Post-Edit ON/OFF
10. Constraint Aware Editing ON/OFF
11. Update connectivity information when design is modified ON/OFF
12. Incremental Check Against Source ON
13. Selection Granularity ON/OFF
14. InfoBalloon ON/OFF
15. Display Stop Level
16. Create Snap Mode
17. Edit Snap Mode
18. Dimming ON/OFF
19. Display Net Names ON/OFF
20. Snap Wires to: Manufacturing Grid
21. Dynamic Measurement ON/OFF
22. Highlight Aligned Edges ON/OFF
23. Layer Scope

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1*, for details on these features.

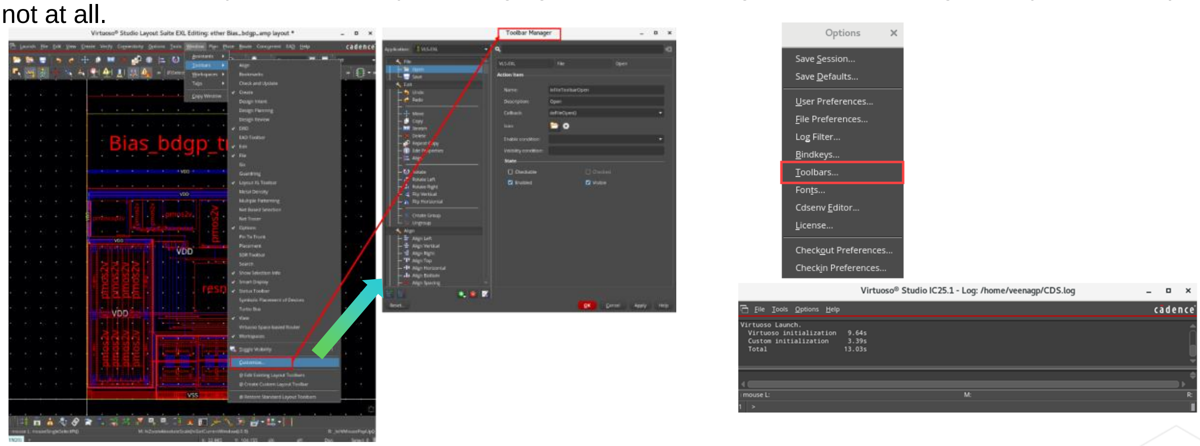
Customizing the Toolbar Graphical Interface Using the Toolbar Manager

From the layout canvas, choose **Window – Toolbars – Customize**

or

CIW – Options – Toolbars to display the Toolbar Manager.

You can customize your toolbars by rearranging the icons, adding icons, or deleting icons you use frequently or not at all.



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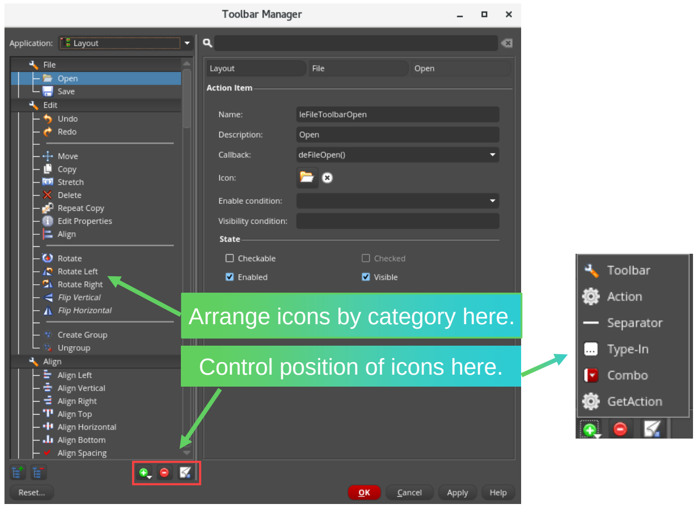
To maximize the working area in the canvas, you can re-organize the default toolbars to better reflect your working style.

Refer to *<CDS_Install_Dir>/share/cdssetup/dfII/toolbars/byApplication/* directory for the toolbars files.

Toolbar Manager: Editing the Layout Toolbars

The Toolbar Manager lets you make incremental and local changes to the Virtuoso application toolbars.

You can add, edit, or remove toolbars and toolbar items of applications whose toolbars are defined in Virtuoso Design Editor toolbar files.



Using the Toolbar Manager

The **Toolbar Manager** lets you make incremental and local changes to the Virtuoso application toolbars. You can add, edit, or remove toolbars and toolbar items of applications whose toolbars are defined in Virtuoso Design Editor toolbar files.

Note: The Toolbar Manager cannot modify toolbars created programmatically using SKILL APIs, such as *hiCreateToolbar*.

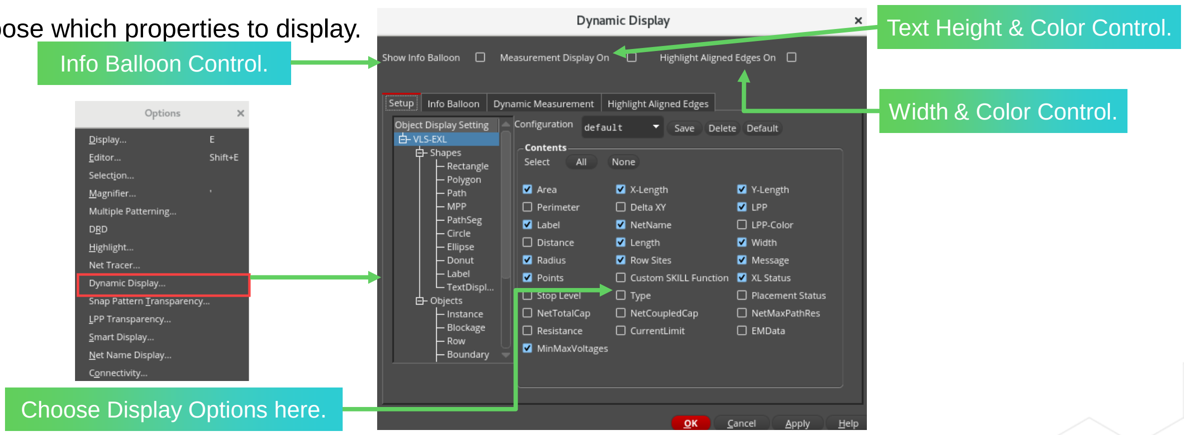
The **Toolbar Manager** saves your changes in the *applicationName.overlay* files in the *.cadence* directory present in your home directory. When you launch an application, the Virtuoso design environment creates toolbars based on the *applicationName.toolbars* file, overlaid by the *applicationName.overlay* file. This method ensures the seamless implementation of your changes even after you upgrade the Virtuoso installation.

Refer to the *Virtuoso Studio Design Environment User Guide, Product Version IC25.1*, for more details on these features.

Displaying the Dynamic Display Form

In the layout canvas, choose **Options – Dynamic Display** to display the Dynamic Display form to control the dynamic display of the objects.

- Quickly describe the object in question with an Info Balloon.
- Choose which properties to display.



Dynamic Display Form

The Info display balloons are controlled by choosing **Options – Dynamic Display**. The available properties include Area, X-Length, Y-Length, Perimeter, Delta XY, LPP, Label, NetName, Distance, Length, Width, Radius, Row Sites, and Message. Points and hooks for using a custom SKILL function are also available. This lets you access the properties quickly and easily without having to press any keys.

By clicking the *Shapes* and *Objects* categories, you can filter the information that is displayed in the canvas as you pass the cursor over an object.

Show Info Balloon: When on, information balloons will appear displaying information about an object or shape when the pointer is moved over the existing object. The type of information can be customized and is object dependent (environment variable: balloonOn).

Measurement Display On displays information about objects and shapes dynamically, during the creation or editing of objects. The type of information can be customized and is object dependent (environment variable: measurementDisplayOn).

Highlight Aligned Edges On highlights edges of the shapes aligned to the pointer in horizontal and vertical directions (environment variable: highlightAlignedEdges).

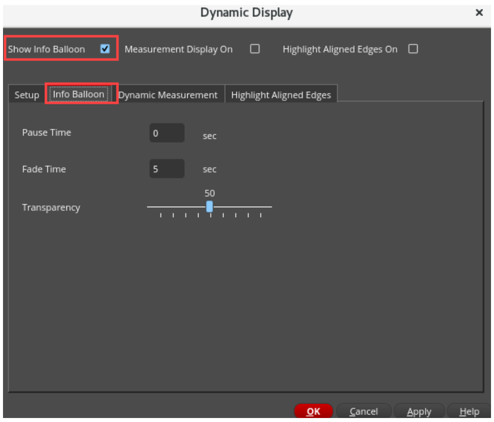
The Dynamic Display form comprises the following tabs:

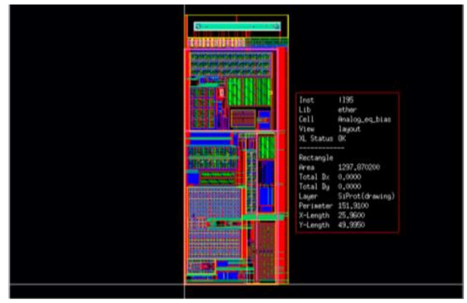
- Setup
- Info Balloon
- Dynamic Measurement
- Highlight Aligned Edges

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1*, for details on these features.

Dynamic Display Form: Info Balloon Tab

- By selecting the Info Balloon tab in the Dynamic Display form, you can set the *Pause Time*, *Fade Time*, and *Transparency* values.
- With the *Show Info Balloon* option selected, an Info Balloon appears when the cursor passes over an object.





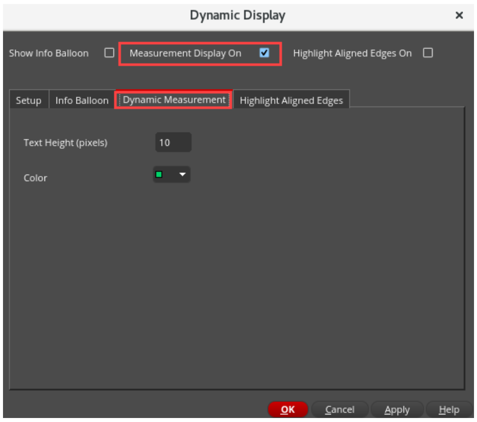
Info Balloon Tab

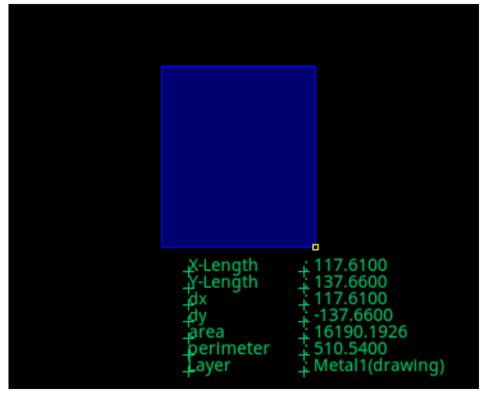
By selecting the Info Balloon tab in the Dynamic Display form, you can set the *Pause Time*, *Fade Time*, and *Transparency* values.

- **Pause Time** indicates the amount of time before the information balloon will appear when the pointer is over an object. Valid values are between **0** and **10** (environment variable: balloonPauseTime).
- **Fade Time** indicates the amount of time (in seconds) for the information balloons to fade away after the balloon appears. Valid values are between **.001** and **100** (environment variable: balloonFadeTime).
- **Transparency** indicates the transparency of the balloon information box. When set to **0**, the box is completely transparent. When set to **100**, objects behind the box are not visible. The default is **50** (environment variable: balloonTransparency).

Dynamic Display Form: Dynamic Measurement Tab

- By selecting the Dynamic Measurement tab in the Dynamic Display form, you can control the *Text Height* and *Color*.
- When *Measurement Display On* option is selected, while creating a rectangle, wire, or polygon, Dynamic Measurement provides real-time geometric values.





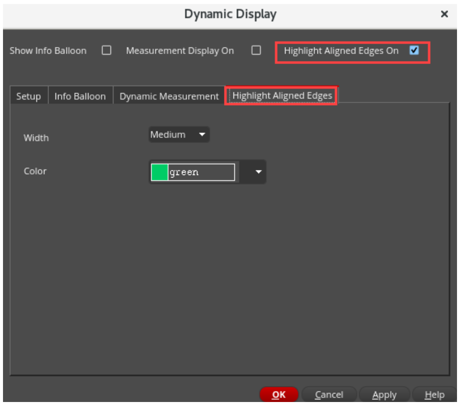
Dynamic Measurement Tab

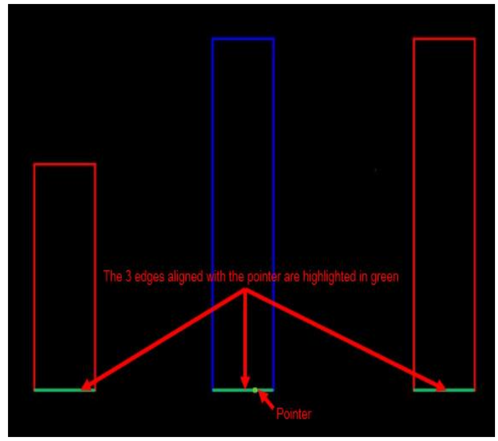
By selecting the Dynamic Measurement tab in the Dynamic Display form, you can control *Text Height* and *Color*.

- **Text Height (pixels)** indicates the height of the text displayed during dynamic measurement. Valid values are any integer between **4** and **50** (environment variable: measurementTextHeight).
- **Color** indicates the color of the text displayed during dynamic measurement (environment variable: dynamicMeasurementTextColor).

Dynamic Display Form: Highlight Aligned Edges Tab

- By selecting the Highlight Aligned Edges tab in the Dynamic Display form, you can control the *Width* and *Color*.
- When *Highlight Aligned Edges On* option is selected, the edges of the shapes aligned to the pointer in horizontal and vertical directions are highlighted.





Highlight Aligned Edges Tab

By selecting the Highlight Aligned Edges tab in the Dynamic Display form, you can control *Width* and *Color*.

- **Width** indicates width of the highlight that is displayed on the edges of the shapes aligned. You can choose from *Thin*, *Medium*, or *Thick* (environment variable: highlightAlignedEdgesWidth).
- **Color** indicates the color of the highlight displayed on the edges of aligned shapes (environment variable: highlightAlignedEdgesColor).

Module Summary

In this module, you learned how to

- Use the Layers, Objects, and Grids Assistants
- Customize the drawing layers
- Define and use layer sets
- Use custom bindkeys
- Save user-specified options as defaults
- Build your own customized toolbar
- Use Dynamic Display: Info Balloon, Dynamic Measurement, and Highlight Aligned Edges

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Labs

Lab 3-1 Using the Palette Assistants

Lab 3-2 Loading and Verifying the Bindkeys

Lab 3-3 Saving the Defaults

Lab 3-4 Using the Toolbar Manager

This page does not contain notes.

Quiz: The Palette Containers in Virtuoso

Virtuoso Layout Suite supports which of the following types of Palette Containers?

Select all that apply.

A. Single Assistant Container
B. Multiple Assistants Container
C. Single Window Container
D. Multiple Windows Container
E. All of the above

Answers: A, B, and C

Virtuoso Layout Suite supports the Single Assistant Container, Multiple Assistants Container, and Single Window Container.

Answers: A, B, and C

Virtuoso Layout Suite supports the Single Assistant Container, Multiple Assistants Container, and Single Window Container.

Quiz: The Palette Assistant in Virtuoso

The **Palette** Assistant is comprised of which of the following panels?

Select all that apply.

A. Layers
B. Objects
C. Grids
D. Instances
E. Nets
F. All of the above

Answers: A, B, and C

Palette is a tool used to view and manage the display context of layer, object, and grid items by editing their properties.

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Answers: A, B, and C

Palette is a tool used to view and manage the display context of layer, object, and grid items by editing their properties.

Quiz: Displaying the Area in Layout

What is the command used to open the cellview data only within the area you describe by entering the selection shape?

Select the correct choice.

A. Set Area Display
B. Select By Area
C. Select By Line
D. Extend selection to object
E. None of the above

Answer: A

In the layout canvas, choosing the *View – Area Display – Set* menu command opens the Set Area View Level form which can be used to open the hierarchy in the defined area.

Answer: A

In the layout canvas, choosing the *View – Area Display – Set* menu command opens the Set Area View Level form which can be used to open the hierarchy in the defined area.

Quiz: The Dynamic Display Form in Layout

What are the options available in the **Dynamic Display** form?

Select all that apply.

A. Show Info Balloon

B. Measurement Display On

C. Highlight Aligned Edges On

D. Display Edges On

E. Display Highlight Edges On

F. All of the above

Answers: A, B and C

The Dynamic Display form has the Show Info Balloon, Measurement Display On, and Highlight Aligned Edges On options.

Answers: A, B and C

The Dynamic Display form has the Show Info Balloon, Measurement Display On, and Highlight Aligned Edges On options.

Quiz: The Technology File in Virtuoso

Which panel lists the layer-purpose pairs that are defined in the *leLswLayers/techLayerPurposePriorities* subsection of the technology file?

The **Layers** panel lists the layer-purpose pairs defined in the *leLswLayers/techLayerPurposePriorities* subsection of the technology file.

Which panel lists the layer-purpose pairs that are defined in the *leLswLayers/techLayerPurposePriorities* subsection of the technology file?

- The **Layers** panel lists the layer-purpose pairs defined in the *leLswLayers/techLayerPurposePriorities* subsection of the technology file.

Quiz: The Panel to Control the Visibility and Selectability of Objects in Virtuoso

Which panel can be used to control the visibility and selectability of objects such as instances, pins, and vias?

To control the visibility and selectability of objects such as instances, pins, and vias, you can use the **Objects** panel.

Which panel can be used to control the visibility and selectability of objects such as instances, pins, and vias?

- To control the visibility and selectability of objects such as instances, pins, and vias, you can use the **Objects** panel.

Quiz: Loading the Bindkey File Automatically in Virtuoso

Which file can be used to load the Bindkey File automatically?

You can load the Bindkey File from the **.cdsinit** Startup File automatically.

Which file can be used to load the Bindkey File automatically?

- You can load the Bindkey File from the **.cdsinit** Startup File automatically.

Quiz: Controlling the Appearance of Objects and the Behavior of Commands in Layout

Which form controls the appearance of objects and the behavior of commands in the layout cellview?

The **Display Options** form controls the appearance of objects and the behavior of commands in the layout cellview.

Which form controls the appearance of objects and the behavior of commands in the layout cellview?

- The **Display Options** form controls the appearance of objects and the behavior of commands in the layout cellview.

Module 4

Basic Layout Commands

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Module Objectives

In this module, you

- Save and restore your design window
- Use the Dynamic Measurement feature
- Use selection options to streamline your design efforts
- Use the Quick Align and Align toolbar features

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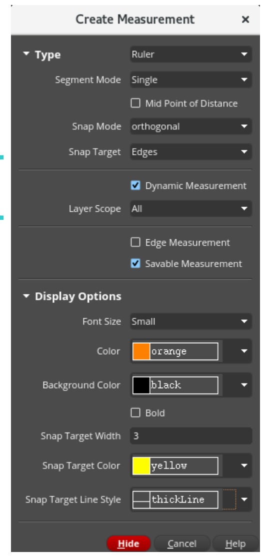
The Create Measurement Form

Choose **Tools – Create Measurement** menu command.

The **Create Measurement** form with all the fields expanded:

- Enable Dynamic Measurement
- Specify Layer Scope:

All	Measurement considers all layers, for example, between M1 and M2
Active	Only active layer in the Palette is considered
Edge	Only the layer underneath the cursor position is considered
Palette	List of LPPs in Palette is considered
Selectable	Only selectable LPPs are considered



- Whether to create a Ruler or Distance or both
- Single/Multiple/Auto Segment Mode
- Find Mid Point of Distance
- Snap Mode
- Snap Target Edges or Points

- When Edge Measurement is On, length of the edge is displayed rather than distance between the edges
- Savable Measurement creates rulers that can be saved in the database
- Display Options are controlled here

The Create Measurement Form

Type lets you select the display type of the ruler (environment variable: rulerDisplayType). **Ruler** displays the ruler in the design window. **Distance** displays the distance between two objects on the design window. **Both** displays both the ruler and the distance between two objects on the design window.

Dynamic Measurement, if selected, enables you to dynamically display measurement between the pointer and the closest edges. If you move the pointer between two shapes, the distance between the two edges is displayed. If you move the pointer on an edge of a shape, the edge is highlighted in yellow. Also, the distances between the highlighted edge and the two closest edges are displayed.

Layer Scope lets you select the layers to be considered for dynamic measurement. You can select *All*, *Active*, *Edge*, *Palette*, or *Selectable* layers.

Edge Measurement, if selected, lets you enable the edge measurement mode for rulers. Edge measurement mode allows you to click the edge of an object and a measurement is drawn with the two vertices as the two ends. The first click lets you select the edge and the second click lets you place the measurement on the canvas (environment variable: rulerEdgeMode). **Note:** The Edge Management check box is disabled when the *Snap Target* is None or when the *Segment Mode* is *Multiple* or *Auto*.

Savable Measurement saves the persistent rulers in a cellview when you save the cellview. It is on by default. If this option is off, all the rulers existing in the cellview are deleted when you save the cellview (environment variable: saveRulers).

Display Options let you define the ruler and snap target visualization settings.

Font Size lets you select the font size of the ruler labels. You can choose from *Small*, *Medium*, or *Large* (environment variable: rulerFontSize).

Color lets you select the font color of the ruler labels.

Background Color lets you select the background color of the ruler labels (environment variable: rulerLabelBackgroundColor).

Bold, if selected, displays the ruler labels in bold face (environment variable: rulerFontBold).

Snap Target Width enables you to specify the width of the snap target highlight. The default is 0 (environment variable: targetWidth).

Snap Target Color enables you to select the color of the snap target highlight. The default Environment variable is targetColor.

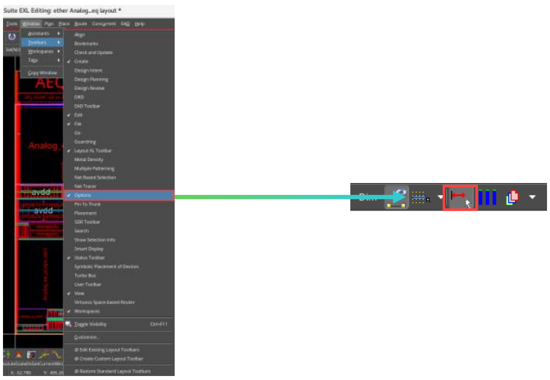
Snap Target Line Style enables you to select the type of snap target highlight. The default is thickline (environment variable: targetLineStyle).

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

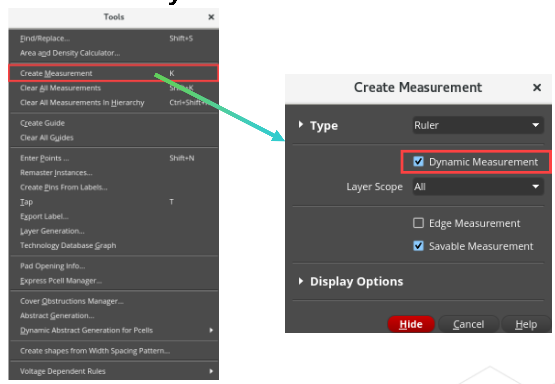
Enabling the Dynamic Measurement

The **Dynamic Measurement** option, if selected, enables you to dynamically display measurement between the pointer and the closest edges in the Create Measurement form.

1. From **Window – Toolbars**, activate the **Options** toolbar and use the **Dynamic Measurement** icon.



2. Invoke the **Tools – Create Measurement** (bindkey “**<Key>K**”) menu command and enable the **Dynamic Measurement** button.



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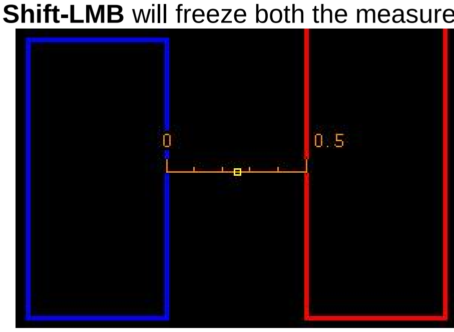
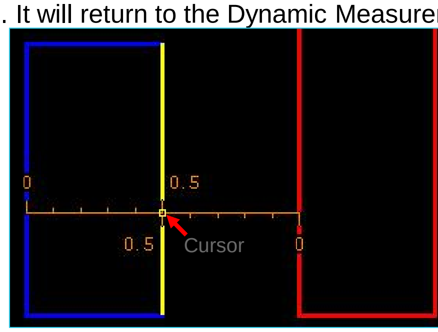
The **Dynamic Measurement** option, if selected, enables you to dynamically display measurement between the pointer and the closest edges in the Create Measurement form.

Using the Dynamic Measurement

1. Invoke the *Create Measurement* form by choosing the **Tools – Create Measurement** menu command or through bindkey “**<Key>K”** and enable the *Dynamic Measurement* option.
2. Move the cursor between 2 edges.
3. The distance between these 2 edges is displayed. Freeze the measurement by pressing **Shift-LMB**. It will return to the Dynamic Measurement mode.
4. Move the cursor to an edge. The edge will be highlighted in yellow. Distances between this edge and the two closest edges are displayed.
5. Pressing **Shift-LMB** will freeze both the measurements. It will return to the Dynamic Measurement mode again.





LMB will create a traditional ruler. To delete all measurements, use **Shift-K**.

In the Create Measurement form, **Dynamic Measurement**, if selected, enables you to dynamically display measurement between the pointer and the closest edges.

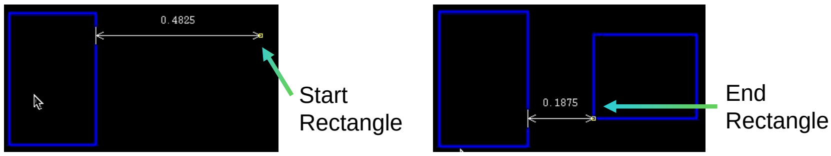
If you move the pointer between two shapes, the distance between the two edges is displayed.

If you move the pointer on an edge of a shape, the edge is highlighted in yellow. Also, the distances between the highlighted edge and the two closest edges are displayed.

Dynamic Measurement in Create/Edit Commands

Once enabled, **Dynamic Measurement** is displayed in the **Create/Edit** commands. For example,

1 Create Rectangle



Measurement between the closest shape and cursor position is displayed.

2 Move Command

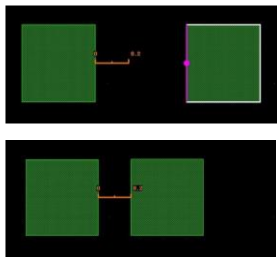
As you move the cursor, **Dynamic Measurement** shows the distance between the initial and new positions, and the spacing between the rectangle and the closest shape.

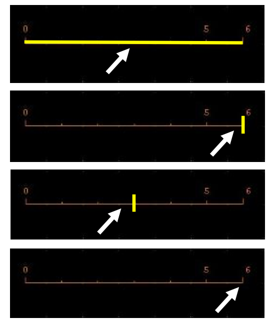
Create Rectangle command: Measurement between the closest shape and cursor position is displayed.

Move command: As you move the cursor, **Dynamic Measurement** shows the distance between the initial and new positions, and the spacing between the rectangle and the closest shape.

Smart Snapping of Ruler in the Quick Align Command

1. Smart snapping in the Quick Align command now supports snapping of the rulers.
2. User can select the ruler, the end points, the end edges, or the centre line of a ruler as the reference or the target.
3. This feature is useful when the user tries to move or stretch an object and align it to a ruler.





You can align objects by using the **Edit – Quick Align** command or the **Edit – Advanced – Align** command.

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Aligning Objects

You can align objects by using the **Edit – Quick Align** command or the **Edit – Advanced – Align** command. The two commands enable you to perform similar tasks but differ in their usage. You can perform the following tasks by using these commands: Align objects to a point, Align objects to an edge, Align objects to a layer, and Align objects to a target axis.

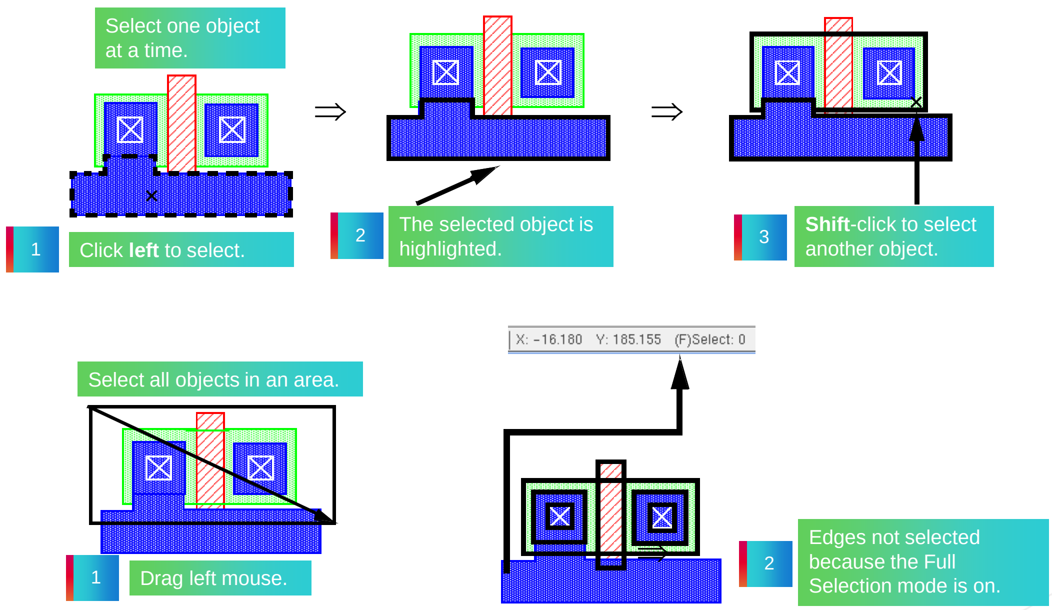
Both the commands work in both pre- and post-selection modes.

The Quick Align command enables you to quickly align objects, in a more interactive manner than the Align command. In comparison with the Align command, the Quick Align command provides the following additional features:

- Enables smart snapping of edges, centerlines, and points within the specified aperture
- Enables cyclic highlighting of the reference and target objects within the aperture distance from the pointer by using the *Spacebar* key
- Enables the alignment with respect to reference or target objects across the hierarchy
- Supports the spacing rules defined in the technology file
- Enables alignment by stretching partially selected objects

The Quick Align command enables you to align an edge, a centerline, or a point to another edge, centerline, or point, whereas the Align command aligns an object bounding box to the bounding box of another object.

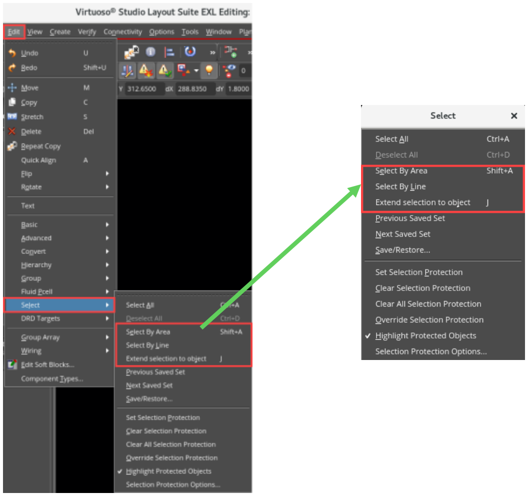
Selecting Objects



- To select one object, **left**-click inside the shape. This also deselects all other previously selected shapes.
- To select additional objects, press the **Shift** key and **left**-click inside each additional shape.
- To deselect shapes, press the **Control** key and **left**-click inside a shape.
- To select an edge or corner, select **F4** (partial select) and **left**-click on an edge or corner.
- To select multiple shapes, **left-**click and drag to enclose the shapes.
- **F4** toggles full and partial select. If using full select, only shapes which are completely enclosed will be selected. If using partial select, each edge or corner which is enclosed will be selected.
- To deselect individual shapes, press **Control** and **left**-click inside the shape. This also works for deselecting multiple shapes when you use **Control** and left mouse drag to enclose the shapes to deselect.

Selecting Objects by Defining the Selection Shape

In the layout canvas, choose the **Edit – Select** menu entry to invoke the commands to select objects by defining the selection shape.



You can select objects by defining the selection shape.

Three methods are available:

1. Selecting Objects By Defining an Area.
2. Selecting Objects by Drawing a Line.
3. Fully Selecting Partially Selected Objects.

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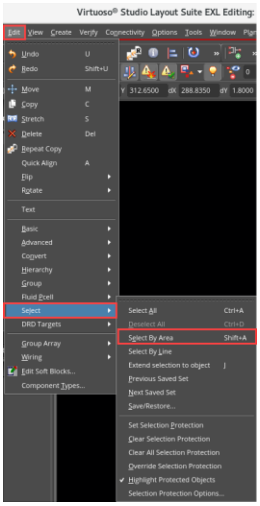
Selecting Objects

You can select objects by defining the selection shape:

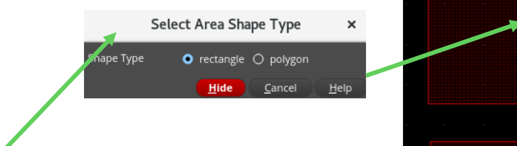
- Selecting Objects By Defining an Area
- Selecting Objects by Drawing a Line
- Fully Selecting Partially Selected Objects

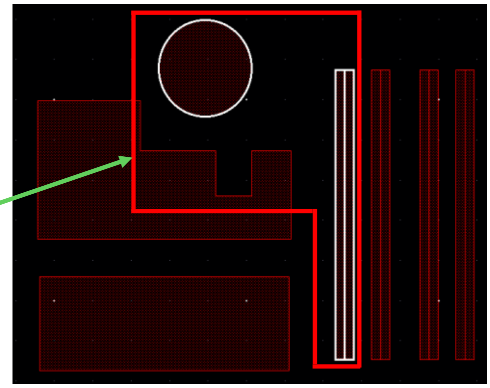
1 Selecting Objects by Area

In the layout canvas, choose **Edit – Select – Select By Area** or press the bindkey **Shift+A** to invoke the Select Area Shape Type form.



You can Select By Area (Rectangle or Polygon).





A drawn polygon completely surrounding objects will select them.

Selecting Objects By Defining an Area

To select objects within an enclosed area:

1. Select **Edit – Select – Select By Area**.
2. Press **F3**. The Select Area Shape Type form appears.
3. Select the shape type as *rectangle* or *polygon*.
4. In the design window, draw the shape selected in the form.

   Any objects enclosed within the drawn shape are selected.

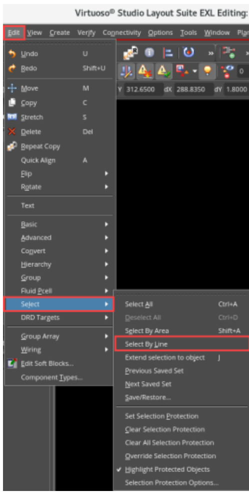
   Note: The selection of fully or partially enclosed objects within the drawn shape is based on the settings in the Area Selection Controls section in the Selection Options Form.
5. You can continue to select more shapes by drawing a rectangle or polygon around them.
6. To add objects to the previous selection, press **Shift** and draw the shape around the additional objects to be selected.
7. To remove objects from the previous selection, press **Ctrl** and draw the shape around the objects to be removed from the selected set.
8. Press **Esc** to cancel the command.

Select Area Shape Type Form

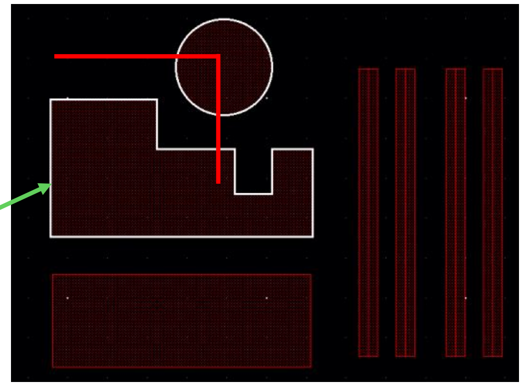
Shape Type enables you to use a *rectangle* (selected by default) or a *polygon* to define an area for selecting objects.

2 Selecting Objects by Line

In the layout canvas, choose **Edit – Select – Select By Line** to activate the Select By Line command.



You can select by Line.



A drawn line intersecting any object will select it.

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Selecting Objects by Drawing a Line

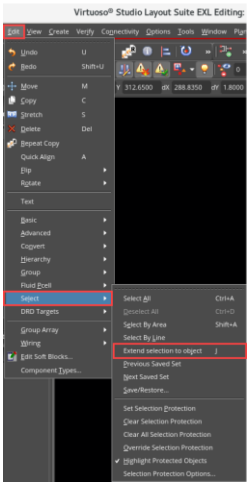
To select objects by using a line:

1. Select **Edit – Select – Select By Line**.
2. In the design window, click on the first point of the line.
3. Click at the next point of the line.
4. You can continue to click to define more points of the line.
5. **Double**-click to end the line.

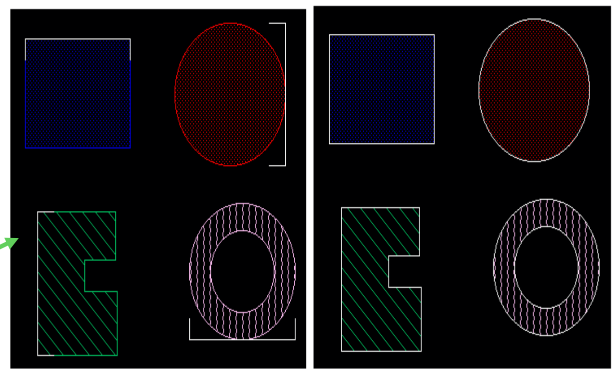
   Any objects overlapping the line are selected.
6. You can continue to select more shapes by drawing a line to overlap them.
7. To add objects to the previous selection, press **Shift** and draw a line to overlap the additional objects to be selected. Keep Shift pressed until you **double**-click to end the line.
8. To remove objects from the previous selection, press **Ctrl** and draw a line to overlap the objects to be removed from the selected set. Keep **Ctrl** pressed until you **double**-click to end the line.
9. Press **Esc** to cancel the command.

3 Selecting Objects by Extending Selection to Object

In the layout canvas, choose **Edit – Select – Extend selection to object** or press the bindkey **J** to activate the Extend selection to object command.



You can fully select objects that are partially selected.



After selecting the partially selected objects, invoking the command will select them fully.

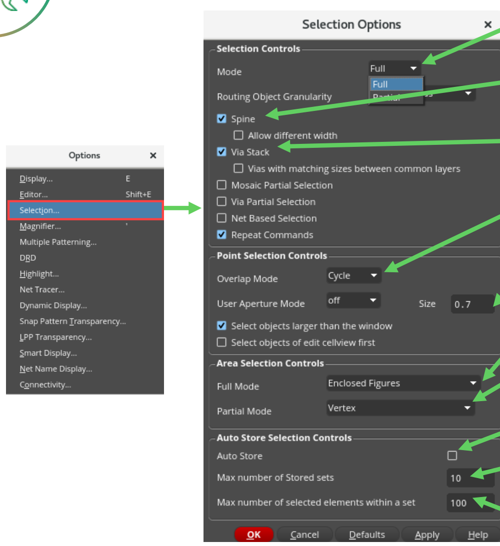
Fully Selecting Partially Selected Objects

To fully select objects that are partially selected:

1. Select **Edit – Select – Extend selection to object [J]**.
2. In the design window, select the partially selected object that you want to fully select.
3. Press **Esc** to cancel the command.

Invoking the Selection Options Form

In the layout canvas, choose **Options – Selection** to open the Selection Options form.



Mode: Full or Partial.

Spine: If unselected, only the pathSeg clicked on will be selected.

Via Stack: Allows selection with a single click.

Overlap Mode: Cycle Select or Toggle Select.

User Aperture Mode: If on, sets selection granularity.

Enclosed, Crossed, or Both Figures.

Vertex, Enclosed, Crossed, or Both Edges.

Auto Store: ON/OFF.

Limits the number of sets which can be retained by the Save/Restore Selection Set form.

Limits the number of objects you can enter into one selection set.

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Selection Options Form

This form comprises the following sections: *Selection Controls*, *Point Selection Controls*, *Area Selection Controls*, and *Auto Store Selection Controls*.

Selection Controls

- **Mode** controls whether the selection is full or partial.
- **Spine** selects all the contiguous *pathSegs* that have the same width as the first-clicked *pathSeg*.
- **Via Stack** allows the selection of a via stack with a single click.

Point Selection Controls

- In **Overlap Mode**, the **Toggle** mode allows you to change selection between two objects. The **Cycle** mode allows you to select overlapped objects one by one by using the left mouse button to select any object and continuing to select them by clicking on the original object until you reach the largest overlapped object.
- **User Aperture Mode** defines a box around the initial point. You can select multiple points within this aperture range and the software will treat it as one point. The default is five pixels. You can specify the size of the box in the **Size** field.

Area Selection Controls

- **Full Mode** controls the area selection in the Full selection mode.
- **Partial Mode** controls the area selection in the Partial selection mode.

Auto Store Selection Controls

- The **Auto Store** section stores multiple sets of selections you have made. You can reload into the cellview these selections without having to reselect the objects. The radio button controls whether or not you can store the selections.
- **Max number of Stored sets** limits the number of sets which can be retained by the Save/Restore Selection Set form.
- **Max number of selected elements within a set** limits the number of objects you can enter into one selection set.

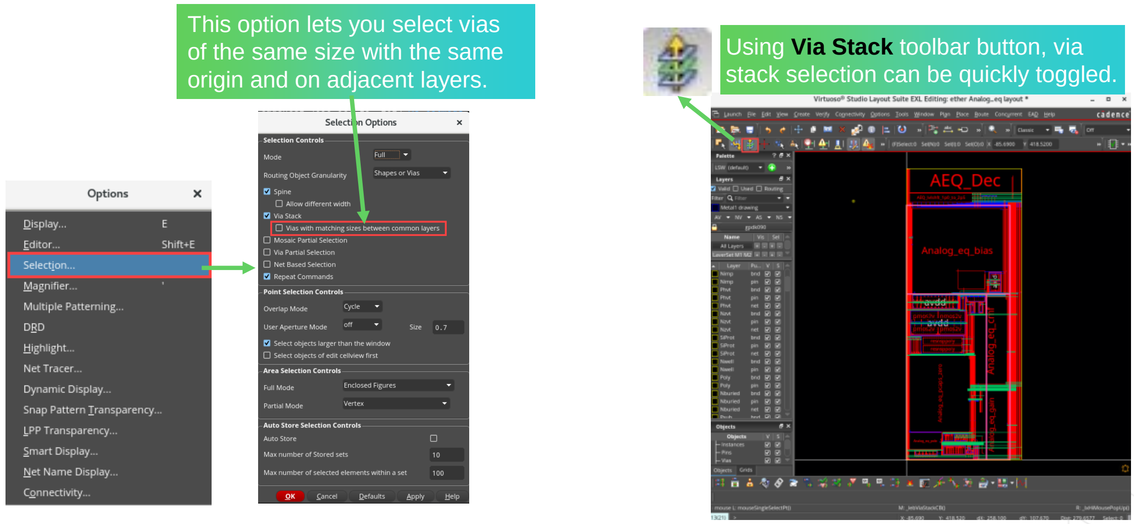
Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for more details on these features.

Via Stack Selection

In the layout canvas, choose **Options – Selection**. The Selection Options form can be used to determine whether the stacked vias must be completely overlapping or not.

This option lets you select vias of the same size with the same origin and on adjacent layers.

Using **Via Stack** toolbar button, via stack selection can be quickly toggled.



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Via Stack allows the selection of a via stack with a single click (environment variable: viaStackSelection).

Vias with matching sizes between common layers lets you select vias of the same via stack when there are two or more super-imposed via stacks on adjacent layers with the same origin. Vias of the same via stack have the same size on common layers. If this option is off, multiple super-imposed via stacks are selected as a single via stack and are available for editing as a single via stack. This option is available only when *Via Stack* is selected.

For each pair of adjacent vias in a via stack, the bounding box of the top layer in the lower via is equal to the bounding box of the bottom layer in the higher via. This refers to the matching sizes of the vias between the common layers and is used for identifying vias of the same via stack. For example, the two vias M1M2 and M2M3 in the via stack M1M3 are on adjacent layers and share the same origin. In addition, the bounding box of Metal2 shape in M1M2 is equal to the bounding box of the Metal2 shape in M2M3. This testifies that the two vias belong to the same via stack (environment variable: viaStackSelectionMatchingSizes).

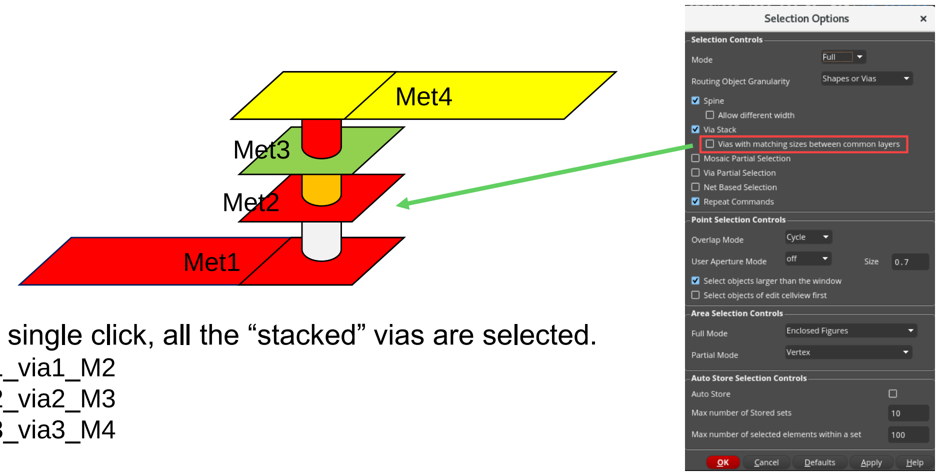
Mosaic Partial Selection allows partial selection of mosaics.

Via Partial Selection allows partial selection of vias. You can control the status of this option using the SKILL® command *hiGetCurrentWindow()~>selectPartialVia*.

Repeat Commands allows the commands to automatically repeat, if you first select the command and then the object. This option does not affect commands if you first select an object, then the command.

Example: Via Stack Selection

With **Via Stack** option selected, a single click will select the group of vias.



With a single click, all the "stacked" vias are selected.

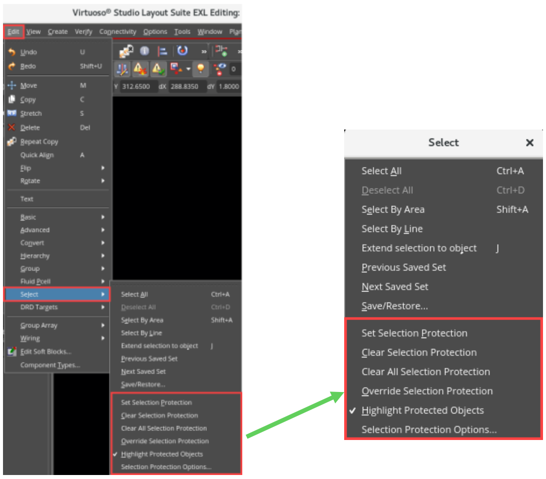
1. M1_via1_M2
2. M2_via2_M3
3. M3_via3_M4

Activating the **Via Stack** selection with a single click will select all the "stacked" vias. In the example depicted here, a single click will select the group of three vias:

1. M1_via1_M2
2. M2_via2_M3
3. M3_via3_M4

Accessing the Protection Commands

In the layout canvas, choose **Edit – Select** menu entry to invoke the commands to set individual objects as selectable or unselectable (selection protection).



The protection commands are:

1. Set Selection Protection
2. Clear Selection Protection
3. Clear All Selection Protection
4. Override Selection Protection
5. Highlight Protected Objects
6. Selection Protection Options

You can use these commands to set individual objects as selectable or unselectable (selection protection).

Using the Protection Commands

You can use the commands on the **Edit – Select** menu to set individual objects as selectable or unselectable (selection protection). To control the selectability of objects based on the LPP they are on, see *By Using the Layers Assistant*.

The options mentioned below are available from the **Edit – Select** menu.

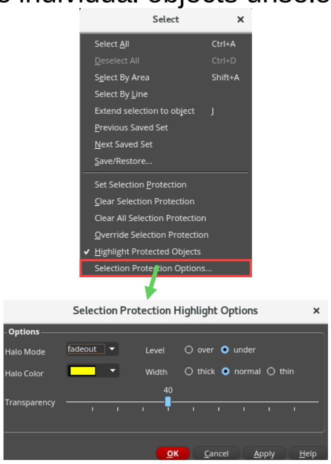
- *Set Selection Protection* – Protects objects from being selected.
- *Clear Selection Protection* – Starts an enter function that removes protection from selected objects. Start this function, then either click or use area selection to specify the objects you want to make selectable again.
- *Clear All Selection Protection* – Makes all currently protected objects selectable again. **Note:** Only the objects that were marked protected by using the Set Selection Protection command are made selectable. The selectability of objects set using the Layers assistant is not affected.
- *Selection Protection Options* – Configures the settings for highlighting the objects that are marked protected. This command opens the Selection Protection Highlight Options form.

Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide*, *Product Version IC25.1*, for details on these features.

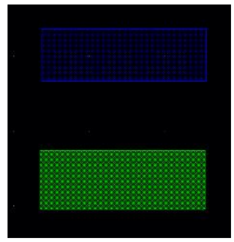
Invoking the Selection Protection Highlight Options Form

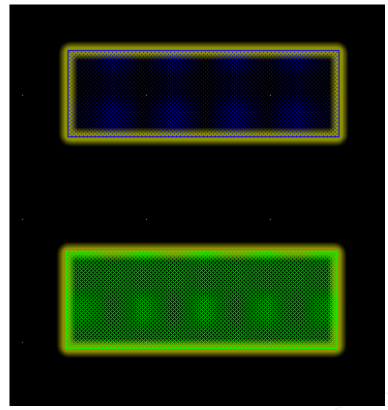
In the layout canvas, choose **Edit – Select – Selection Protection Options** menu entry to invoke the Selection Protection Highlight Options form.

You can make individual objects unselectable.



Shapes are highlighted after setting protection.





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Selection Protection Options – Configures the settings for highlighting the objects that are marked protected. This command opens the Selection Protection Highlight Options form.

When you set protection, the highlight is placed on the shape and the shape will not be selectable, therefore it will not be editable.

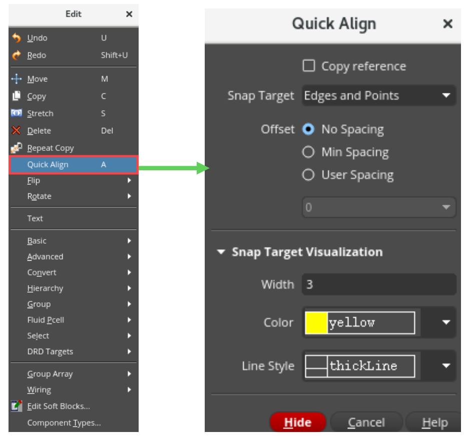
To edit a protected object, the user can either “clear the selection protection” for that object or for all objects in the current design.

Invoking the Quick Align Form

In the layout canvas, choose **Edit – Quick Align** or press the bindkey **A** to invoke the Quick Align form.

Two methods of **Quick Align** are available:

1. Quick Align: Single Object
2. Quick Align: Multiple Objects



Using the Quick Align Command

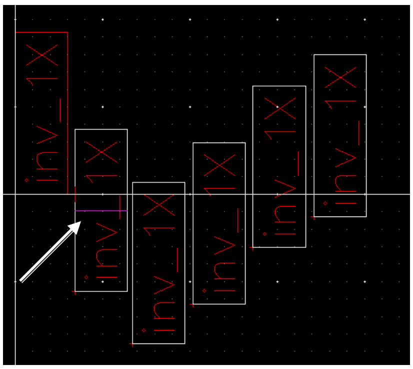
The *Quick Align* command enables you to use objects, instances, and groups as the reference set and also use them as the target to which the reference set is to be aligned. By using the Quick Align command, you can select the edges, centerlines, and points on objects in an instance or a group, which may be present at any level in the hierarchy.

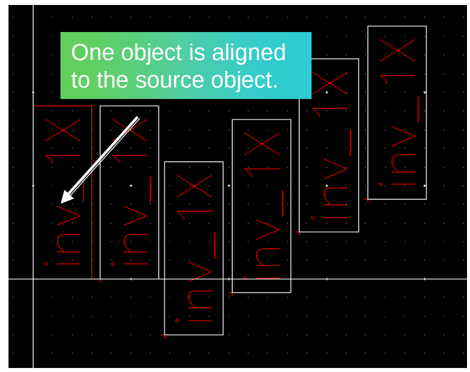
Note: The reference and target object can be the same object in the Quick Align command.

1 Quick Align: Single Object

To align the selected set retaining original spacings and offsets, after invoking the **Quick Align** command:

- Select objects to align.
- **Single**-click on the source alignment edge.
- **Single**-click on the destination alignment edge.





Quick Align Form

Copy reference, if selected, enables you to copy an object during alignment (environment variable: quickAlignMode).

Note: The *Copy* option will only copy fully selected objects. It will not copy partially selected objects.

Offset

- **No Spacing** does not apply any spacing between the reference edge, centerline, or point and the target edge, centerline, or point (environment variable: quickAlignSpacingType).
- **Min Spacing** applies the minimum spacing value of the reference object layer, as specified in the technology file, while aligning the reference set to the target edge, centerline, or point (environment variable: quickAlignSpacingType).
- **User Spacing** applies the spacing you specify to the distance between the reference edge, centerline, or point and the target edge, centerline, or point. You can specify an integer (environment variable: quickAlignSpacingType, quickAlignUserSpacing).

Snap Target Visualization Settings

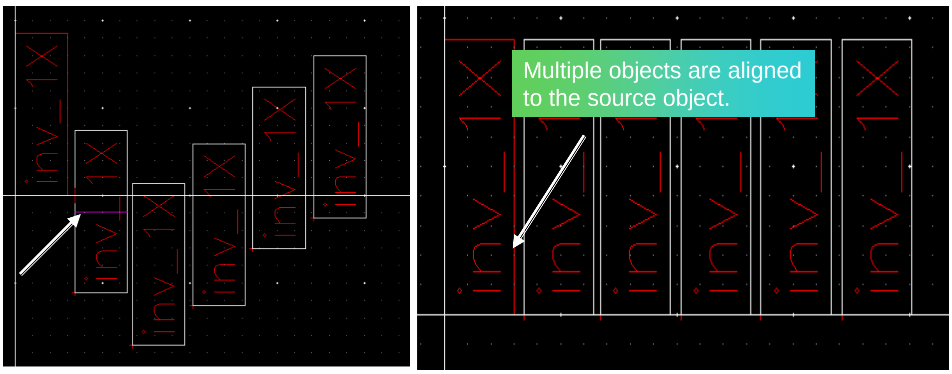
Use this section to customize the appearance of the edge and point highlights that appear on the potential reference and target edges and points.

- **Width** enables you to specify the width of the highlight. The default is 0 (environment variable: targetWidth).
- **Color** enables you to select the color of the highlight. The default is yellow (environment variable: targetColor).
- **Line Style** enables you to select the type of highlight. The default is solid (environment variable: targetLineStyle).

2 Quick Align: Multiple Objects

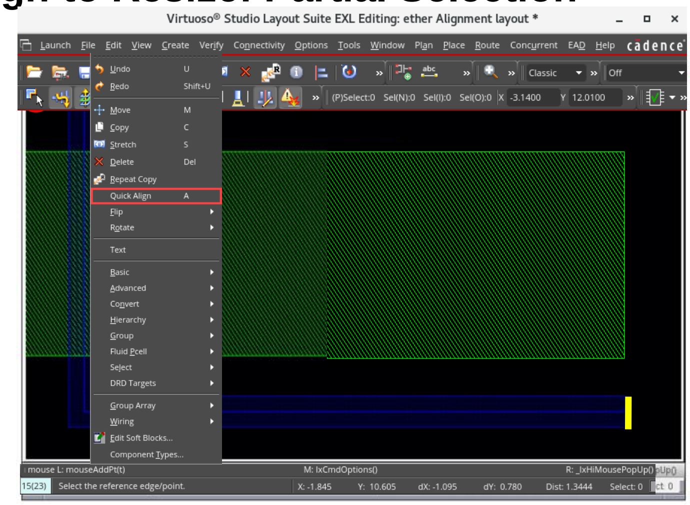
To align the entire selected set to the destination edge, after invoking the **Quick Align** command:

- Select objects to align.
- **Single**-click on the source alignment edge.
- **Double**-click on the destination alignment edge.



Refer to the *Virtuoso Layout Suite XL: Basic Editing User Guide, Product Version IC25.1*, for details on these features.

Using Quick Align to Resize: Partial Selection

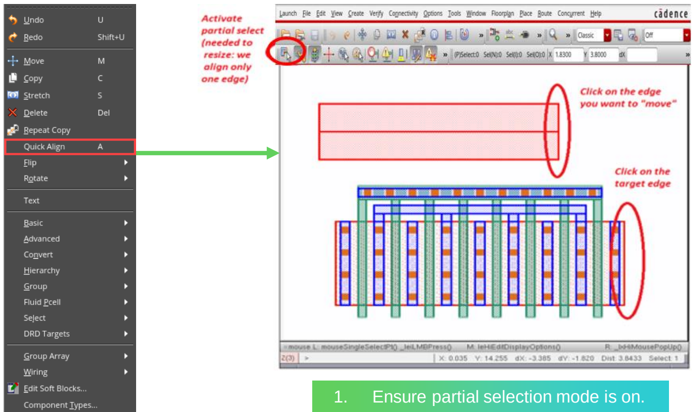


1. Ensure partial selection mode is on.
2. Invoke Quick Align.
3. Click on the edge you want to move.
4. Click on the target edge.

The poly is resized.

Note: The layers you want to click must be selectable. So in the example here, both Metal1.drawing and Poly must be selectable for the command to work.

Example: Using Quick Align to Resize: Partial Selection



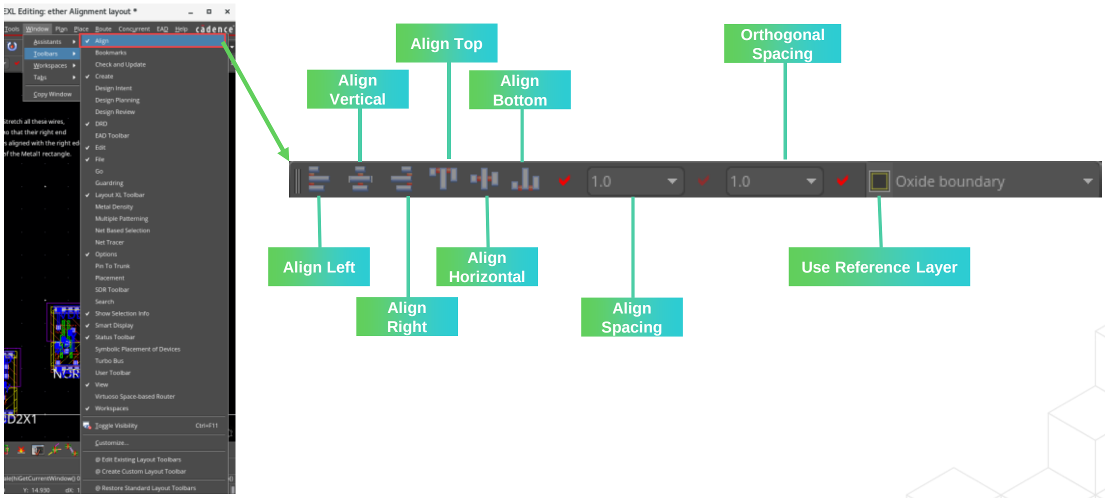
1. Ensure partial selection mode is on.
2. Invoke Quick Align.
3. Click on the edge you want to move.
4. Click on the target edge.

   The poly is resized.

Note: The layers you want to click must be selectable. So in the example here, both Metal1.drawing and Poly must be selectable for the command to work.

Invoking the Alignment Toolbar

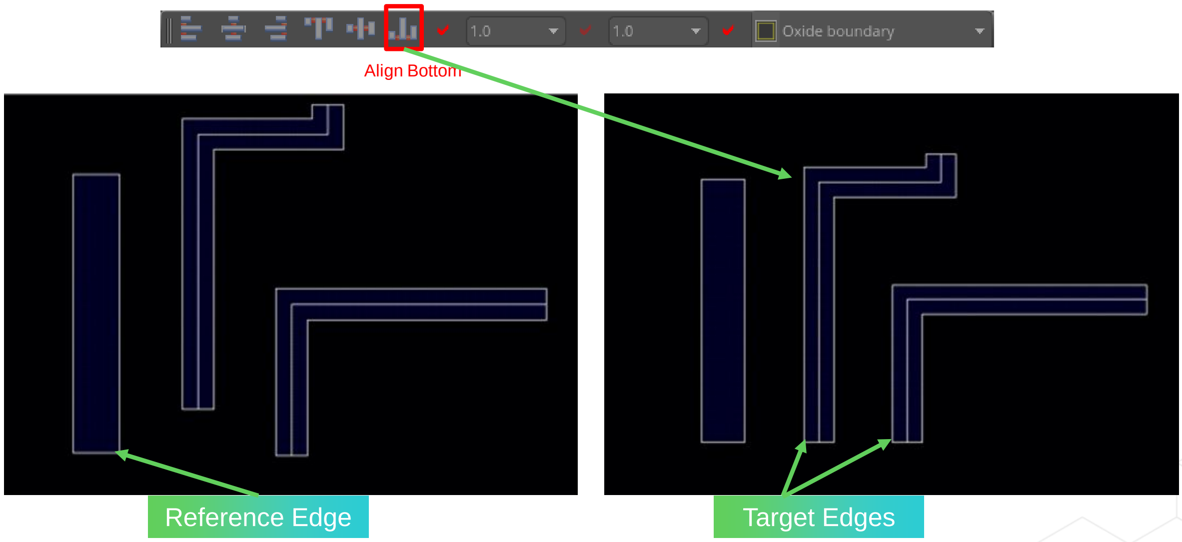
In the layout canvas, choose **Window – Toolbars – Align** to display the Align toolbar.



If the Alignment toolbar is not visible in your workspace, choose the **Window** – **Toolbars** – **Align** menu command to enable the Align toolbar in the layout canvas.

Using the Alignment Toolbar

The **Align** toolbar is used to quickly align layout objects, including shapes, labels, and instances.

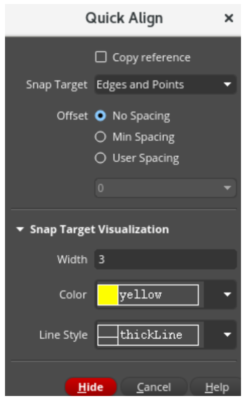


The two numeric fields included in the toolbar can be used to define the relative spacing between the aligned objects and the “orthogonal” spacing respectively. The latter can be used to build “stair” structures.

The spacing values can also be negative values. They are useful to abut transistors or standard cells when no connectivity is available for instance.

Quick Align and Align Toolbar

Choose **Edit – Quick Align** or press bindkey **A**.



Choose **Window – Toolbars – Align**.

Using the Quick Align Command

The *Quick Align* command enables you to use objects, instances, and groups as the reference set and also use them as the target to which the reference set is to be aligned. By using the Quick Align command, you can select the edges, centerlines, and points on objects in an instance or a group, which may be present at any level in the hierarchy.

Note: The reference and target objects can be the same object in the Quick Align command.

Aligning Objects Using the Align Toolbar

For performing quick alignment operations by using the *Align* command, use the Window – Toolbars – Align toolbar. Using the toolbar, you can perform left, right, top, and bottom alignment of the selected objects. You can also align them vertically or horizontally. The toolbar also provides the option of specifying alignment and orthogonal spacing values.

Module Summary

In this module, you learned how to

- Save and restore your design window
- Use the Dynamic Measurement feature
- Use selection options to streamline your design efforts
- Use the Quick Align and Align toolbar features

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Labs

Lab 4-1 Viewing Your Designs

Lab 4-2 Selecting and Deselecting the Objects

Lab 4-3 Using the Alignment Commands

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Quiz: The Option to View Distance in the Create Measurement Form in Layout

Which option, if selected, enables you to view the distance between the pointer and the closest edges in the Create Measurement form?

Enter the answer in the blank field.

____

Answer: Dynamic Measurement

The Dynamic Measurement option, if selected, enables you to view the distance between the pointer and the closest edges in the Create Measurement form.

Answer: Dynamic Measurement

The Dynamic Measurement option, if selected, enables you to view the distance between the pointer and the closest edges in the Create Measurement form.

Quiz: Selecting Objects by Defining the Selection Shape in Layout

What are the methods available to select objects by defining the selection shape?

Select all that apply.

A. Selecting Objects By Defining an Area
B. Selecting Objects By Drawing a Line
C. Fully Selecting Partially Selected Objects
D. Selecting Objects By Drawing a Polygon
E. All of the above

Answers: A, B, and C

You can select objects by defining the selection shape. Three methods are available:

- Selecting Objects By Defining an Area
- Selecting Objects by Drawing a Line
- Fully Selecting Partially Selected Objects

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Answers: A, B, and C

You can select objects by defining the selection shape. Three methods are available:

- Selecting Objects By Defining an Area
- Selecting Objects by Drawing a Line
- Fully Selecting Partially Selected Objects

Quiz: The Selection Protection Commands in Layout

What are the Selection Protection commands available in the Select menu in the layout canvas?

Select all that apply.

A. Set Selection Protection
B. Clear Selection Protection
C. Clear All Selection Protection
D. Override Selection Protection
E. Highlight Protected Objects
F. Selection Protection Options
G. Set Clear Protection
H. All of the above

Answers: A, B, C, D, E and F

The Selection Protection commands available in the Select menu in the layout canvas are:

- Set Selection Protection, Clear Selection Protection, Clear All Selection Protection, Override Selection Protection, Highlight Protected Objects, and Selection Protection Options.

Answers: A, B, C, D, E and F

The Selection Protection commands available in the Select menu in the layout canvas are:

- Set Selection Protection, Clear Selection Protection, Clear All Selection Protection, Override Selection Protection, Highlight Protected Objects, and Selection Protection Options.

Quiz: The Toolbar Used to Arrange the Objects in Layout

Name the toolbar you can use to perform left, right, top, bottom, vertical, and horizontal arrangement of the selected layout objects, including shapes, labels, and instances.

Select the correct choice.

A. Align
B. View
C. Create
D. Search
E. Options
F. None of the above

Answer: A

You can use the Align toolbar to perform left, right, top, bottom, vertical, and horizontal arrangement of the selected layout objects, including shapes, labels, and instances.

Answer: A

You can use the Align toolbar to perform left, right, top, bottom, vertical, and horizontal arrangement of the selected layout objects, including shapes, labels, and instances.

Quiz: The Bindkey to View Full Hierarchy in Virtuoso

Which bindkey can you use to view all levels of the hierarchy in the layout canvas?

You can use the **Shift+F** bindkey to view all levels of the hierarchy in the layout canvas.

Which bindkey can you use to view all levels of the hierarchy in the layout canvas?

- You can use the **Shift+F** bindkey to view all levels of the hierarchy in the layout canvas.

Quiz: The Bindkey for Create Measurement Command in Layout

What is the equivalent bindkey that can be used for the **Create Measurement** command?

You can use the bindkey “**<Key>K**” to invoke the **Create Measurement** command.

What is the equivalent bindkey that can be used for the **Create Measurement** command?

- You can use the bindkey “**<Key>K**” to invoke the **Create Measurement** command.

Quiz: The Measurement Types in the Create Measurement Form in Layout

What are the measurement types available in the **Create Measurement** form?

In the Create Measurement form:

- **Type** lets you select the display type of the ruler (environment variable: *rulerDisplayType*). It has 3 options: **Ruler**, **Distance**, and **Both**.
- **Ruler** displays the ruler in the design window.
- **Distance** displays the distance between two points on the canvas.
- **Both** displays both the ruler and the distance between two points on the canvas.

What are the measurement types available in the **Create Measurement** form?

In the Create Measurement form,

- **Type** lets you select the display type of the ruler (environment variable: *rulerDisplayType*). It has 3 options: **Ruler**, **Distance**, and **Both**.
- **Ruler** displays the ruler in the design window.
- **Distance** displays the distance between two points on the canvas.
- **Both** displays both the ruler and the distance between two points on the canvas.

Quiz: The Quick Align Command in Layout

?

What are the two methods available in the **Quick Align** command?

Quick Align: Single Object and Quick Align: Multiple Objects methods are available in the **Quick Align** command.

What are the two methods available in the **Quick Align** command?

- Quick Align: Single Object and Quick Align: Multiple Objects methods are available in the **Quick Align** command.

Module 5

iPegasus™ DRC and Fill for Virtuoso® Studio

The module on iPegasus DRC and Fill for Virtuoso Studio.

Module Objectives

In this module, you

- Identify the challenges in the advanced node signoff
- Explain the iPegasus™ Advanced Node support
- Set up and run the iPegasus DRC and FILL for Virtuoso® Studio
- Use the appropriate models of the iPegasus DRC and FILL
- Create and manage snapshots for the iPegasus DRC and FILL
- Navigate the iPegasus results in the Annotation Browser and RV

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In this module, you will identify the challenges associated with advanced node signoff. You will learn about iPegasus Advanced Node support and how to set up and run iPegasus DRC (Design Rule Check) and FILL for Virtuoso Studio. Additionally, you will use the appropriate models for iPegasus DRC and FILL. You will also create and manage snapshots for both iPegasus DRC and FILL. Finally, you will navigate and debug the results of iPegasus runs using the Annotation Browser and Results Viewer.

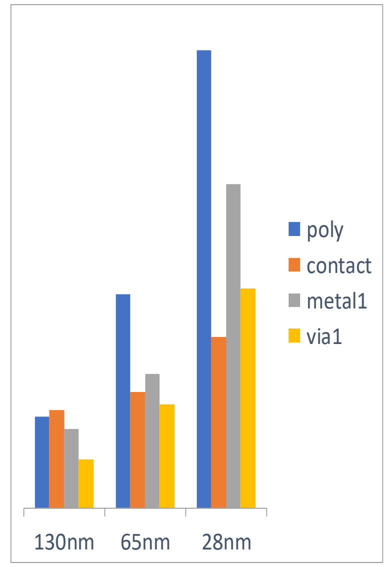
Challenges in the Advanced Node Signoff

Problem: Challenges in the advanced nodes raise the complexity of designs and physical verification. There is an exponential explosion in the DRC count and device complexity.

- Demand for higher performance and lower power
- Raising cost
- Reduced Time-to-Market – shrinking schedules

Solution: InDesign verification

- iPegasus, VIPVS
- In-design DRC and fill with foundry-certified signoff decks
- Native integration to the Virtuoso® workflow
- Interactive checking and instant error reporting



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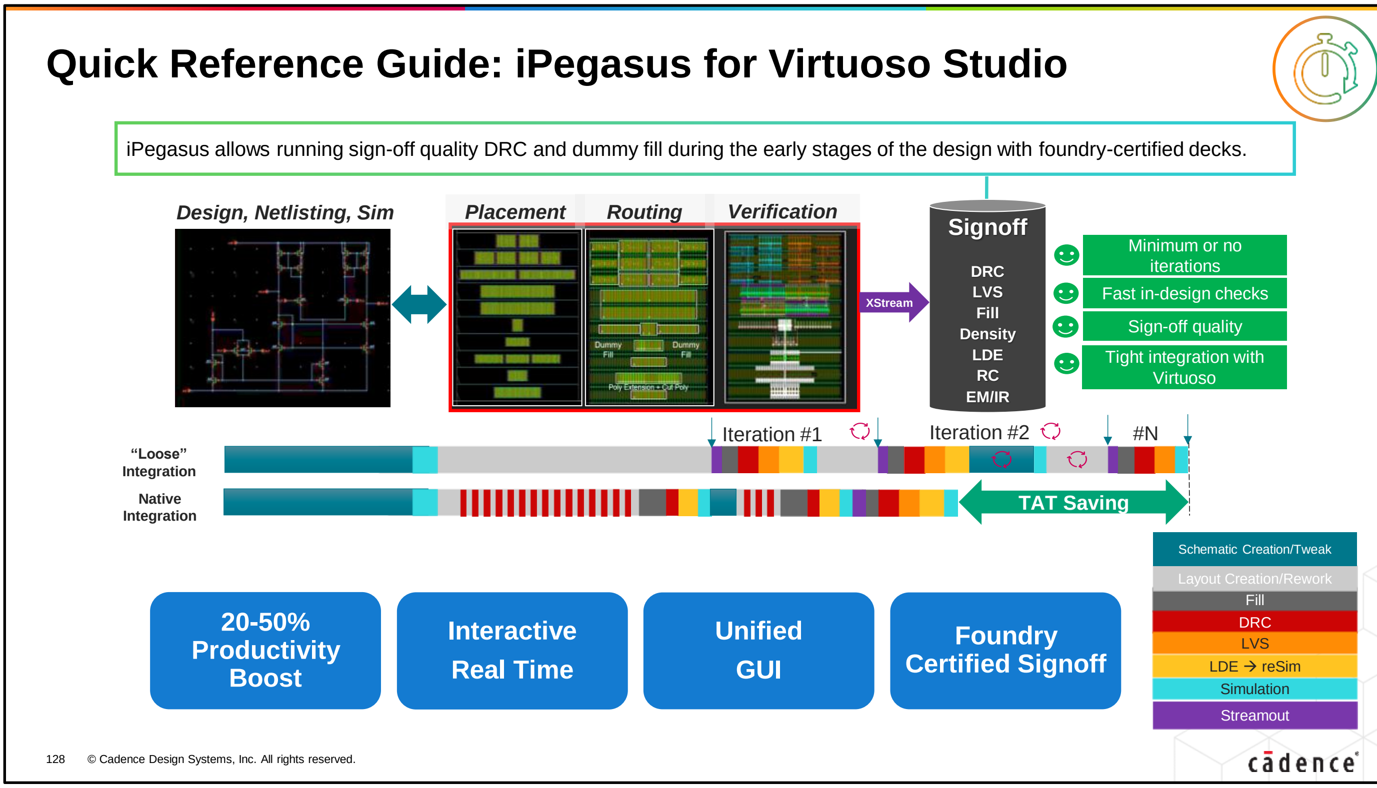
Significant challenges come with Advanced Node Signoff. While on the one hand, we have the market pushing for better PPA numbers and tighter tape-out timelines. On the other side, increasing complexity in advanced nodes leads to challenges like an exponential explosion in the DRC count and complex rules related to factors like FinFET devices, DPT, or Layout-Dependent Effects. There is a demand for cost-effective, faster, and more efficient design solutions.

The unavailability of sign-off quality in-design verification and dummy fill flow makes designers heavily dependent on sign-off tools towards the end of design closure, which significantly decreases productivity and increases overall design turnaround time.

In-design verification solutions such as Virtuoso Studio iPegasus and Virtuoso interactive PVS are meant to address such challenges.

These solutions aim to provide early and instant access to sign-off quality DRC and metal fill checks based on the sign-off foundry rule. The goal is to provide a helpful model when the user runs sign-off quality DRC checks at any stage of layout development and then easily views and debugs the violations. This helps avoid “surprises” at final sign-off physical verification checks. In other words, you don't need to wait for the final signoff verification run results to assess the health of the design.



Virtuoso Studio offers Pegasus Verification tools as part of the layout editor, so you can have a signoff-quality interactive in-Design verification experience, for every shape and route you place, without additional licenses. The Virtuoso integrated Pegasus feature is known as iPegasus. Integration of the Pegasus Verification System, with the AI-based Virtuoso Studio, enables faster turnaround times, and more predictable design cycle times with DRC, metal fill, and DFM signoff.

Here is a graphical representation demonstrating the advantages of Virtuoso iPegasus Physical Verification and Fill, over the traditional signoff solutions. In the traditional mode, the physical verification starts only after the placement and routing stages, and it goes for several iterations before sign-off. Not performing the physical verification at early stages leads to a conservative design 'wasting the silicon area', or a higher number of iterations of DRC during the design closure, and an unpredictable impact of inserted metal and dummy fills.

With the Virtuoso iPegasus model, you perform the faster and signoff quality in-design physical verification checks and fill trials, as soon as the placement begins, sort out any surprises early in the design cycle, and thus significantly reduce the iterations and overall turn-around time.

Layout designers stay inside the Virtuoso environment thru their entire design cycle, Physical Verification, and Fill without breaking their workflow. With a unified GUI, we are providing two-pronged In-Design physical verification - DRD checker to address design constraint overrides, net and net group overrides, and application overrides; and sign-off quality DRC checking thru cloud-ready Pegasus Physical Verification checker.

Submodule 5-1

iPegasus DRC for Virtuoso Studio

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Submodule – iPegasus DRC for Virtuoso Studio.

Submodule Objectives

In this submodule, you

- Signoff your designs with the iPegasus DRC for Virtuoso Studio
- View various fields in the iPegasus DRC toolbar
- Set up iPegasus DRC Run Options form
- Exclude Cells from the iPegasus DRC Runs
- Create snapshots and run the iPegasus signoff DRC check
- Navigate the iPegasus DRC errors in the Annotation Browser and Results Viewer
- Define and override the bloat value for the iPegasus DRC runs

Submodule Objectives.

In this submodule, you will engage in several key tasks. First, you will sign off on your designs using the iPegasus DRC for Virtuoso Studio. You will then explore the various fields available in the iPegasus DRC toolbar and set up the iPegasus DRC Run Options Form. Additionally, you will learn how to exclude specific cells from the iPegasus DRC runs. As part of the process, you will create snapshots and run the iPegasus sign-off DRC check. You will also navigate iPegasus DRC errors using the Annotation Browser and Results Viewer. Finally, you will define and override the bloat value for the iPegasus DRC runs.

What Is the iPegasus DRC for Virtuoso Studio?

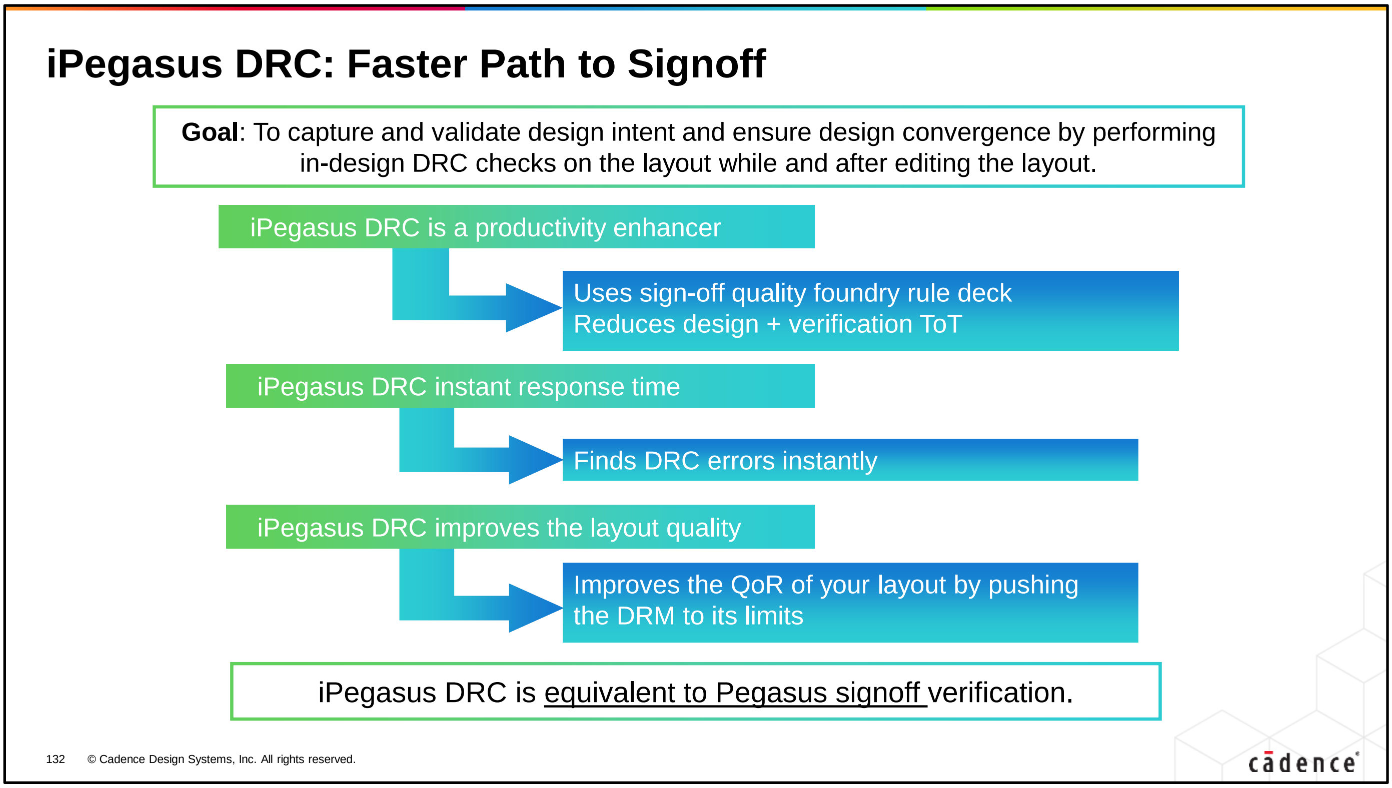
Features

The iPegasus DRC for Virtuoso is a Pegasus-based, integrated, in-design, and on-demand signoff DRC check performed on the layout during the design phase using a foundry-qualified rule deck.

- Captures and validates design intent and ensures design convergence.
- **Performance Gain:**
  - 2X faster - No OA-to-GDS streamout.
  - Edit the layout while running DRC.
- **iPegasus DRC Toolbars:**
  - Quicker DRC closure with Virtuoso integrated toolbar.
  - snapshots - customized pre-compiled rulesets.
  - Flexible Filters – Run over selected area, rules, and layers.
  - Optional density and connectivity checks.

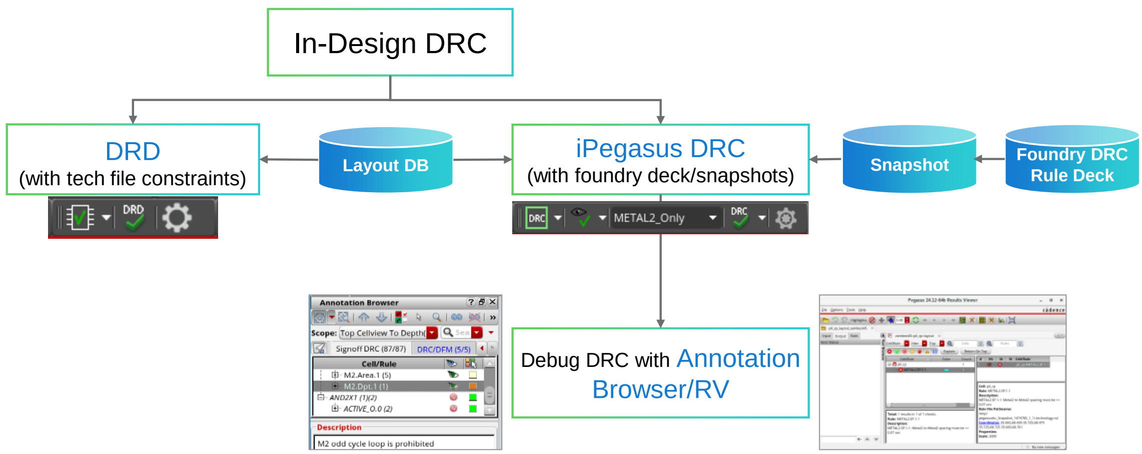
- **Advanced Node Challenges:** Coloring (MPT) Rules Supported
- **Annotation Browser and Results Viewer:**
  - Multiple Debug options.
  - Improved error visualization.
- iPegasus DRC license NOT required for smaller designs (process-node dependent)!

The Virtuoso Studio integrated DRC model features two DRC engines: DRD and iPegasus DRC signoff checker. Both engines come equipped with dedicated toolbars within Virtuoso Studio. Virtuoso iPegasus DRC provides a comprehensive Pegasus-based interactive sign-off quality DRC check for your designs. It utilizes foundry-qualified rule decks early in the design cycle to reduce overall turnaround time, ensuring faster and higher-quality layouts. iPegasus captures and validates design intent by conducting interactive DRC checks on the layout immediately after any edits are made. In terms of performance, Virtuoso iPegasus DRC is demonstrated to be twice as fast as its predecessor, significantly shortening the edit-check-fix cycles. Unlike traditional signoff modes, the iPegasus model does not require a stream-out process, allowing DRC to be run directly on the layout database without the need for OA-to-GDS data translation. With "Virtuoso iPegasus DRC," users can edit the layout while performing DRC checks and can run DRC on either editable or read-only cell views. iPegasus DRC is managed through an integrated toolbar in Virtuoso, which features link to an options form GUI. "Virtuoso iPegasus DRC" offers area-specific checks, which can be selected from the toolbar or the options form. The form provides additional options for optimizing the DRC run based on the rules and layers selected. The "Virtuoso iPegasus DRC signoff checker" utilizes a pre-compiled design rule deck known as "Snapshots," eliminating runtime overhead for rule deck compilation and speeding up the DRC process. The Virtuoso iPegasus DRC expanded verification package includes optional density and connectivity checks, and coloring rules are also supported within the utility. Annotation Browser serves as the default debugging environment for Virtuoso iPegasus DRC, providing improved error visualization and a streamlined debugging experience. Additionally, users can activate the Result Viewer from the options form. Lastly, if you hold Virtuoso XL, EXL, or MXL licenses, a separate Virtuoso iPegasus DRC license is not required for most runs. However, there are some limitations: users can run iPegasus DRC if the checked design area falls within specific limits based on the process node, while larger designs necessitate the full Virtuoso iPegasus DRC license.



The “Virtuoso iPegasus DRC” offers various features that expedite the sign-off process. Its primary goal is to capture and validate design intent, ensuring design convergence by performing in-design signoff quality DRC checks right after making layout edits. By reducing the number of DRC sign-off runs and minimizing pre-tape-out layout revisions, “Virtuoso iPegasus DRC” significantly enhances productivity. It pushes the Design Rule Management (DRM) to its limits, providing instant reports of DRC errors that help improve the quality of your layout. However, it is important to note that a final sign-off physical verification DRC is still recommended.

Flowchart: In-Design DRC in Virtuoso Studio



Let us look at the “In-Design DRC Flow”.

The DRC toolbar is separated into two parts – iPegasus DRC and DRD.

What is DRD?

DRD stands for Design Rule Driven DRC. DRD is an in-design verification tool that supports and works on technology file constraints. In other words, DRD picks Design constraints from the technology library and uses them instead of a foundry rule deck. When enabled, DRD provides visual and physical feedback about design constraint violations while you interactively create and edit a layout, to facilitate correct-by-construction layout design.

On the other hand, the iPegasus DRC offers a Pegasus-based, in-design interactive sign-off DRC check on your design, using the foundry-qualified rule deck.

The iPegasus supports running DRC only for a selected set of checks. For that, you need snapshots– the customized subset of rules. You may create your snapshots using the GUI options, or they are delivered as part of PDK.

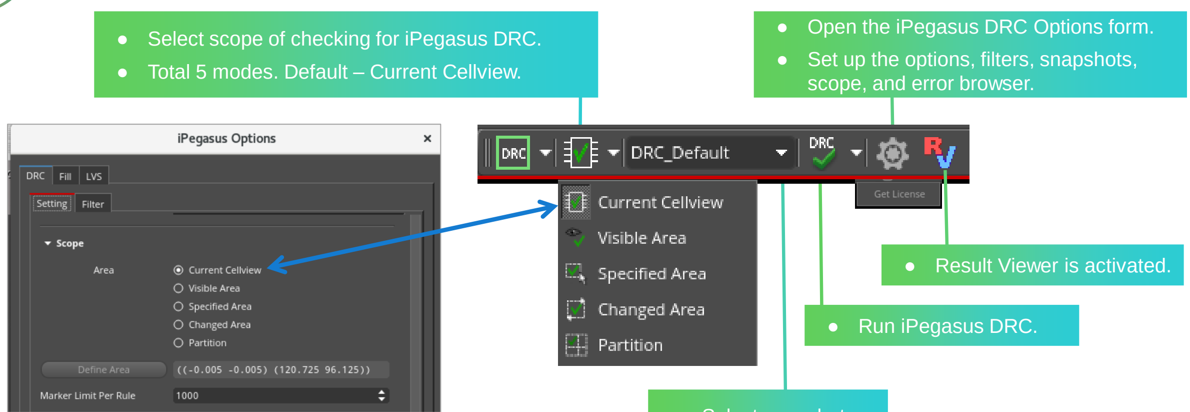
You must use a combination of DRD and the iPegasus for efficient verification and design closure.

Finally, select the scope of the run and start your iPegasus DRC.

Once the run is complete, there are two options to view and analyze the violations: the Annotation browser and the Results Viewer. In both cases, use the available filtering and searching features to highlight the violations in the layout. Debug and correct them. Rerun the iPegasus DRC and repeat the process as and when required.

iPegasus DRC Toolbar

Virtuoso Environment → Toolbars → iPegasus DRC toolbar



iPegasus DRC Options form

Let’s take a closer look at the iPegasus DRC Toolbar.

The DRC toolbar is active when DRC is selected from the dropdown selector at the left end of the common iPegasus Toolbar, which is also the default selection.

Let's examine the various fields in the iPegasus DRC toolbar. Click the right-most settings icon to invoke the iPegasus DRC options form. As the name indicates, you must set up your run using the various fields in this form.

Use the second field from the left to set the scope of the run. The five drop-down options allow you to run DRC on a range of selections, from the Visible Area to the Current Cellview. The Current Cellview mode is the default selection. In addition, the Changed Area option checks only the edited portion of the layout, and the Partition mode is supported when the layout is split between designers.

You may select the scope of the run from the settings tab of the iPegasus DRC options GUI or the toolbar, and both will be in sync.

You may run the signoff DRC on the complete foundry rule deck, or a customized subset called snapshot. You may create your snapshots using the GUI options, or they are delivered as part of PDK. Available snapshots are listed in the snapshots field on the toolbar. By default, the Snapshots are sorted alphabetically. You may select a snapshot from the list and start the signoff DRC run.

Use the Run iPegasus DRC field to start your signoff DRC run. To run iPegasus DRC over a large area, you need the tool license. Procure it using the drop-down option next to the Run iPegasus DRC field.

At the end of the run, iPegasus DRC lets you view and debug violations through the Annotation Browser or Results Viewer. Though Annotation Browser is the default browser, you can activate the Results Viewer as well from the iPegasus DRC options form. Once enabled, the RV icon will appear in the toolbar.

Cadence ASK Video (20481899): [Introduction to the DRD User Interface]

iPegasus DRC Run Options Form

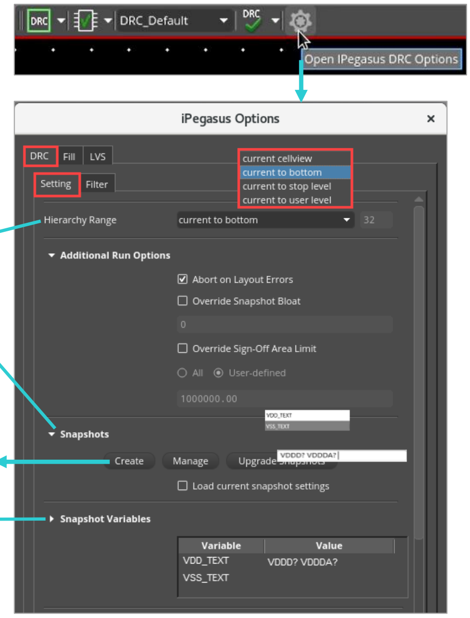
Setting Tab – 1

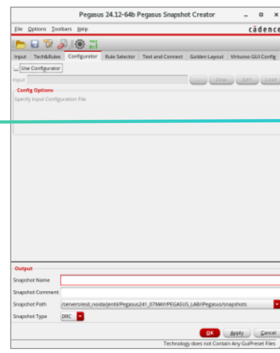
iPegasus toolbar (DRC) → Open iPegasus DRC Options → iPegasus Options form → DRC tab → Setting tab

- Hierarchy Range – Depth for the iPegasus DRC runs.

- Snapshot – the pre-compiled deck of user-selected DRC rules.
- Click Create to invoke the Snapshot Creator form.
- Options to manage and upgrade existing snapshots.

- Advanced nodes – Customize snapshot PG variables for connectivity DRC checks.





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To set up the iPegasus DRC run, use the unified iPegasus options form, which can be accessed via the iPegasus toolbar. Once you select DRC from the iPegasus tool selector drop-down, and click the ‘Open iPegasus DRC Options’ field, the options form opens with the DRC tab selected.

The iPegasus DRC options form contains two sub-tabs: Settings and Filter. We'll begin by discussing the various configurable features available in the Settings tab.

The Hierarchy Range setting specifies how many hierarchy levels will be included in the iPegasus DRC checking process. There are four options for the hierarchy range: current cell view, current to bottom level, current to stop level, and current to user level. By default, the user level is set to a depth of 32.

Next, let’s go to the snapshot field.

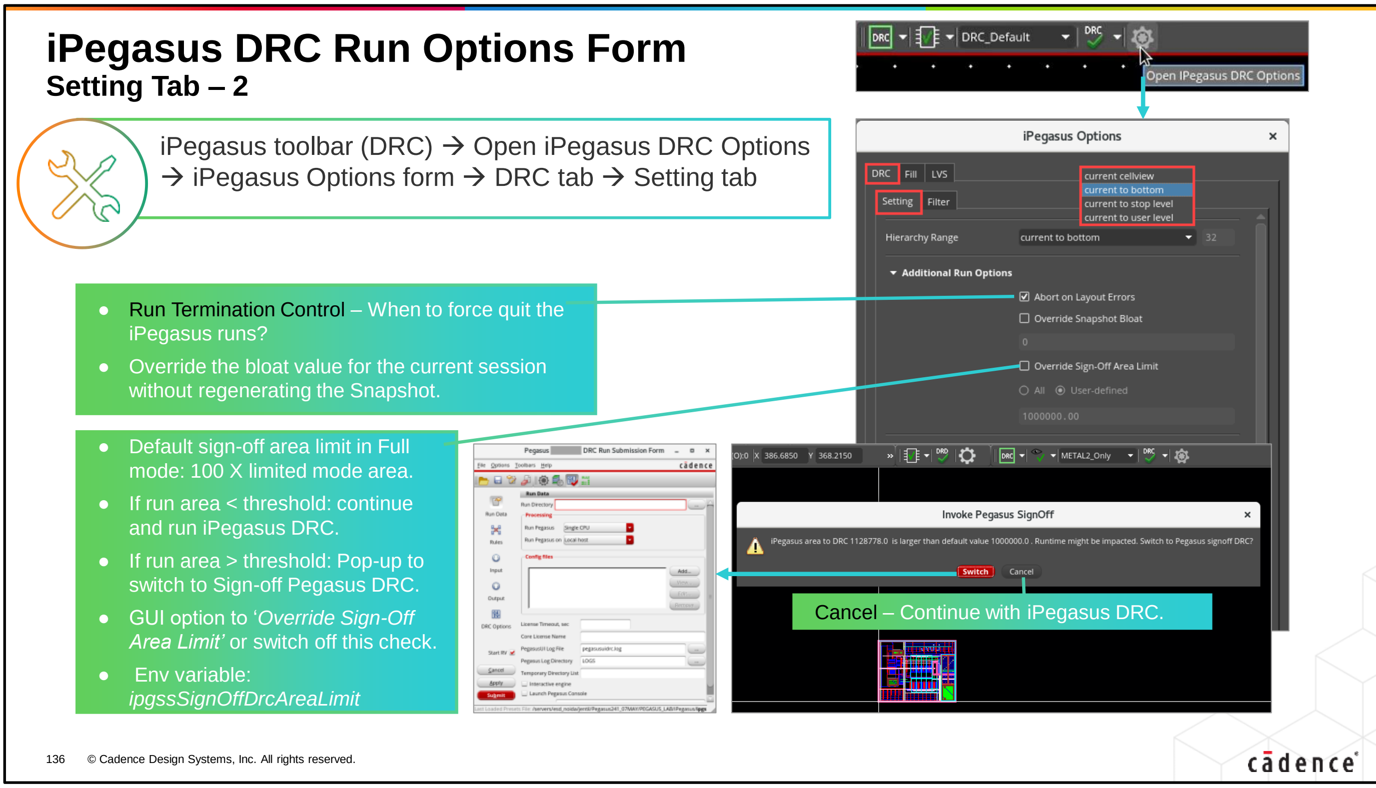
Snapshots are the pre-compiled deck of user-selected DRC rules.

The “Snapshot” field lets users create, manage, and upgrade Snapshots.

Click Create to invoke the Snapshot Creator form to create and save snapshots.

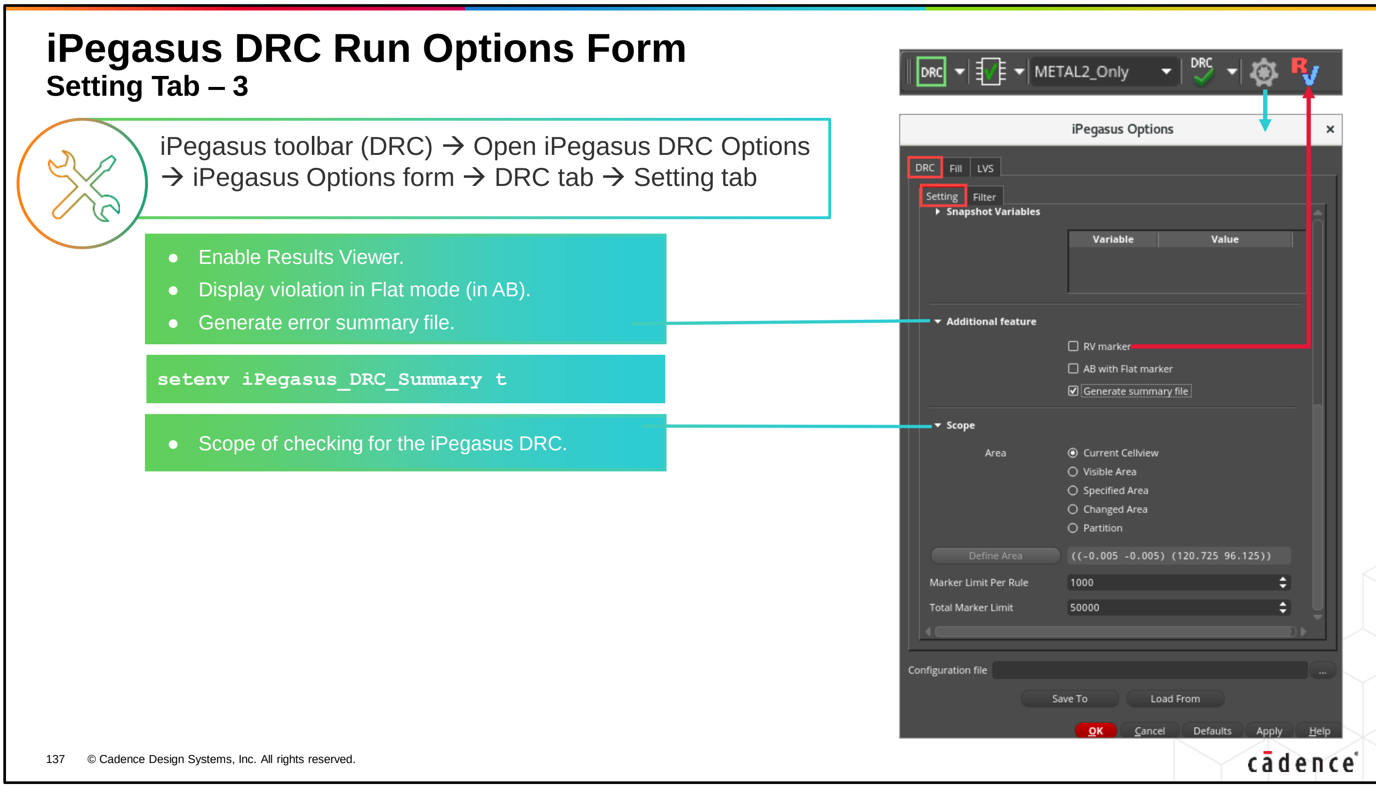
The content of the Snapshot setting field can be customized. This use model intends to give the CAD administrator control over the snapshot accessibility to target users.

In advanced node runs, the iPegasus Snapshot utility creates a new file called "drc.var," which includes power and ground variables formatted as variable and value. Virtuoso reads this drc.var file from the snapshot and fills in the "Variable" and "Value" fields in the Snapshot Variables section of the iPegasus DRC options form. During the next iPegasus DRC run, the PG values from the Snapshot Variables form will be utilized. These values can be adjusted in the settings; however, changes made will not affect the limited or full mode of iPegasus DRC.



In the Advanced Run Options section, the "Abort on Layout Errors" option is enabled by default. This option will terminate the iPegasus DRC run if it identifies any missing cell masters or mismatched rule precision values in relation to the input database precision. Additionally, the "Override Snapshot Bloat" field can be used to adjust the bloat value for the current iPegasus DRC session without the need to regenerate the Snapshot.

iPegasus runs DRC in two modes: Limited and Full. In Full mode, a warning appears if the DRC run area is too large, affecting performance. The threshold for this warning is 100 times the node's Limited mode area allowance. If the area is below the threshold, the run proceeds normally. Exceeding the threshold triggers an interactive window with options to 'Switch' to a standard Pegasus DRC signoff run or 'Cancel' the form to continue with the iPegasus DRC run. Selecting the switch option will display the Pegasus DRC run submission form. You can adjust or disable this area check in the iPegasus DRC Run Options Form using the 'Override Sign-off area limit' field.



The annotation browser serves as the default error browser for iPegasus DRC. In the Additional Features section, you can select the RV Marker field to display DRC markers in the result viewer. When the RV option is selected, the RV icon will show up in the toolbar. Additionally, the form provides options for displaying DRC markers in a flat hierarchy and for generating a DRC run summary file. A summary file with the name 'iPegasusDRC.summary' will be dumped in the working directory at the cost of additional runtime.

The Scope section defines the area that iPegasus DRC will consider during the run. There are five options available:

1. Current Cellview: Checks all shapes in the currently editable cell view.

2. View Area: Checks all shapes within the current viewable area.

3. Specified Area: This option checks all shapes in a user-defined area of the design. When this option is selected, the Define Area field becomes active, allowing you to specify the region you want to highlight in a pink box for the iPegasus DRC run. You will need to provide the lower-left and upper-right coordinates for this region.

4. Changed Area Mode: This dynamic DRC check enables you to make multiple layout edits while tracking these modifications between each DRC run. In this mode, DRC checks are conducted only on the modified sections of the layout.

5. Partition Area Mode: This mode allows the DRC to be run on Virtuoso Concurrent Layout Editing (CLE) partitions, enabling multiple designers to work on the same cell view simultaneously. This approach helps to parallelize efforts and improve productivity within the layout design teams. Please note that the Partition mode is applicable only for design partitions and not for the original cell layout view.

Note that the scope of the run can be set from the iPegasus toolbar as well, and both will be in sync.

The iPegasus DRC computes the final checking area for all modes by incorporating the bloat value defined in the current active snapshot. The Marker Limit specifies the maximum number of markers that can be created per rule, with a default value of 1,000. Additionally, the total number of markers per run is capped at 5,000 by default. Both these values are set to achieve DRC violation from iPegasus as close to a standalone Pegasus run. However, both values can be changed as needed.

iPegasus DRC Run Options Form

Filter Tab – 1

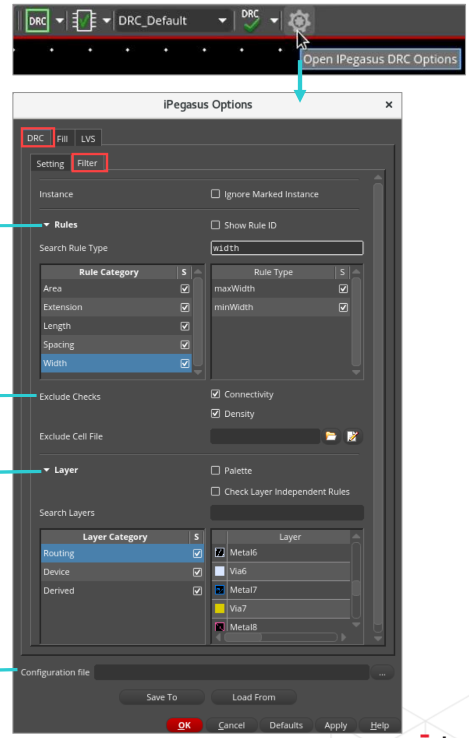
iPegasus toolbar (DRC) → Open iPegasus DRC Options → iPegasus Options form → DRC tab → Filter tab

- More controls over iPegasus DRC rules.
- Search Rule Type – specifies strings to filter the rules.
- Rule category to LHS and filtered rule type list to RHS.

- Exclude Connectivity and Density checks.
- Exclude selected cells and skip the DRC checks.

- Palette - What-You-See Is What-You-Check
- Layer – filtering options for layers.

- Configuration File – saves the current configuration to a file and loads it for future runs – saving time.



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The Filter tab allows you to select criteria, such as Rules and Layers, when running iPegasus DRC. You can use the filtering features to control the DRC checks for an entire category or for individual selections.

Let's begin with the dynamic Rules filtering feature within the selected snapshot. This section is divided into two parts: Rule Category and Rule Type. The "S" stands for Selectability; click on it to toggle the status of the entire list of displayed line items. Select a rule category to view all the rules supported by iPegasus DRC within that category. The Rule Type section on the right lists the rules filtered by a search string or by selecting one or more categories. Each line item on both sides includes selection checkboxes.

You can toggle the display from Rule Type to Rule ID by using the Show Rule ID checkbox. The search rule type field allows you to specify strings for filtering.

Connectivity and Density checks typically run at the final design stages and are time-consuming. By default, both checks are excluded from the run. You may alter the status. Use the 'Exclude Cell File' field to input a file that lists the cells to exclude from the current run.

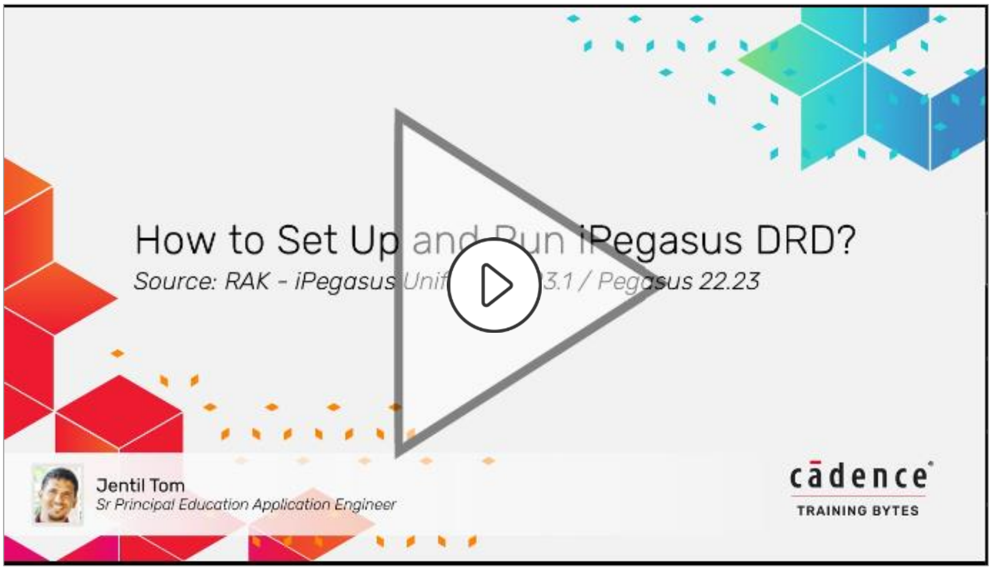
The Layer section provides options for filtering layers from the DRC run. In palette mode, which is active by default in limited mode runs, iPegasus DRC only monitors those layers whose visibility is enabled in the Virtuoso Layers panel of the Palette Assistant for violations. Consequently, only violations associated with the visible layers in the layout suite will be displayed.

The Check Layer Independent Rules option allows running DRC with all such rules from the snapshot in full mode. Rules to identify acute angles or orthogonal shapes fall into this category.

The Search Layers feature lets you specify strings to filter the layers. The Layer Category lists three categories of layers: Routing Layers, Device Layers, and Derived Layers. The Layer section on the right displays the layers filtered by a search string or by selecting one or more layer categories.

Use the configuration file feature to save your current settings and load them for future use, which will help reduce setup time. The "Browse" button allows you to locate and select an existing configuration file, as well as specify where to save a new configuration file. The "Save To" option enables you to save the current settings from the Options form to the designated configuration file. Meanwhile, the "Load From" field allows you to load settings from a specified configuration file into the Options form.

Demo: How to Set Up and Run iPegasus DRD

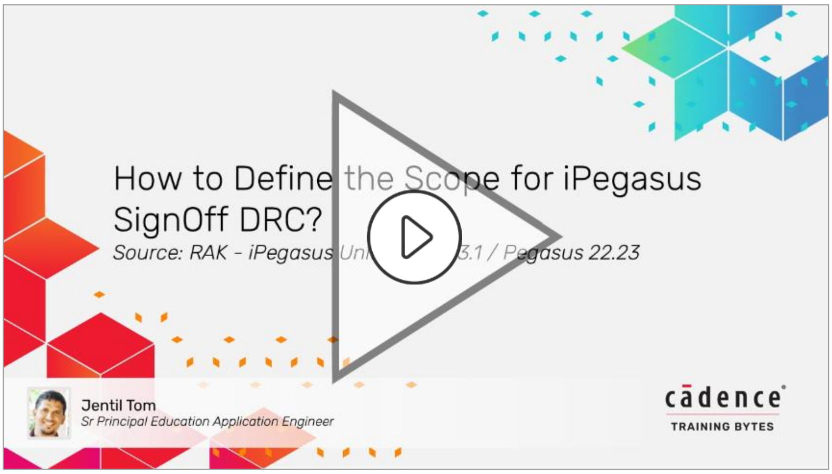


Video Play Time: **4.16** minutes

Click the Play button to start the video.

Click the link How to Set Up and Run iPegasus DRD? to play the video on support.cadence.com.

Demo: How to Define the Scope for iPegasus SignOff DRC



Video Play Time: **3.49** minutes

Click the Play button to start the video.

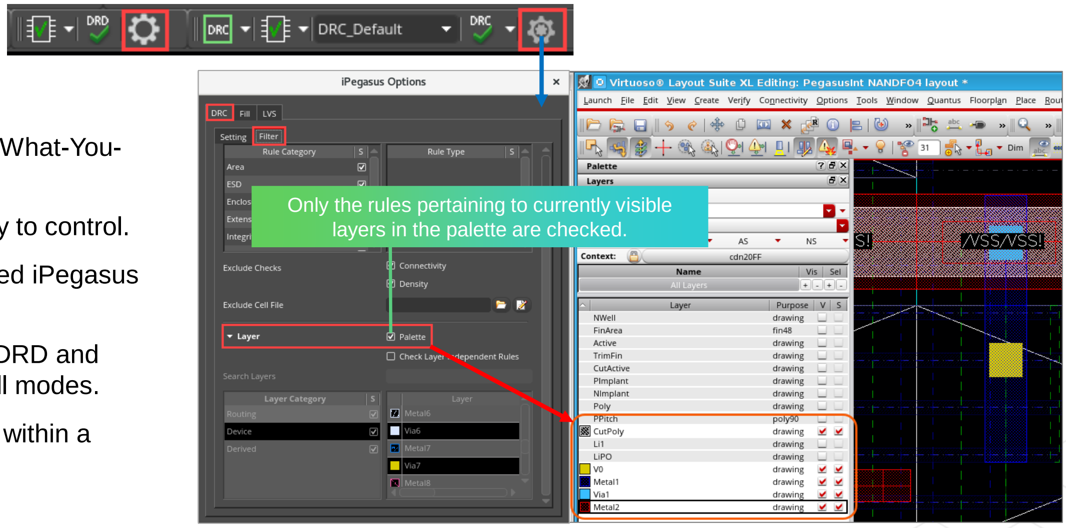
Click the link [How to Define the Scope for iPegasus SignOff DRC?]() to play the video on support.cadence.com.

Palette Feature in DRD and iPegasus DRC for Virtuoso Studio

DRD/iPegasus DRC Options form → Filters tab → Layer → Palette

Features:

- What-You-See Is What-You-Check.
- Scalable and easy to control.
- Always on in limited iPegasus DRC mode.
- On/Off control in DRD and iPegasus DRC full modes.
- Additional control within a snapshot.



The palette feature is an intelligent tool integrated into the DRD and iPegasus DRC systems. It selects design rules based on the visibility of layers in the layout suite, operating on the principle of "What You See Is What You Check."

To enable the palette feature, open the DRD or iPegasus DRC Run Options form and check the 'Palette' option in the Layer section under the Filters tab. Once activated, only the rules that pertain to the currently visible layers in the Virtuoso layout suite's layer palette will be considered in subsequent runs.

For instance, if Metal 1 is visible in the palette, checks for M1-to-Via1 and M1-to-M2 will still be conducted, even if Via1 and M2 are not currently visible in the suite.

Advantages of the Palette Feature:

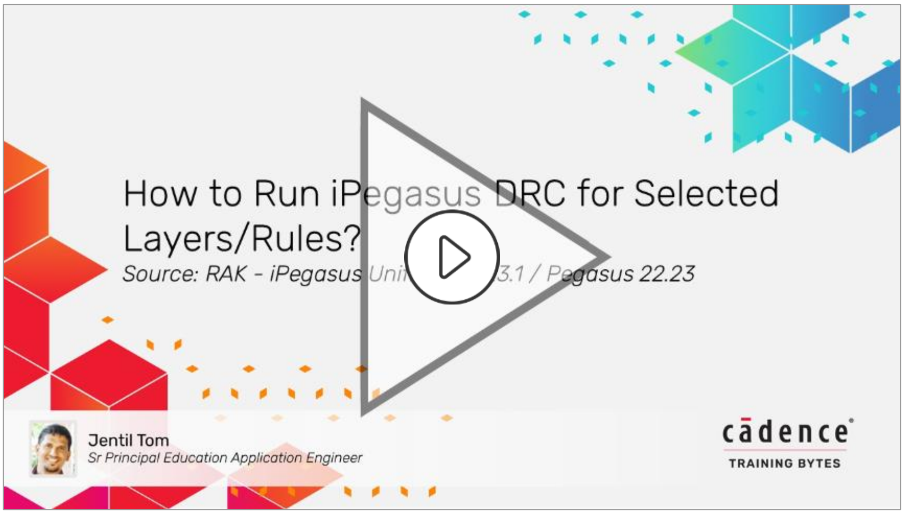
The palette feature is scalable and easy to manage.

Please note that in iPegasus XL/EXL/MXL mode, the DRC runs with the Palette feature always on. This means that if the Palette option is checked, iPegasus DRC will respect that setting. However, if the Palette option is unchecked, Virtuoso will internally activate it regardless, and you will see a corresponding warning displayed in the CIW.

However, the Palette control is possible in DRD and iPegasus DRC in full mode.

Additionally, the palette feature provides an extra layer of control within a snapshot.

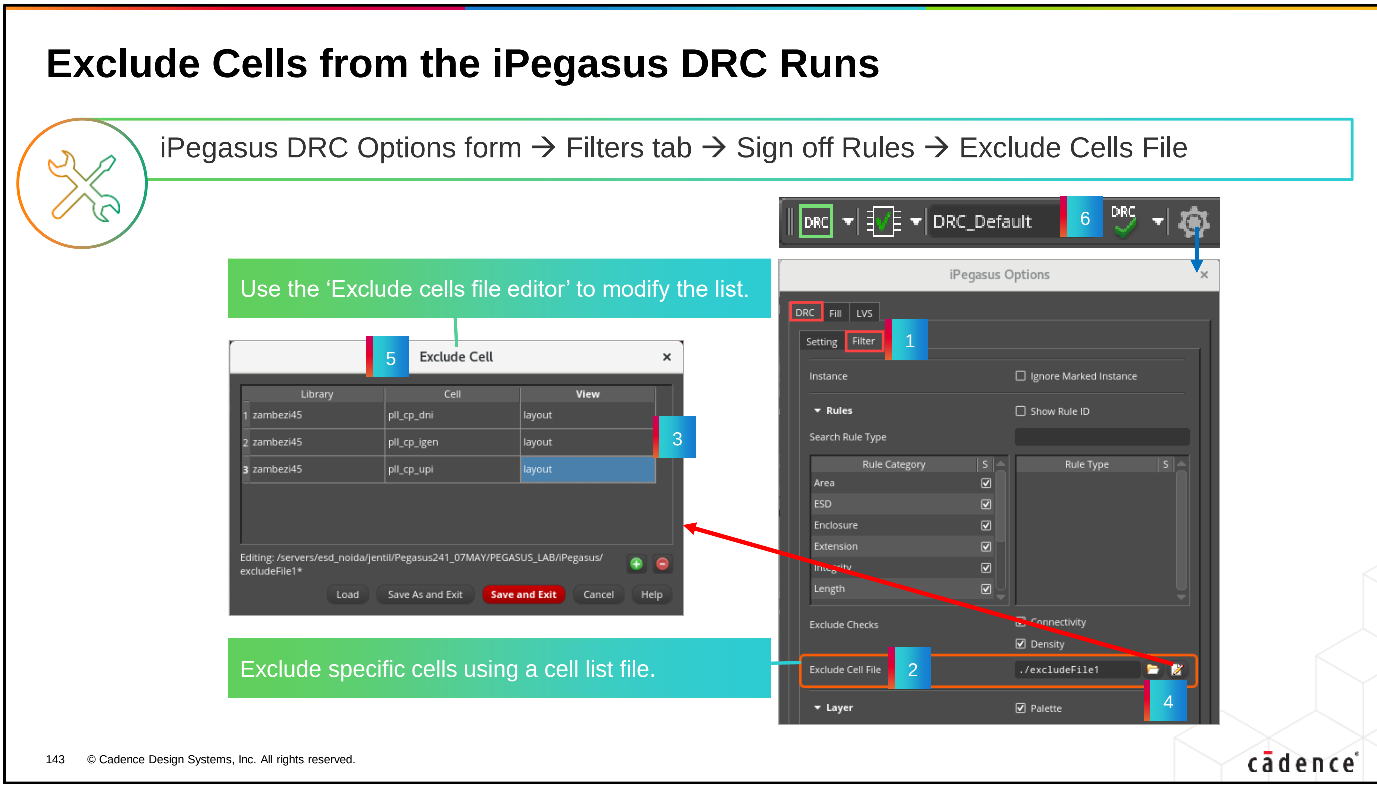
Demo: How to Run iPegasus DRC for Selected Layers/Rules



Video Play Time: **4.09** minutes

Click the Play button to start the video.

Click the link [How to Run iPegasus DRC for Selected Layers/Rules?]() to play the video on support.cadence.com.



How to Exclude Cells from a Virtuoso iPegasus DRC Run?

To exclude specific cells from a Virtuoso iPegasus DRC run, follow these steps:

1. Open the Virtuoso iPegasus DRC Options form and navigate to the Filters tab.

2. In the Rules section, you will find the option labeled 'Exclude Cells File.' This allows you to filter the rules displayed by iPegasus DRC on the canvas.

3. Create a file that lists the cells you want to exclude, ensuring the syntax follows the order of library, cell, and view. Save this file on your system.

4. Browse to locate and load this file in the 'Exclude Cells File' field. Once the file is loaded, the corresponding file name and path will automatically populate the field.

5. You can also use the Exclude Cells file editor to create a new Exclude Cells file or edit an existing one. To add a new entry, click the plus sign and then double-click on the Library, Cell, and View fields to make the appropriate selections for the cell you wish to exclude. After making your changes, save the file and exit.

6. During the iPegasus DRC run, the tool will check for cell availability, cell duplication, and the syntax of cell declarations. If any information is lacking, warning messages will be displayed in the CIW window. If everything is in order, the specified cells will be successfully excluded, and DRC errors will not be indicated for those cells.

Significance of Snapshots

- **Snapshots** – created by users or the CAD team.
  1. iPegasus DRC/Fill Options form → 'Create' Snapshot.
  2. CAD admins can remove snapshot fields with shell variables.

- Advantages of using a snapshot:
  - Reflects application intent
  - Eliminates run-time overhead by rule deck compilation
  - Faster switching between snapshots
  - Reusability

- By default, items in the snapshot combo field are sorted alphabetically.

In practical scenarios, Snapshots are created by users or the CAD team. Users can generate Snapshots using the options available in the "iPegasus Options" form. The CAD team may create them for users and control access to these Snapshots based on certain variables.

For both iPegasus DRC and fill runs, all approved Snapshots will be listed under the "Snapshot combo" field in the respective "iPegasus Toolbars." By default, items in the Snapshot combo field are sorted alphabetically.

During the design phase, using Snapshots for signoff DRC and fill checks offers several advantages over running a complete rule deck. Snapshots facilitate targeted DRC and fill runs that align with the application's intent. Since Snapshots are pre-compiled rule decks, they save compilation time. Additionally, Snapshots allow for faster switching between different sets and promote the reusability of custom rule sets. As a result, Snapshots save valuable run time and lead to optimized signoff runs.

How to Create and Manage Snapshots for the iPegasus Runs

A snapshot is a pre-compiled DRC/fill deck that is loadable only by Pegasus.

Snapshots allow the iPegasus direct access to design rules during signoff checks without recompiling. Let us see how to create and use snapshots to run DRC/Fill.

Let’s explore how to create and manage Snapshots for iPegasus signoff runs.

A Snapshot captures DRC or fill rules in a pre-compiled binary format that only Pegasus can interpret. This format allows iPegasus to access design rules directly, without needing to recompile them from the foundry rule deck. This capability helps save compilation time, enhances flexibility, and ensures optimal usage of Virtuoso. In this section, we will discuss how to identify, create, manage, and utilize Snapshots.

Step 1: Create Snapshots – Four ways

Create Snapshots → Manage / Upgrade Snapshots → Run iPegasus with Snapshots

There are four ways to create a snapshot:

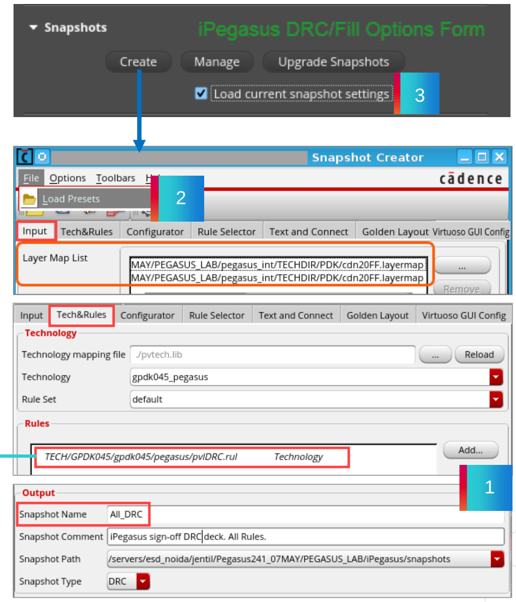
1. Interactive creation via Snapshot Creator
   - Minimum inputs: Layermap file, Rule Deck, Snapshot name
   - Built-in intelligence to validate the inputs
2. Preset file with config setup
   - Snapshot Creator: Load Presets icon / File → Load Presets
3. Load current snapshot settings
4. Command line

```
pegasusGenSnapshot -name <snapshotName>
<options> rulefile1 [... rulefileN]
```

Alternative way to invoke Snapshot form:

```
pegasusgui -pegasusintSnapshot
```

Source of snapshot – At least one readable file must be specified.



The first step is to create a Snapshot, which can be done in several ways. One option is to use the "Snapshot Creator" form, which allows you to create the Snapshot interactively. To invoke this form, click the "Create" button under the Snapshot section in the "iPegasus DRC-Fill Options" form.

The Snapshot Creator consists of multiple tabs, and at a minimum, you will need to provide three inputs: the Layermap file, the Rule Deck, and a name for the Snapshot. The Snapshot Creator form has built-in intelligence to check the validity of your inputs before generating Snapshots. You should enter the "Layer Map" files in the input tab, the Rule Deck in the Tech and Rules tab, and the snapshot name in the output field located at the bottom half of the form. Once you have entered all required information, click "Apply" or "OK." The Snapshot will then be created and listed under the "Snapshot field" on the iPegasus DRC or fill toolbar.

Additionally, you can configure the Snapshot Creator form using a preset file. To load a preset, click the "Load Presets" icon on the File toolbar, or select "Load Presets" from the dropdown menu under "File." This action will load a pre-saved configuration into the Snapshot Creator form.

You also have the option to load the Snapshot Creator form with the settings of the current active Snapshot by selecting the appropriate checkbox available on the iPegasus options form. This feature allows you to create a Snapshot based on an existing reference Snapshot setting and is enabled by default when iPegasus detects an active Snapshot. Another way to create a Snapshot is through the command line. Using the command provided, you can directly open the Snapshot form in standalone mode.

Step 1: Create Snapshots – GUI Setup

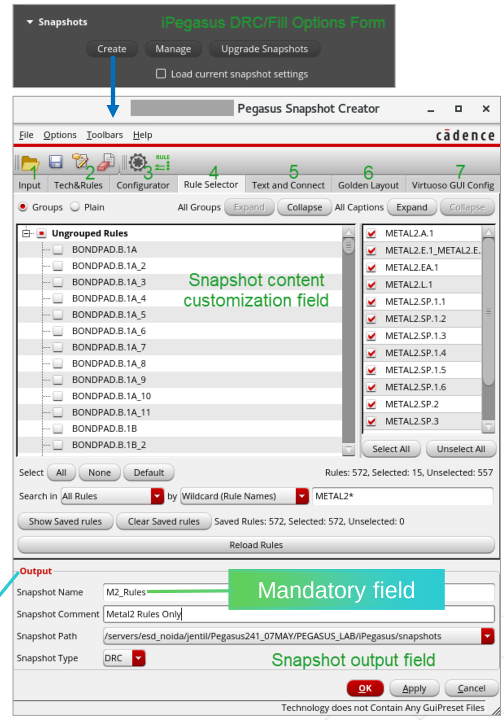
Create Snapshots → Manage / Upgrade Snapshots → Run iPegasus with Snapshots

iPegasus DRC/Fill Options form → Setting → Snapshots → Create → Pegasus Snapshot Creator

The Snapshot Content Customization section has six tabs:

1. **Input:** Polygon handling info
2. **Tech&Rules:** Technology design rules deck
3. **Configurator:** To override the default rule-deck setting
4. **Rule Selector:** Interactive rule selection
5. **Text and Connect:** Text-based connectivity
6. **Golden Layout:** For graphical DRC
7. **Virtuoso GUI Config:** Customize the default state of snapshots.

- Snapshot name, comment, location, and type

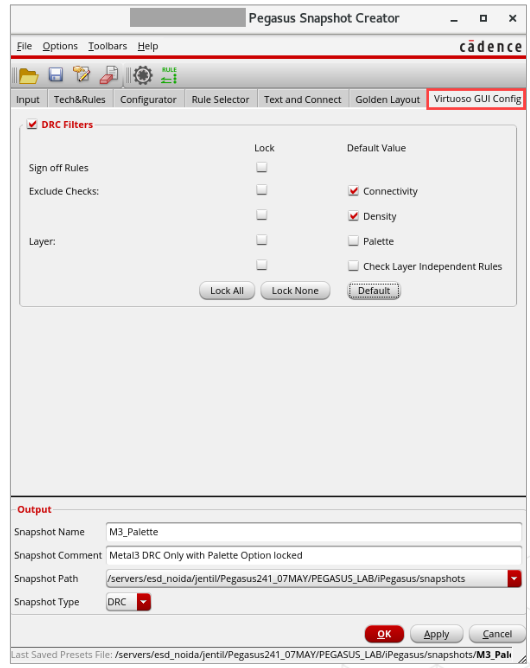


Let's take a closer look at the GUI-based Snapshot Creation process. The Pegasus Snapshot Creator form is adapted from the "Pegasus run submission form," but it includes fields specifically for Snapshot creation. This form can be divided into two main sections: the upper half, which is the "Snapshot Content Customization" field, and the lower half, the "Snapshot Output Setup" field. The "Content Customization Field" consists of seven tabs: Input, Tech and Rules, Configurator, Rule Selector, Text and Connect, Golden Layout, and Virtuoso GUI Config. The Snapshot output fields, however, remain consistent across all tabs. The "Input" tab provides polygon handling information between Virtuoso and Pegasus. It includes a mandatory layer map list and an optional object name table. By default, Pegasus retrieves the Layer map file from the chosen technology. The "Tech and Rules" tab specifies the primary rule deck for Snapshot creation. At least one readable rule deck file must be provided in the technology or rules section. If no valid rule deck is specified, “iPegasus” will indicate missing required inputs by highlighting the fields in red. The "Configurator" tab allows users to override the default rule-deck setting with a "configuration file." Meanwhile, the "Rule Selector" tab enables interactive selection of the rules for Snapshot creation, including wildcard features for blanket selections. The "Text and Connect" tab defines text-based connectivity criteria for iPegasus, and the "Golden Layout" tab addresses golden layout pattern files used for graphical DRC. The "Virtuoso GUI Config" tab offers additional options to customize Snapshot behavior. The "Output" field includes the final name for the Snapshot, a comment about the Snapshot, the directory path where the Snapshot will be saved, and the option to select the Snapshot type. A Snapshot name is required, and users can choose between two Snapshot types from the drop-down menu: DRC or Fill, depending on their application needs. The bloat value is automatically determined by the tool based on the selected rules. Once you have completed the form, click “Apply” or “OK” to create the Snapshot, which will then populate under the appropriate Snapshot fields for sign-off DRC or fill runs. After the snapshots are generated, you can close the form.

Step 1: Create Snapshots – GUI Setup

Create Snapshots → Manage / Upgrade Snapshots → Run iPegasus with Snapshots

iPegasus DRC/Fill Options form → Setting → Snapshots → Create → Pegasus Snapshot Creator → Virtuoso GUI Config tab



- DRC Filter - To customize the behavior of snapshots.
- Configure the default settings for Connectivity, Density, Palette, Layer-Independent Rules, and Rules filters.
- Lock option to make selections non-editable.
- Applicable only in full mode.
- Precedence over all other settings.

cādence

Let's take a closer look at the "Virtuoso GUI Config" tab. This tab allows you to customize the behavior of your snapshots further.

In the DRC Filters section, you can configure the default settings for Connectivity, Density, Palette, Checking of Layer-Independent Rules, and Rules filters. Additionally, you can make these settings non-editable by selecting the lock option. All the selections will be valid for that specific snapshot only.

Once you select this snapshot in the iPegasus toolbar, the configured settings will be reflected in the iPegasus options form, and you cannot change the locked settings from the iPegasus options form.

It's important to mention that DRC Filter Options will only be applicable in full mode and will be ignored in limited mode to prevent conflicts. Furthermore, the DRC Filter Options take precedence over all other settings.

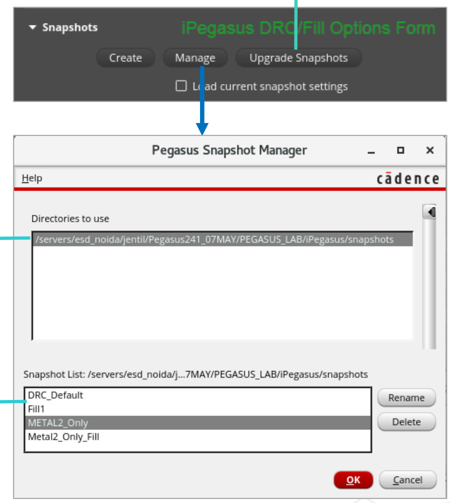
Step 2: Manage and Upgrade Snapshots

Create Snapshots → Manage / Upgrade Snapshots → Run iPegasus with Snapshots

iPegasus DRC/Fill Options form → Setting → Snapshots → Create → Snapshot Creator form → Manage → Pegasus Snapshot Manager

- Upgrade Snapshots to align with advanced nodes.

1. Select the directory to use.
2. Select the snapshot(s).
3. Rename/delete snapshots.

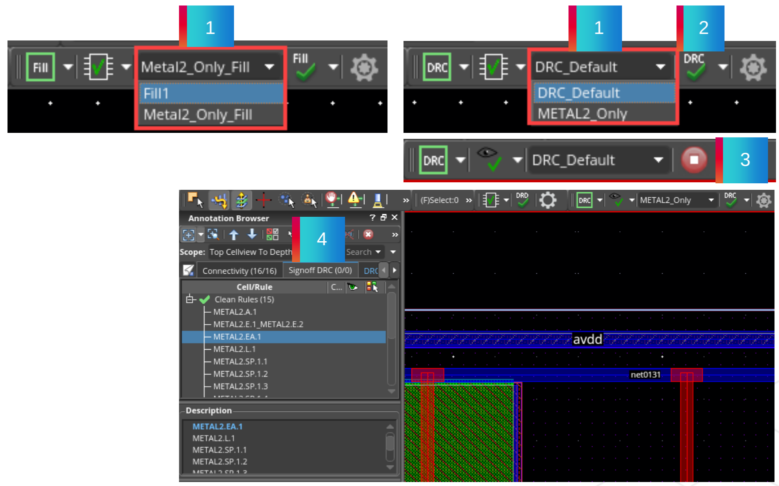


The iPegasus tool offers options for managing and upgrading snapshots. To access the Pegasus Snapshot Manager, click the “Manage” button in the Snapshot section of the “iPegasus DRC-Fill Options” form. This will display all available snapshot directories. Once you select a directory, the snapshots contained within it will appear in the box below. You can choose to delete or rename existing snapshots. After making any changes, be sure to click OK to apply them. Additionally, iPegasus supports upgrading snapshots as the process nodes evolve. To initiate this upgrade process, click the “Upgrade Snapshots” button located in the Snapshot section of the “iPegasus DRC-Fill Options” form.

Step 3: Run iPegasus DRC with Snapshots

Create Snapshots → Manage / Upgrade Snapshots → Run iPegasus with Snapshots

1. Select the snapshot (optional).
2. Run iPegasus DRC/Fill.
3. Run-in-progress.
4. Debug DRC / Simulate with Fill.
5. Repeat till the design is clean.



You are prepared to run the iPegasus signoff process using snapshots. The approved snapshots will be listed alphabetically under the "Snapshot Combo" field in the corresponding iPegasus Toolbars. To start the iPegasus DRC or fill run, select a snapshot from the list and click the respective icons on the iPegasus toolbar to initiate the run. While the run is in progress, the signoff checker icon turns red. The icon returns to its original form once the run is complete, and the results will be displayed in the Annotation Browser and Result Viewer, depending on the configuration.

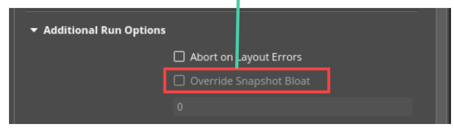
Review and debug the results, and repeat the process until the design is clean.

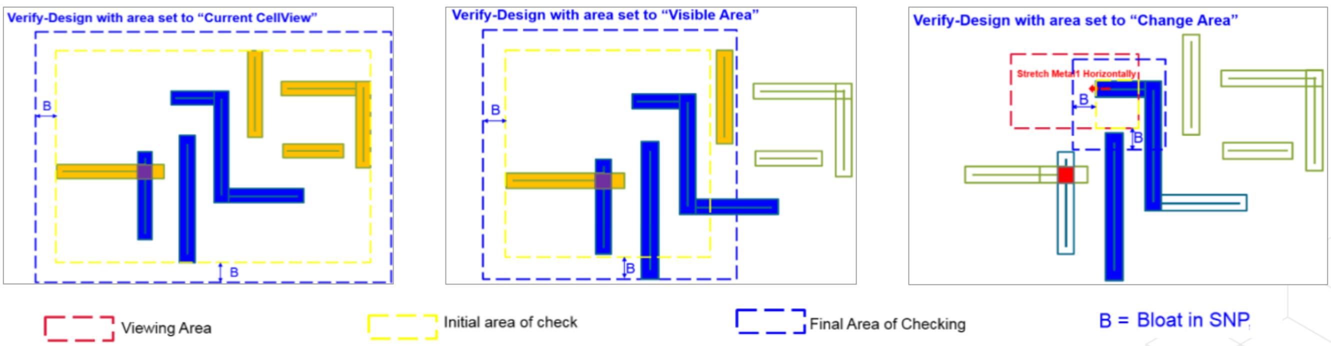
What Is Bloat Value?

Bloat value defines the buffer region (in X-Y direction) around the current editing area, which sums up the *effective region of checking* for Interactive runs.

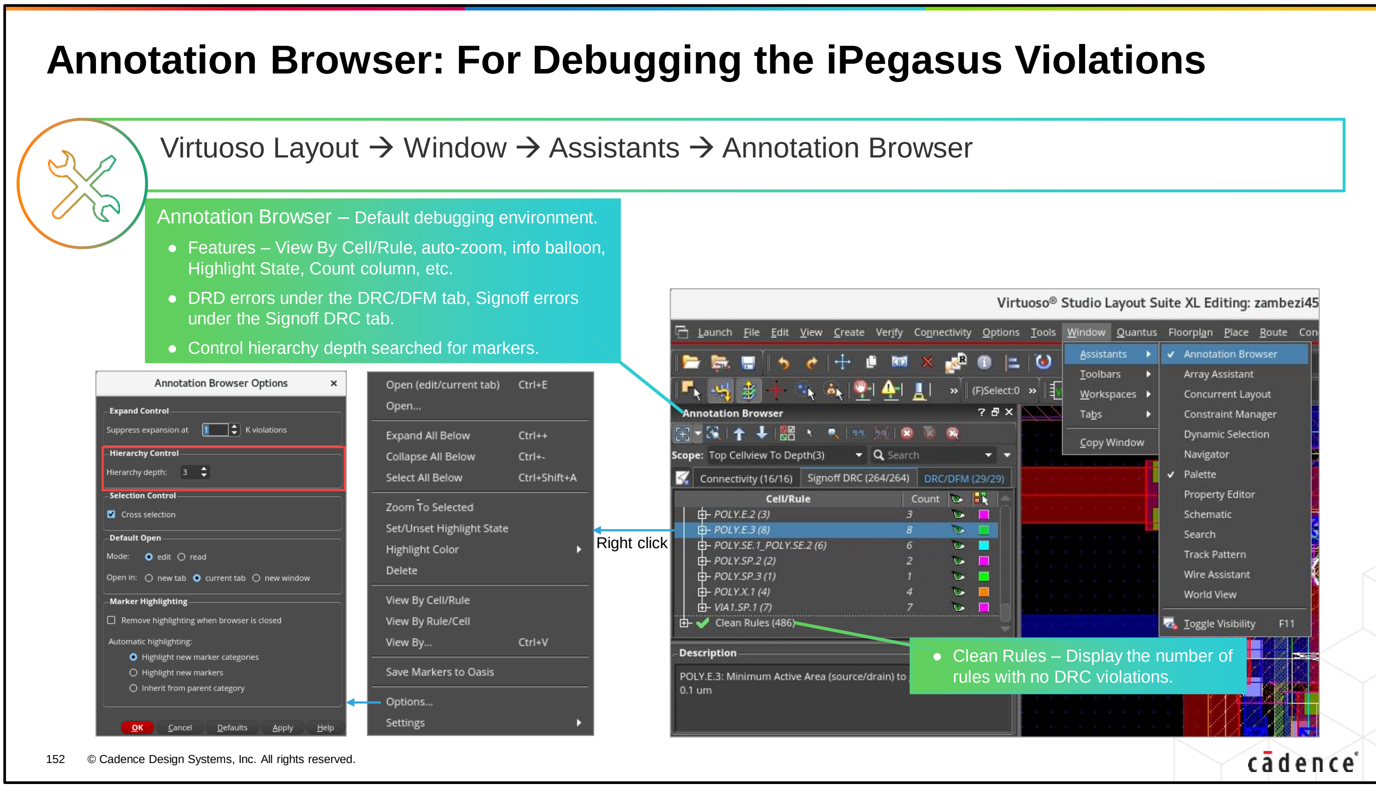
Effective Region of checking = Current editing area + Bloat value

iPegasus DRC Option form → Additional Run Options→ Override Snapshot Bloat





The Virtuoso iPegasus DRC tool allows you to perform Design Rule Checks (DRC) on specific areas of a cell view. The Bloat value is calculated based on the rule constraints and is stored in a snapshot. The maximum Bloat value defines the final checking area by creating a buffer region around the current editing area in both the X and Y directions. As a result, the effective DRC window consists of the current editing area plus the Bloat value during the iPegasus run. You can select the scope of the DRC run from the iPegasus DRC toolbar. Accompanying images illustrate the effective area being checked—accounting for the Bloat value—for various available area mode selections. These images display the final "area of checking" for the modes: current cell view, visible area, and changed area. In these images, 'B' represents the Bloat value defined in the snapshot. The Pegasus engine automatically determines the Bloat value based on the source rule deck. However, users have the option to override the Bloat value for the current session by using the "Override snapshot bloat" field found in the Additional Run Options section of the iPegasus DRC Options form. By default, this field is disabled, but you can enable it in full mode by selecting "Get License" from the DRC toolbar.



The Annotation Browser is the default tool for browsing iPegasus errors. However, for iPegasus DRC runs, users can select the Results Viewer option via a checkbox in the iPegasus options form.
If the Annotation Browser is not visible, you can access it from the Virtuoso layout suite by selecting Window from the drop-down menu, then Assistants, and finally Annotation Browser.
When activated, the Annotation Browser displays violations across various tabs. The iPegasus DRC results can be found in the "Signoff DRC" tab, fill results under the "Signoff Fill" tab, and the DRD results appear in the "DRC DFM" tab.
The Annotation Browser provides reports on markers, allowing you to view and analyze Design Rule Check (DRC), Antenna, Density, Electrical Rule Check (ERC), and fill violations. Selected markers are automatically zoomed in and highlighted in the layout, enabling you to start debugging.
The Annotation Browser has several additional features. You can view violations in two formats: "view-by-cell" and "view-by-rule." The scope of the browser allows you to see hierarchical markers in the hierarchy view, from the top cell down to the specified depth.
The count column shows the number of violations for each rule and can be sorted by the number of violations. There is also a "Clean Rules" option that displays the number of rules with no DRC violations.
To use the info balloon feature, activate it from the Dynamic Display form, which can be accessed via the options drop-down menu. This feature helps list details related to the layout suite, such as the tool name, a brief message, and the violation description.
You can control the highlight state and the color used to render markers on the layout canvas in the browser pane.

Annotation Browser: Advanced Features

- Improved data storage during AB hierarchical traverser.
- CDS Environment Variables:
  1. To fix column sizes in AB.

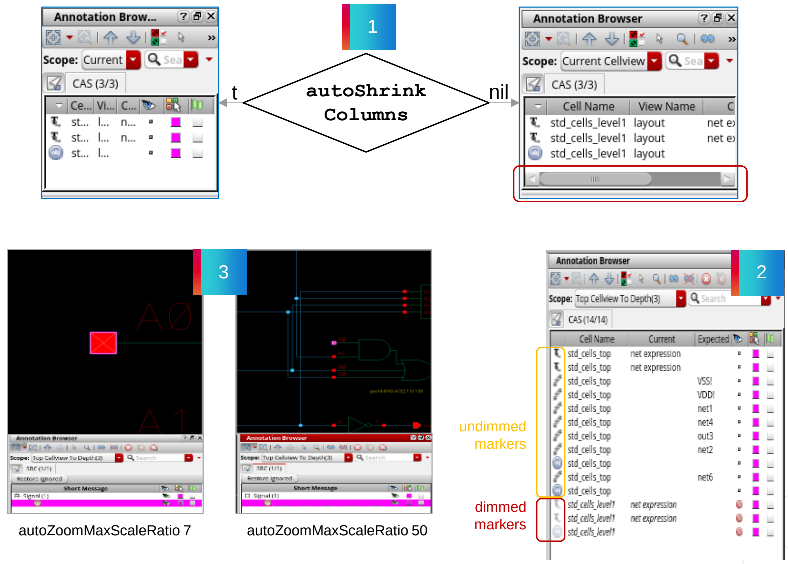
     `autoShrinkColumns t | nil`

  2. To allow descending/ascending cell view by **double**-clicking the dimmed marker.

     `descendOnDoubleClick t | nil`

  3. To define the max zoom factor.

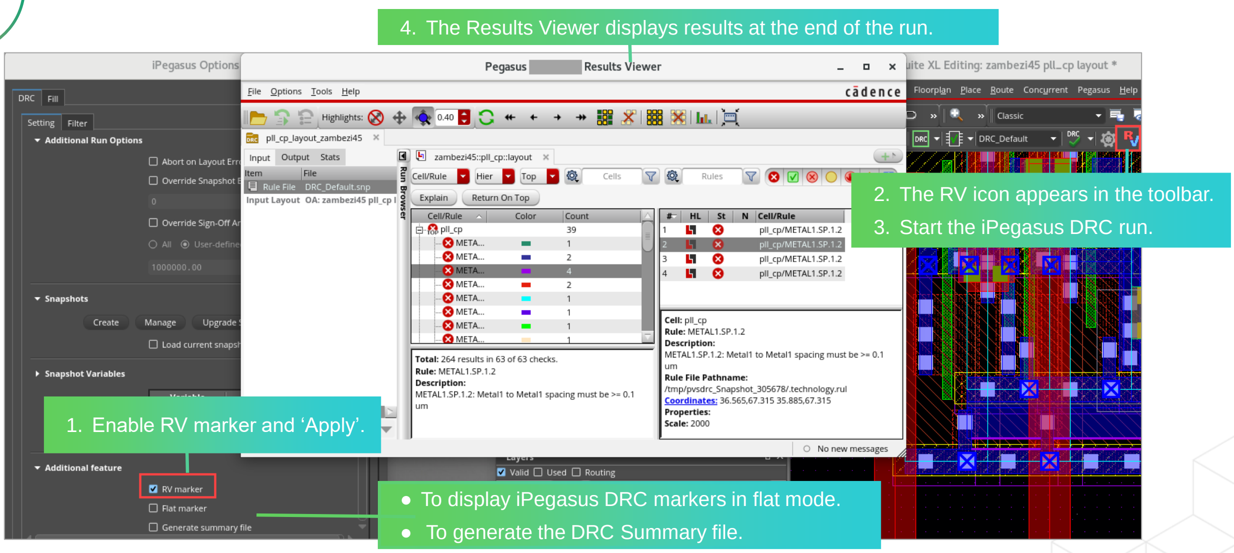
     `autoZoomMaxScaleRatio 7 | <number>`



Let's explore some of the advanced features of the Annotation Browser.
The new algorithm for the Annotation Browser improves how data is stored during hierarchy traversal in large designs, enabling faster navigation through hierarchical paths.
There are specific CDS environment variables designed to address certain tasks. One such variable is "auto Shrink Columns," which helps maintain the size of individual columns even when the window of the Annotation Browser is resized. By default, this variable is set to true (denoted as Boolean t), meaning that when the Annotation Browser is resized smaller, the columns will also shrink. To prevent automatic column resizing, you can set this variable to nil (false).
There are two types of markers in the Annotation Browser: "undimmed" (dark) markers, which indicate cells in the currently edited view, and "dimmed" (light) markers, which represent cells in the top or bottom hierarchies that are not in the current edited view. By setting the CDS environment variable "descend On Double Click" to true, you can enable the functionality to navigate into the cell view associated with a dimmed marker simply by double-clicking it. This action will cause the Annotation Browser to descend into the cell view containing the marker and automatically zoom in. The default setting for this variable is nil (false).
Additionally, the default auto-zoom feature in the Annotation Browser is limited to 7 times the size of the marker box. However, you can adjust this by using the environment variable "auto Zoom Max Scale Ratio," which allows you to increase the maximum zoom level for a better overview of the marker. While the default value is 7, you can set it to a higher zoom factor according to your preference.

Results Viewer: For Debugging the iPegasus DRC Violations

iPegasus DRC Options Form → Additional Features → RV marker



The results from the iPegasus DRC can be viewed in the Results Viewer as well.

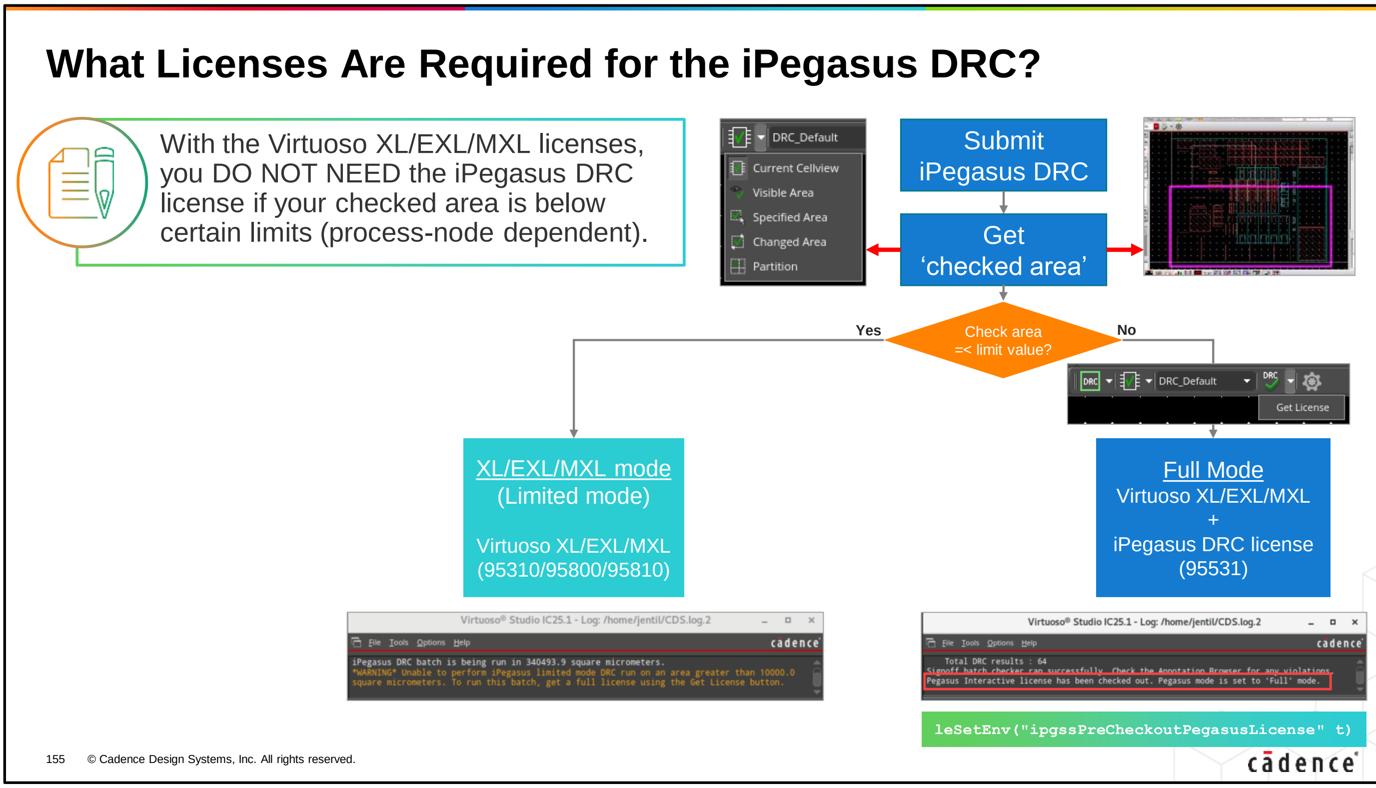
To enable the RV option, select the RV marker from the Additional Features section in the iPegasus DRC Options Form and click "Apply." The RV icon will only appear in the toolbar if the RV option is activated.

Start the run.

If you select the RV option, the run will display markers in the Results Viewer. If you do not select the RV option, the markers will be visible in the annotation browser instead.

Select the Flat marker from the Additional Features section in the iPegasus DRC Options Form to display the DRC markers flat. By default, this option is not selected, and iPegasus displays the markers hierarchically.

You have the option to generate a DRC Summary file. If you choose to select the "Generate Summary File" option, iPegasus DRC will create a summary of the DRC run. When this option is enabled, a warning message will inform you about the potential impact on performance.



What licenses are required to run iPegasus DRC for Virtuoso Studio?

iPegasus supports window-based DRC, which allows you to run DRC on a selected checked area. To better understand the different modes of iPegasus DRC, it's important to define a checked area.

The checked area refers to the specific area on which the DRC is applied. More specifically, it is the area sent by iPegasus DRC to the Pegasus engine, excluding the bloat extension. The size of the checked area varies depending on the area mode selection in iPegasus and is independent of the process nodes.

Virtuoso Studio is available in three license versions: XL, EXL, and MXL. The license for in-design DRC is known as Cadence iPegasus DRC for Virtuoso Studio, and this license is backward compatible with older versions of Pegasus Interactive and Virtuoso i-PVS.

The iPegasus DRC can operate in two modes - the XL-EXL-MXL mode and the full mode.

The XL-EXL-MXL mode, also referred to as the limited DRC mode, does not require an iPegasus DRC license. In this mode, if you possess a Virtuoso XL, EXL, or MXL license, and the checked area falls within specified limits, you can verify your designs for DRC with certain restrictions, without the need for an iPegasus DRC license.

However, if the checked area exceeds the set limit, an iPegasus DRC license is required. This mode is known as Full mode, allowing you to run DRC regardless of the checked area's size. The size of the checked area will be displayed, enabling the user to choose the appropriate iPegasus mode based on design requirements.

Let's consider a real design scenario:

Open your design in a fresh setup of Virtuoso Studio. During this initial stage, the iPegasus DRC license is not checked out, and the system operates in XL-EXL-MXL mode. You can make area selections and start the sign-off DRC. The run will proceed if the checked area is within the specified limits, and the results will be loaded into the selected viewer. If the checked area is too large, the run will not begin, and the Command Interpreter Window will indicate that the tool cannot perform iPegasus limited mode DRC on an area larger than the limit. To run this batch, you will need to obtain a full license by selecting the "Get License" option from the run iPegasus drop-down on the toolbar.

Once you switch to Full mode, the "Get License" field will become inactive, and you cannot revert to XL-EXL-MXL mode during the same session. You may also use the environment variable, iPegasus PreCheckout Pegasus License, to check out the license.

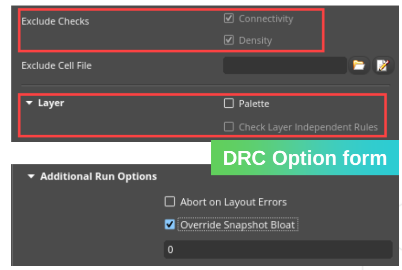
Once the iPegasus DRC license is checked out, you can click the "Run iPegasus DRC" icon to initiate the iPegasus DRC run in Full mode. All relevant details, including the checked area, Pegasus license mode, and run status, will be displayed in the CIW window.

iPegasus DRC for Virtuoso Studio: License Modes

Comparison

Features	iPegasus DRC XL/EXL/MXL Mode	iPegasus DRC Full Mode
iPegasus DRC for Virtuoso Studio license (#95531)	Not checked-out	Checked-out
DRC Area scope	Limited area (*node-based*)	Unlimited area (*with 100x threshold warning*)
Design use model	Limited area check	Block/chip level
CPUs	4	4
Connectivity check	Off	On/Off (*controllable*)
Density check	Off	On/Off (*controllable*)
Palette feature	ON	On/Off (*controllable*)
Layer independent rules check	Off	On/Off (*controllable*)
Snapshot support	Yes	Yes
Override Snapshot Bloat value	No	Yes
Foundry Deck Support	PVS and Pegasus	PVS and Pegasus

Window-Based DRC (*XL/EXL/MXL Mode*)	
Process Node	Area Limit
2 - 7 nm	25um2
10 - 22 nm	100um2
28 - 180nm	10000um2



DRC Option form

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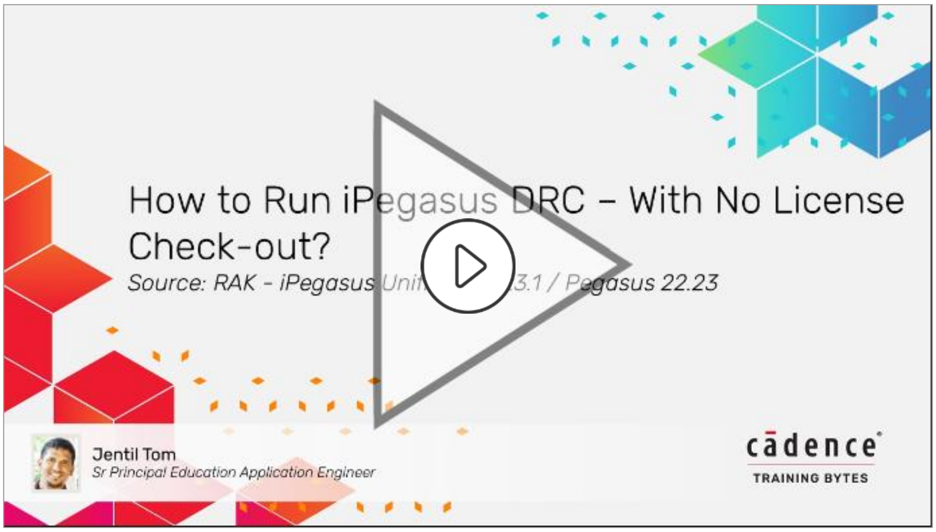
The table compares two modes of iPegasus DRC: XL-EXL-MXL Mode and Full Mode.
XL-EXL-MXL Mode, often referred to as “limited mode,” offers restricted access to certain functionalities, features, and areas. In contrast, Full Mode provides complete access to all functionalities. However, a valid Pegasus path must be defined in both modes.
In XL-EXL-MXL Mode, you do not need an iPegasus DRC license to run signoff DRC on a checked area, as long as it falls within specific limits. This area limitation is based on the process node. For further details, refer to the table on the right-hand side.
There are two brackets for advanced nodes and one for mature nodes. The limited mode area allowance is set at 25 square microns for nodes of 7 nanometers and below. You can verify your DRC without an iPegasus license if the checked area is 25 square microns or smaller. For nodes between 8 and 22 nanometers, the area allowance is 100 square microns or less. For mature nodes of 23 nanometers and above, the checked area is limited to 10,000 square microns or below.
XL-EXL-MXL Mode is suitable for smaller designs and selected area checks without needing an iPegasus DRC license. For example, if you want to run DRC on a design at 45 nm technology over an area of 7,850 square microns, you will not need an iPegasus license.
However, you will require a checked-out iPegasus DRC license in Full Mode, which allows higher scope and more features for in-design DRC checks. Full Mode is suitable for block and chip level signoff.
However, it's important to note that even in full mode runs, the iPegasus DRC will issue a warning message if the DRC run area is too large, which may impact performance. The threshold area size is set to 100 times the limited mode area allowance of the respective node. Now, let's look at how specific features differ between limited and full modes. Under the Filters tab of the DRC Options form, density and connectivity checks are disabled in limited mode, and the corresponding checkboxes cannot be modified. In contrast, these checks become active and selectable in Full Mode, which can be enabled by selecting "Get License" from the DRC toolbar. The Palette feature instructs iPegasus to run DRC only on the layers visible in the Virtuoso Layer Palette. This feature is set to be internally always ON in limited mode, whereas it can be controlled in full mode runs. However, the Layer Independent Rules check is available only in Full Mode. In Full Mode, you can override the bloat value set for the selected snapshots, while this can’t be done in Limited Mode.

Demo: How to Run iPegasus DRC – With No License Check-out

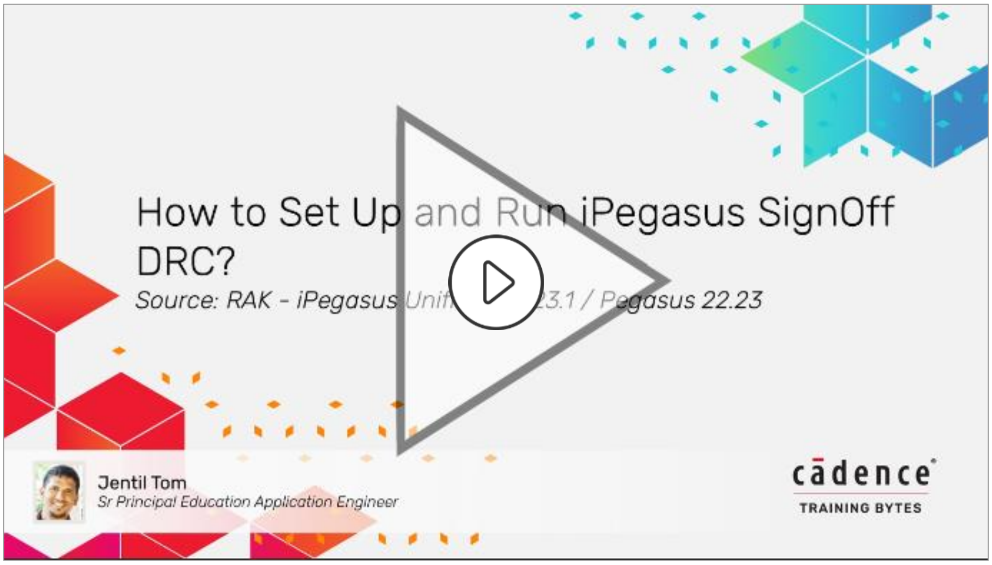


Video Play Time: **4.43** minutes

Click the Play button to start the video.

Click the link How to Run iPegasus DRC – With No License Check-out? to play the video on support.cadence.com.

Demo: How to Set Up and Run iPegasus SignOff DRC



Video Play Time: **5.46** minutes

Click the Play button to start the video.

Click the link [How to Set Up and Run iPegasus SignOff DRC?] to play the video on support.cadence.com.

iPegasus DRC Versus Pegasus Batch Signoff DRC

	iPegasus DRC	Pegasus Batch Signoff DRC
Virtuoso Integration	Tightly integrated	Loosely integrated
Use model	In-design DRC on partial layouts	DRC run on complete layouts
Stage of run	Early design stages	Final signoff verification
Connectivity/Density	Disabled (default)	Enabled
Rule groups – snapshot	Enabled	Disabled
Accuracy	Signoff quality	Signoff quality
Hierarchy Range	Current cell view only	Flat or Hierarchical
Marker limit per rule	5000 (default)	1000
Total Marker Limit	5000 (default)	Unlimited

Let's compare the Virtuoso-integrated iPegasus DRC versus the Pegasus Batch signoff verification.

Virtuoso iPegasus and the Pegasus Batch mode have two different use models.

Virtuoso iPegasus DRC is an in-design feature that provides access to signoff quality DRC checks at the early stages of layout development.

iPegasus DRC could be used during the design implementation phase or Signoff Correction to prevent the user from inadvertently creating errors.

However, at the end of the design phase, the final sign-off Pegasus physical verification is recommended.

What are the appropriate rules for iPegasus DRC?

iPegasus DRC supports almost all DRC rules. However, the Virtuoso iPegasus exempts a few expensive rules that require complete connectivity or demand larger layout areas from the default runs. The flow doesn't recommend connectivity-related Antenna or Density rules, area fill-dependent density, DFM, or Lithography checks during the initial runs. For example, it doesn't make sense to run density checks on partial layouts with no dummy fills. When the layout is ready, change the setup and run the complete rule deck.

The idea behind Virtuoso iPegasus is to run a set of DRC rules based on layout readiness, get results quickly, fix issues, then move on.

The customized rule groups, called snapshots, can be enabled or disabled in Virtuoso iPegasus, based on layout development steps.

Pegasus Batch runs usually run on complete layouts. At the end of the layout development, the Pegasus Batch run must be performed for final signoff verification with the complete rule deck.

On accuracy, both Pegasus Batch and Virtuoso iPegasus are 100% correlated. Both flows use a foundry-certified signoff quality rule deck for verification.

On the hierarchy range of the run, while Pegasus Batch can be run on flat or hierarchical, the iPegasus runs DRC on the current cell view in flat mode.

The default values can be modified in the Virtuoso iPegasus GUI on the number of markers.

Frequently Asked Questions

iPegasus DRC

In the iPegasus DRC, how to change the default bloat value for a snapshot?

Use the field iPegasus DRC Options form – Setting – Additional Run Options – Override Snapshot Bloat.

During the iPegasus DRC runs, how to change the number of markers and errors displayed in the error browser?

Alter the following values:

1. iPegasus DRC Options form – Setting – Scope – Marker Limit Per Rule / Total Marker Limit.
2. Equivalent CDS environment variables: drdVioLimitPerRule, drdBatchVioLimit.

Here are a few frequently asked questions on Virtuoso iPegasus and their answers.

The bloat value defines the buffer region around the current editing area, which sums up the effective region of checking for Interactive runs.

When running Virtuoso iPegasus, by default, the DRC Viewer will only show 5000 errors per rule, and 5000 errors in total. The two limit values can be changed in the DRC Run Options form, by altering the fields, Marker Limit Per Rule, and Total Marker Limit, respectively.

Submodule 5-2

iPegasus FILL for Virtuoso Studio

Submodule on iPegasus Fill for Virtuoso Studio.

Submodule Objectives

In this submodule, you

- Define iPegasus Fill flow for Virtuoso Studio
- Compare the traditional Fill flow and the iPegasus Signoff Fill flow
- Discuss the features and options in the iPegasus Signoff Fill form
- Set up and generate the iPegasus Signoff Fill
- Exclude selected Cells from iPegasus Signoff Fill
- Merge the fills from multiple iPegasus runs
- Create Symmetrical Fill with iPegasus Signoff Fill

Submodule Objectives.

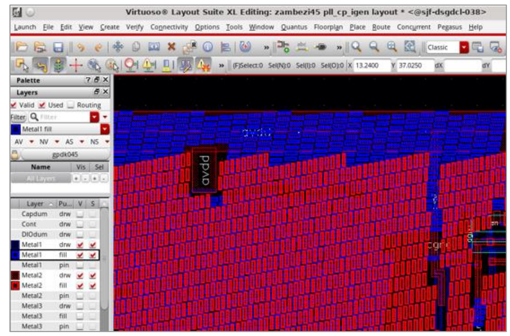
In this submodule, you will Define iPegasus Fill flow for Virtuoso Studio, Compare the traditional Fill flow and the iPegasus Signoff Fill flow, Discuss the features and options in the iPegasus Signoff Fill form, Set up and generate the iPegasus Signoff Fill, Exclude selected Cells from iPegasus Signoff Fill, Merge the fills from multiple iPegasus runs, and Create Symmetrical Fill with iPegasus Signoff Fill.

What Is the iPegasus Fill for Virtuoso Studio?

Features

With the iPegasus Fill for Virtuoso Studio, you can generate dummy fills using the Pegasus signoff engine based on the foundry-certified fill decks.

- Supports a qualified foundry fill deck.
- Complete snapshot flow support – Quicker filling.
- Generated fills are mapped (into the block lib/cell/view) and merged into the OA design.
- Fill with selected layers over the selected area.
- Symmetry fills.
- Incremental fill: Re-fill the modified area.
- Support for blockage layers.
- Net aware feature – For protecting critical nets
- License: Cadence iPegasus FILL for Virtuoso Studio (#95532)



What Is Virtuoso iPegasus SignOff Fill?

The iPegasus SignOff Fill is an advanced fill solution feature, part of the iPegasus package, with which you can generate dummy fills using the Pegasus sign-off engine, based on foundry-certified fill decks.

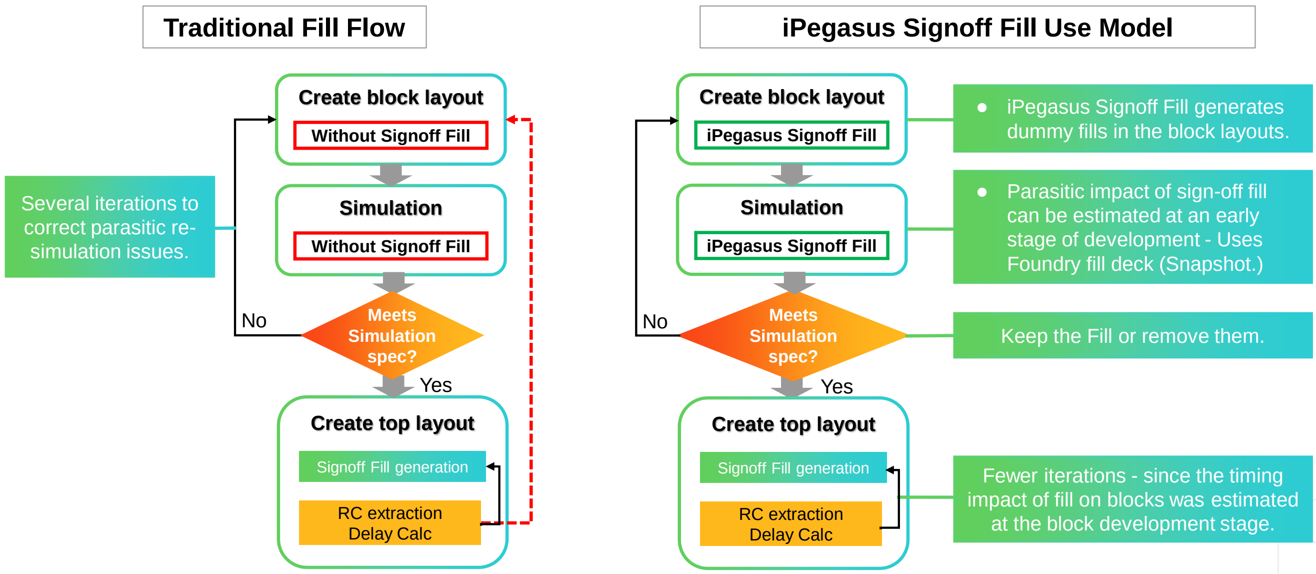
The SignOff Fill step is tightly integrated into the Virtuoso iPegasus flow.

For the iPegasus SignOff Fill, you may generate dummy fills, using complete foundry fill decks, or a customized set of rule decks called snapshots. The SignOff Fill feature is snapshot-based. Since the rule parser stage is eliminated for snapshots, filling on advanced nodes takes less run time.

The tool automatically maps generated dummy fills into the Virtuoso OA database. The generated fills are immediately included in the block library-cell-view. There is no need to merge the fills with the design manually.

The users have the option to select the layers to fill. The user can define the area of signoff fill, using scope options in the fill GUI. The flow supports symmetry fills with respect X or Y axis, or a point. Virtuoso iPegasus SignOff Fill also has blockage support – the blockage layers must be defined with the foundry fill deck.

Flowchart: iPegasus Signoff Fill for Virtuoso Studio Flow



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Let’s compare and see how the Virtuoso iPegasus Signoff Fill Flow fares viz-a-viz the traditional fill flow.

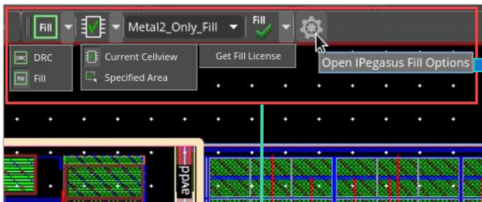
A typical fill flow starts with block layout creation, followed by simulation. If simulation results meet the specifications, the designer proceeds to the top-level layout creation with Signoff Metal Fill. The cycle continues until the final layout with fill. In the traditional Fill flow, several iterations are required, to achieve acceptable simulation results, that meet RC extraction specifications, if issues are found during parasitic re-simulation.

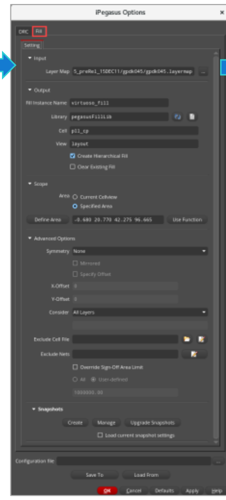
To reduce the number of iterations during cell or block development, you can use the iPegasus Signoff Fill feature, to generate dummy fills inside the block, and estimate the timing impact of dummy fills at early development stages. If simulation results meet the specifications, the designer may keep generated fill, and proceed to top-level layout creation. If not, remove the fill, and repeat the cycle till the timing specs are met. Since the iPegasus Signoff Fill flow helps estimate the parasitic impact of metal Fill, at an early development stage, it results in fewer fill iterations.

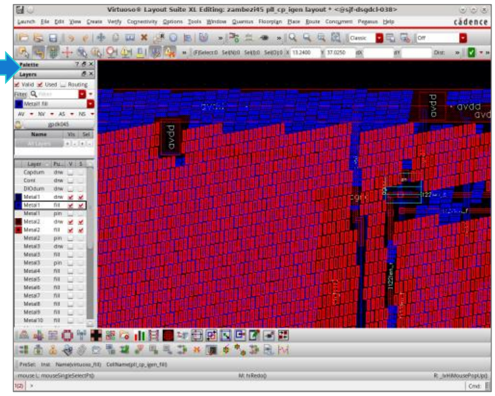
Invoking the iPegasus Signoff Fill Form

Virtuoso environment → iPegasus Fill toolbar → Open iPegasus Fill Options

A standalone functionality. No dependency on the iPegasus DRC toolbar.







iPegasus Fill Toolbar

1. iPegasus tool Selection – DRC/Fill
2. Scope selection
3. Snapshot selection
4. Click ‘Fill’ to generate iPegasus Fill.
5. Open iPegasus Fill options form.

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Let's explore the technology and user interface setup to generate the iPegasus Signoff Fill.

In the iPegasus Fill toolbar, click the rightmost icon to open the iPegasus Fill options form. The toolbar also includes fields for selecting the scope of the run and the snapshot, which are necessary for starting the fill generation process.

To begin the iPegasus Fill process, access the iPegasus Fill options form. From there, configure the various sections within the Fill form before generating the fill structure.

Setting Up the iPegasus Signoff Fill

Set up Signoff Fill form – input, output, scope, advanced options, and snapshot sections.

1. Set Layer map file
2. Enter output fill name, lib, etc.
3. Choose Fill Area (Optional)
4. Select Fill Layer, symmetry, etc.
5. Create Blockages (Optional)
6. Create/select Fill Snapshot(s
7. Generate iPegasus Fill

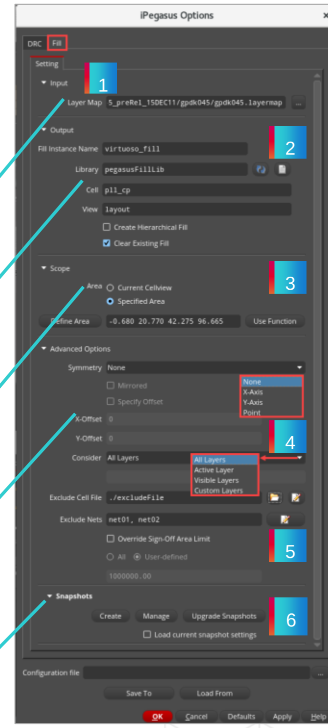
Input required
- layer map file – option to load custom file.

Output Fill
- Fill instance/library/cell/view names.
- Click for Hierarchical fill output.
- Click to clear old fill.

Scope of fill
- Current cell view or user-defined area.
- SKILL function: *pgssUserSelectedFillArea*

Advanced Options
- Fill symmetrically or mirrored along XY axis.
- Set an offset from the XY axis.
- Choose the layer(s) to fill
- Blockage – Exclude cells/nets from fill.

Snapshot creation – GUI and command
- Create/manage/upgrade snapshots.
- Option to save/load configuration.



How to Set Up the iPegasus Signoff Fill Creation?

To set up the iPegasus Fill, configure it using various sections of the iPegasus Fill options form. These sections include Input, Output, Scope, Advanced Options, and Snapshot.

The Layer Map file can be selected from the technology directory. This file defines the mapping between the fill layer and the layer-purpose pair to backport generated dummy fills into OA technology.

The metal fill is created within a cell, overlapping the existing design. In the Output section, specify the instance name, cell name, and the view name for the Metal Fill cell to be created. The default view is the layout. Specify the library where the Metal-Fill cell will be created. If the PDK rule deck supports it, there is an option to create a hierarchical fill output. For example, the GPDK45 rule deck does not support hierarchical fills, but advanced nodes do.

You can choose to clear the existing fill and recreate a new one; otherwise, iPegasus will respect the current Metal Fill and only generate new fills if necessary. Be aware that this feature applies only to the layers considered for the run. For instance, if you select the 'Clear existing fill' option and generate fills only for Metal 2 during a specific run (with the settings adjusted to consider visible layers while making only Metal 2 visible in the layer palette), then only the pre-existing Metal 2 fills will be cleared before generating a new Metal 2 fill. Pre-existing fills for Metal 1 or Metal 3 will remain unaffected.

Regarding the scope of the Signoff fill, you have two options. You may either fill the current cell view or define a specific area to be filled. For specific area selection, you need to enter the area coordinates or define them by selecting the area from the layout using your mouse. You can also specify a complex polygon area using the SKILL function `Pegasus-User-Selected-Fill-Area`. Load this function into the CIW and select some cells in the design. The area for fill generation will be determined by the selected cells on the layout canvas once you click the Use Function button on the Pegasus Signoff Fill form.

In the Advanced Options section, area-specific filling includes an additional feature known as symmetrical fill. This feature allows you to fill in a symmetrical manner along the x-axis, y-axis, or at a point. X and Y axis fill can be related to mirroring, whereas point-based symmetry can be associated with a common-centroid approach. Additional options include ensuring mirrored fill and specifying an X or Y offset for symmetry selections.

The next feature is layer-based filling. Use the options under the "Consider Layers" field to define which layers will be used in the fill run. Four options are available:

Select All Layers to process every layer. Select Active or Visible Layers to generate Signoff fills for the currently active or visible layers on the Virtuoso layer palette, respectively. Select Custom Layers to generate dummy fills only for user-defined layers.

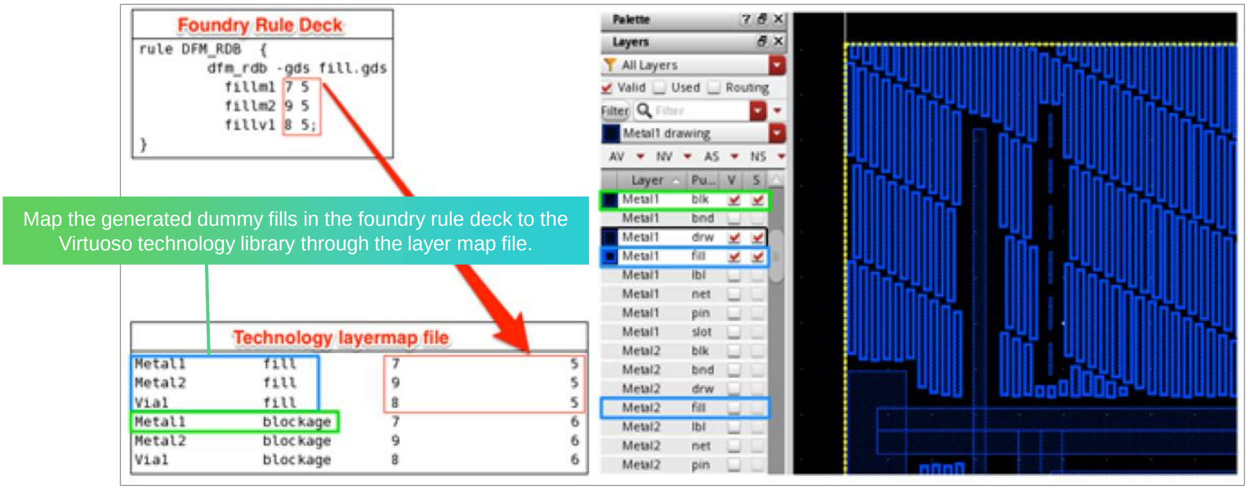
You can use the Exclude Cells field to specify cells that you want to exclude while running the Metal Fill. You may choose a file containing a list of cells, create a new one, or update an existing one through the Exclude Cells File Editor. The Exclude Nets field enables the Net Aware Feature, which helps create fills with an extra offset around critical nets.

Lastly, in the Snapshot section, the iPegasus Signoff Fill feature is snapshot-based. A snapshot captures design rules in a compiled binary format that allows Pegasus direct access to fill rules without needing to recompile them. Click on "Create" to open the Snapshot Creator form. The Snapshot Creator allows you to create a snapshot interactively. The generated snapshots will be listed under the snapshot drop-down menu in the iPegasus toolbar. The options form also provides additional features for managing and updating snapshots.

Once you have made your selections, click 'Apply' in the iPegasus options form to apply the changes.

Finally, return to the iPegasus toolbar, select the snapshot you want to use, and click the 'Run iPegasus Fill' icon to initiate the Signoff fill generation. At the end of a successful run, the fill cell is created under the specified library and gets automatically instantiated over the design. The CIW will display the summary of the generated fill with details such as how many fill shapes have been generated for each Metal layer.

iPegasus Signoff Fill for Virtuoso Studio
Technology Setup



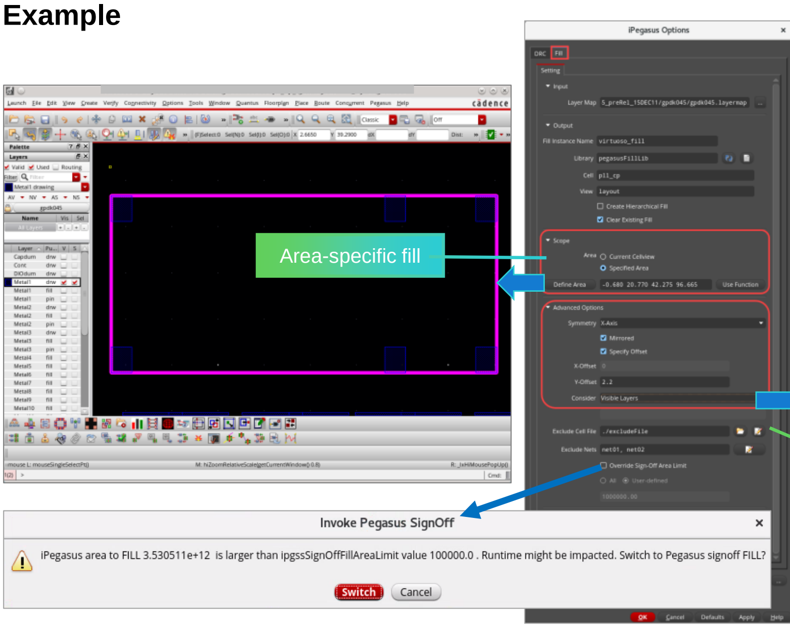
Let’s examine the technology setup for the Signoff fill process.

To integrate the generated dummy fills into the OA technology layer map file, it is essential to establish a clear mapping between the fill layer and its corresponding layer-purpose pair. Proper mapping must exist between the generated dummy fills in the foundry rule deck and the Virtuoso technology library; without this mapping, the generated dummy fills will not appear in Virtuoso. Therefore, a correctly configured technology layer map file and an accurate foundry rule deck are critical for the Signoff fill process.

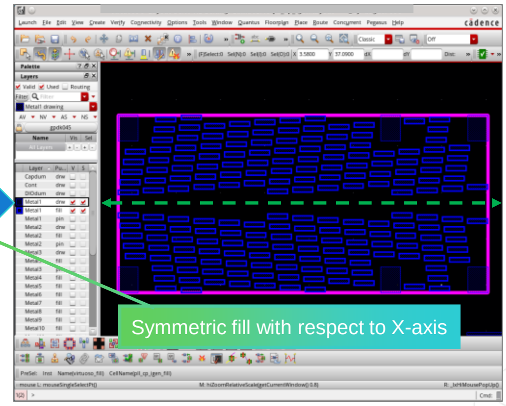
It is important to note that blockage layers should not be considered during the filling process. If blockage layers are present, they must be defined in the layer definition and layer map section of the foundry fill deck, alongside the fill layers.

iPegasus Signoff Fill for the Virtuoso Studio

Example



iPegasus Fill Area Threshold Warning - Ref Table	
7nm & below	25um2 x 100
8nm – 22nm	100um2 x 100
28nm – 180nm	10000um2 x 100



Here are some examples of the advanced features of the iPegasus Signoff Fill.

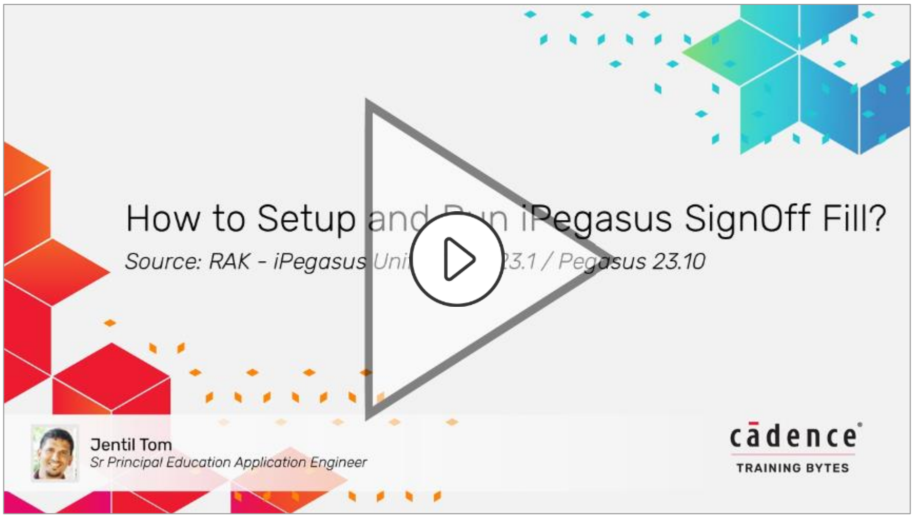
The demo on the left showcases the area-specific Signoff Fill. With the scope definition features in the iPegasus Fill Options Form, you can specify a particular area to be filled. You can either manually enter the area coordinates or use your mouse to select the area on the layout. The selected portion will be highlighted, and the fill will only occur within that area.

iPegasus operates in two DRC modes: Limited and Full. In Full mode, iPegasus Fill issues a warning if the fill run area is too large, which could impact performance. The warning threshold is set at 100 times the node's Limited mode area allowance. If the fill area is below this threshold, the run will proceed as normal. If it exceeds the threshold, an interactive window will appear, offering options to 'Switch' to a standard Pegasus Fill or 'Cancel' the form to continue with the iPegasus Fill run. Choosing the switch option will open the Pegasus Fill run submission form. You can adjust or disable this area check in the iPegasus Fill Options Form by using the 'Override Sign-off area limit' field.

On the right-hand side, the area-specific fill feature includes an additional symmetrical fill capability. In this example, the selected area is set to be filled with Metal 1, symmetrically along the X-axis, with the mirrored option enabled. As shown in the resultant fill output, within the selected region, the Metal 1 layers are filled symmetrically along the X-axis, with the top half being a mirror image of the bottom half. Symmetrical and mirrored fills are crucial for advanced analog blocks, where matched devices require appropriate matching fill structures to ensure signal integrity.

It's important to carefully select the fill area around symmetrical structures. Running symmetry fill on a non-symmetrical structure could lead to DRC violations or overlaps in metal fill.

Demo: How to Setup and Run iPegasus SignOff Fill?

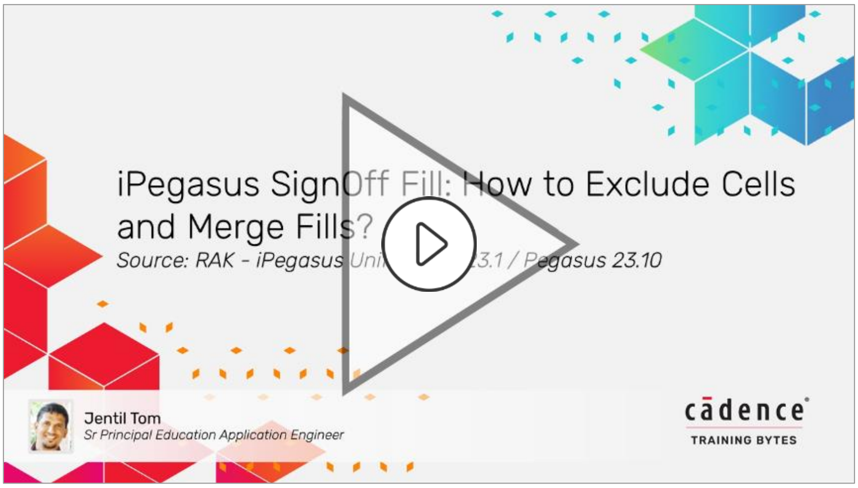


Video Play Time: 3.52 minutes

Click the Play button to start the video.

Click the link [How to Setup and Run iPegasus SignOff Fill?] to play the video on support.cadence.com.

Demo: iPegasus SignOff Fill: How to Exclude Cells and Merge Fills?

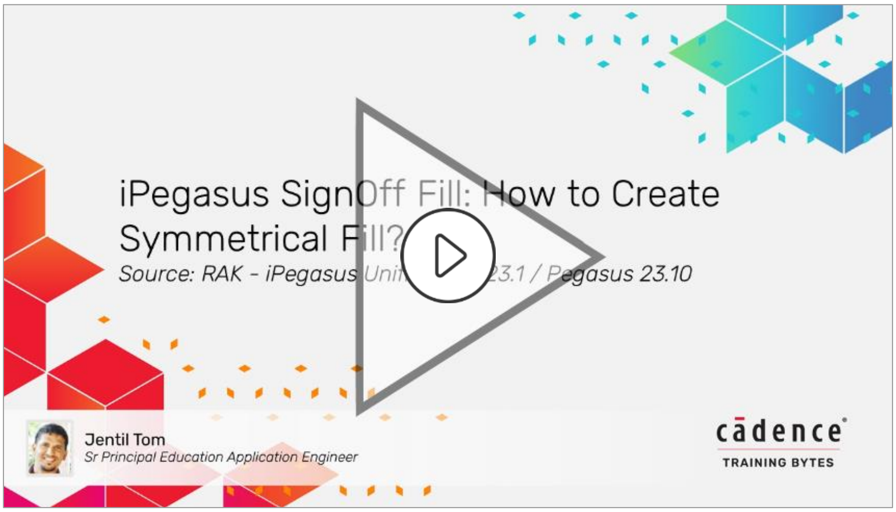


Video Play Time: **3.54** minutes

Click the Play button to start the video.

Click the link [iPegasus SignOff Fill: How to Exclude Cells and Merge Fills?] to play the video on support.cadence.com.

Demo: iPegasus SignOff Fill: How to Create Symmetrical Fill?



Video Play Time: **3.34** minutes

Click the Play button to start the video.

Click the link [iPegasus SignOff Fill: How to Create Symmetrical Fill?] to play the video on support.cadence.com.

Labs

Lab 7-1 Setting Up and Running iPegasus Signoff DRC

Lab 7-2 Setting Up and Running iPegasus Signoff Fill

After visiting the course content, run these labs to test your understanding.

Module 6

Course Conclusions

This page does not contain notes.

Summary

In this course, you learned to

- Manage database libraries
- Analyze and customize the Virtuoso® graphic environment
- Use the Palette Assistant
- Use the custom bindkeys
- Use the Toolbar Manager
- Use the basic commands available in the Virtuoso Layout Suite

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References

Refer to the links below for further details on the topics in this course.

[Virtuoso Layout Viewer User Guide Product Version IC25.1]

[Virtuoso Layout Suite XL: Basic Editing User Guide Product Version IC25.1]

[Virtuoso Studio Design Environment User Guide IC25.1]

[Virtuoso Layout Suite SKILL Reference IC25.1]

[Cadence User Interface SKILL Reference IC25.1]

Virtuoso Layout Viewer User Guide Product Version IC25.1

Virtuoso Layout Suite XL: Basic Editing User Guide Product Version IC25.1

Virtuoso Studio Design Environment User Guide IC25.1

Virtuoso Layout Suite SKILL Reference IC25.1

Cadence User Interface SKILL Reference IC25.1

Module 7

Next Steps

cadence

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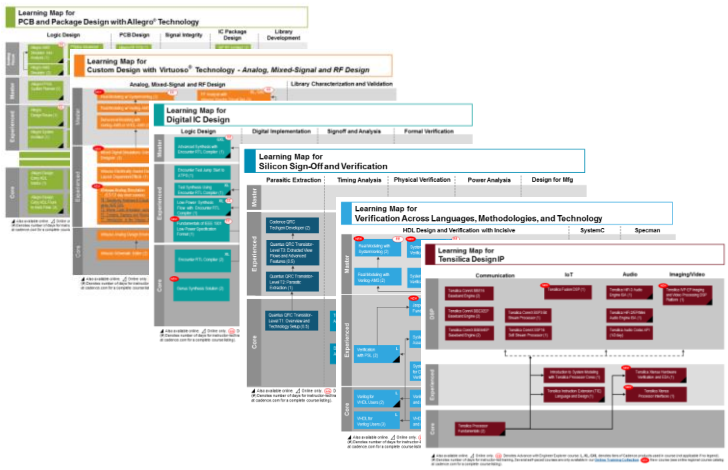
Learning Maps

Cadence® Training Services learning maps provide a comprehensive visual overview of the learning opportunities for Cadence customers.

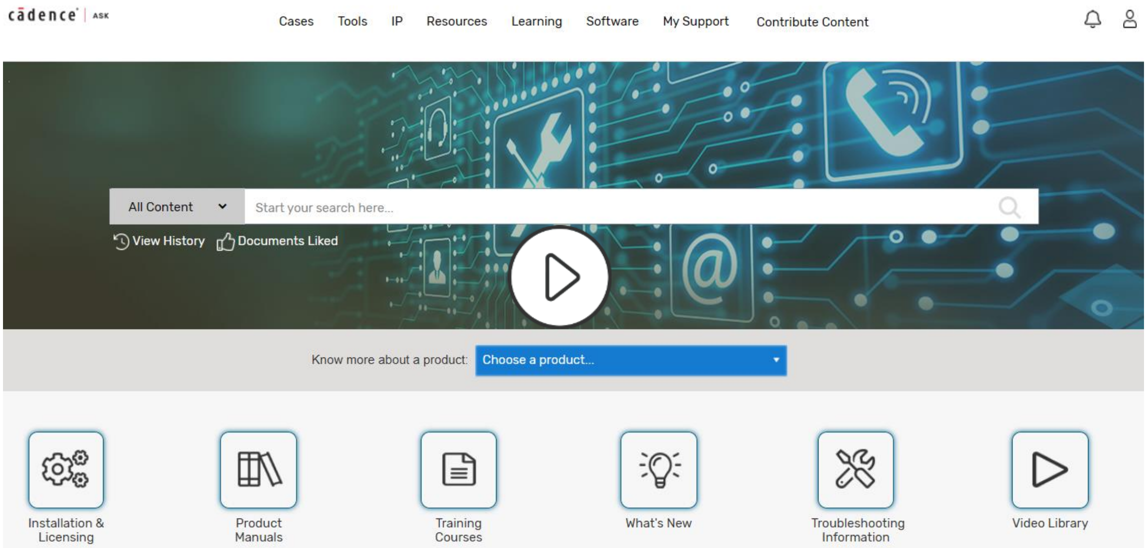
Click here to see all our courses in each technology area and the recommended order in which to take them.



Go here to view the learning maps:

http://www.cadence.com/Training/Pages/learning_maps.aspx

Cadence ASK – Learning and Support Portal



Cadence Ask now includes over 2000 product/language/methodology videos (“Training Bytes”)!

Click here to view a demo of Welcome to Cadence ASK – Learning and Support Portal.

Wrap Up

- Complete Post Assessment, if provided
- Complete the Course Evaluation
- Get a Certificate of Course Completion

Thank you!

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